

The Pennsylvania State University

The Graduate School

**DEVELOPMENT OF THE ANALYTIC CAPABILITIES OF NEAR-ZERO FIELD
MAGNETORESISTANCE SPECTROSCOPY THROUGH EXPERIMENT AND
MODELING OF THE STOCHASTIC QUANTUM LIOUVILLE EQUATION**

A Dissertation in

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by

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ABSTRACT

The primary object of this research is to further develop the technique of near-zero field magnetoresistance (NZFMR) as a spectroscopy technique that approaches the capability of conventional electron paramagnetic resonance (EPR) and electrically detected magnetic resonance (EDMR) in determining the structure and chemistry of performance limiting atomic-scale defects in semiconductor devices and semiconductor materials and insulators. Although the ultimate analytical power of the NZFMR technique may never fully match that of the more conventional resonance techniques, it offers a simplicity in the measurement apparatus which could greatly decrease the cost of its implementation in comparison to EPR and EDMR. This work provides a comprehensive overview of both the qualitative and quantitative models built from the stochastic quantum Liouville equation to describe and model spin-dependent transport in both EDMR and NZFMR measurements providing the explicit theoretical connection between the defects observed in EDMR and NZFMR measurements. This work also demonstrates the experimental utility of NZFMR technique with results from studies of isotopic dilution and forming gas anneals on Si/SiO₂ metal-insulator-semiconductor capacitors. These results show the explicit, experimental connection between the defects observed in EDMR and NZFMR. Lastly, this work presents a method by which to analyze and interpret NZFMR results and then extends that same method to analyze and further interpret EDMR results. Thus, this work also merges the theory and experiment by demonstrating the explicit connection between the defects observed with the EDMR measurement, the NZFMR measurement, and both the qualitative and quantitative theory that describes their responses.

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Twenty-Three Times

Fedor V. Sharov

*Sample, Sample, on a Tee,
I'm on rethread twenty-three.*

*Could you please just make a bond?
I have not done you ANY wrong.*

*My anger rises, red I see.
Blood pressure spikes like an I-V.*

*But wait! It clicked! The wire bond hit!
These settings are a perfect fit!
I've done the gate and now I go,
to stress her; hope it won't blow.*

*But look, oh no! Lo and behold!
It seems the glue just could not hold...
My head, it hurts. My fingers, sore.
I'm now on re-thread twenty-four.*

Chapter 1 Introduction

As the semiconductor industry races to keep pace with Moore's law of doubling the number of transistors that can be placed on a chip, the requirements for advances in the downscaling of transistor technology becomes ever more challenging. Today, devices are fabricated on extremely small scales with new and more complex materials systems. At such small scales, individual atomic scale defects are responsible for the improper function of these devices. Still to this day, devices are characterized by basic electrical measurements and then optimized via fabrication trial and error. Despite being a successful approach to design a working device, the approach can take several years to optimize the devices. To expedite this design process, it is best to understand the physical mechanisms and root-causes which lead to the formation of these atomic scale defects in the first place. With this physical understanding, process optimization could be accelerated while development costs of new technology could be reduced. This requires, new, innovative, and fast root-cause identification methods and tools which are easily can be easily integrated into current testing platforms.

1.1 Atomic Scale Defect Detection through Electron Paramagnetic Resonance & Electrically Detected Magnetic Resonance

Several techniques exist which allow scientists and engineers to study and understand performance limiting defects in fully processed devices. However, electron paramagnetic resonance (EPR) has been the premiere technique used to determine the physical and chemical nature of paramagnetic point defects in semiconductor materials, devices, and insulators [1]–[7]. Quantifying and understanding the nature of these defects is crucial in determining how to eliminate them and further advance modern micro- and nano-electronic devices. However,

when studying fully formed micro- and nanoscale electronic devices, conventional EPR is limited in both its sensitivity and selectivity to defects directly involved in device performance. A derivative technique, electrically detected magnetic resonance (EDMR) has provided great insight into atomic scale defects which are directly involved in device performance [8]–[10]. EDMR takes advantage of spin-dependent currents in the active region of devices to overcome the sensitivity and selectivity shortcomings of conventional EPR.

1.2 Near-Zero Field Magnetoresistance Spectroscopy

Recently, measures of spin-dependent recombination (SDR) currents and spin-dependent trap-assisted tunneling (SDTAT) currents observed through low-frequency EDMR measurements have displayed magneto-resistive effects near zero magnetic fields. This near-zero field magnetoresistance (NZFMR) response has been shown to be independent of the oscillating magnetic field utilized in both conventional EPR and EDMR making the NZFMR measurement much simpler to perform than both the EPR and EDMR measurement. Additionally, the NZFMR response has been shown to often closely scale with the amplitude of the EDMR response in studies of device stressing [11], radiation damage [12]–[14], changes in temperature [15] and bias [10], [13], [15]. The apparent connection of the defects observed in the NZFMR response and in the EDMR response suggest that NZFMR spectroscopy could be a viable new tool to study defects in semiconductors and insulators [13], [16], [17].

1.3 Objectives & Overview

The primary object of this research is to further develop and advance NZFMR as a spectroscopy technique that approaches the analytical power of EPR and EDMR at measuring

and determining performance limiting atomic-scale defects in semiconductor devices, materials, and insulators. Chapter 2 outlines and reviews the basic theory of EPR and the spin-related interactions commonly observed in EPR, EDMR, and NZFMR measurements. Chapter 3 provides a comprehensive overview of the recent developments in both the qualitative and quantitative models used to describe spin-dependent transport observed via both EDMR and NZFMR measurements. Chapter 3 provides the theoretical connection between the defects observed in EDMR and NZFMR. Chapter 4 demonstrates the experimental results of studies of isotopic dilution and forming gas anneals on Si/SiO₂ metal-insulator-semiconductor capacitors. These results show the explicit, experimental connection between the defects observed in EDMR and NZFMR. Chapter 5 presents a method by which to analyze and interpret NZFMR results and then extends that same method to analyze and further interpret EDMR results. Chapter 5 merges theory and experiment together by demonstrating the explicit connection between the defects observed with the EDMR measurement, the NZFMR measurement, and both the qualitative and quantitative theory that describes their responses.

Chapter 2 Spin & Electron Paramagnetic Resonance

2.1 Introduction

This chapter will outline some the basic concepts of spin, spin-related interactions, and how unpaired spins and their interactions with their local environment are measured with electron paramagnetic resonance (EPR).

The observation of the property of spin was first made by Otto Stern and Walther Gerlach in 1922 in which they sent a beam of silver atoms through a spatially varying magnetic field [18]. The silver atoms were deflected before striking their intended target. The area surrounding the target revealed discrete points of accumulation rather than a continuous distribution. In 1925, an interpretation of a particle “spinning” around its own axis was provided by George Uhlenbeck and Samuel Goudsmit in the spirit of the quantum theory provided by Neils Bohr and Arnold Sommerfeld [19]. Particles with non-zero magnetic moments were deflected from their trajectory from the force the experienced from their interactions with the magnetic field gradient present in the experiment.

A similar phenomena of the splitting of an electron’s spin state in a magnetic field, termed the Zeeman effect, gave rise to the first EPR experiment conducted by Evgeni K. Zavoiski in 1944 [20]. Since then, EPR has been utilized to study paramagnetic (unpaired) spins across a wide range of applications in fields ranging from physics, biology, chemistry, materials science, and engineering.

2.2 Intrinsic Angular Momentum

Spin is a property of elementary particles and atomic nuclei. Spin is an intrinsic form

of angular momentum. Consider an electron of mass, m_e , rotating about a point with velocity, $\vec{v}$. The angular momentum of the electron with respect to the point with which it is rotating about is defined as

$$\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m_e \vec{v}. \quad (2.1)$$

If we consider that the electron which is rotating about that same point has some charge, q_e , we can use Ampere's law and define the magnetic moment, $\vec{\mu}$, normal to the plane of the closed current loop as

$$\vec{\mu} = AI\hat{k} \quad (2.2)$$

where $A = \pi r^2$ and the current of the loop is defined as

$$I = \frac{q_e v}{2\pi r}. \quad (2.3)$$

Given an idealized circular orbit, the magnitude of the velocity within the orbit can be defined as

$$|\vec{v}| = \frac{|\vec{L}|}{m|\vec{r}|}. \quad (2.4)$$

Solving for velocity, v , in Equations (2.2) and (2.4), we find

$$v = \frac{2|\vec{\mu}|}{qr} = \frac{|\vec{L}|}{m_e r}. \quad (2.5)$$

If we consider that spin angular momentum is quantized in units of $\hbar$, that is

$$|\vec{L}| = \hbar \quad (2.6)$$

and we can define the gyromagnetic ratio of the electron as

$$\gamma_e = \frac{|\vec{\mu}|}{|\vec{L}|} = \frac{q_e}{2m_e}. \quad (2.7)$$

We can then define the Bohr magneton, as

$$\mu_B = |\vec{\mu}| = \gamma_e |\vec{L}| = \frac{q_e \hbar}{2m_e}. \quad (2.8)$$

Similarly, the nuclear magneton containing a single proton of mass, m_p , can be defined as

$$\mu_N = \frac{q_p \hbar}{2m_p}. \quad (2.9)$$

Since the mass of the proton is approximately 2000 times larger than of the electron, the Bohr magneton defined in Equation (2.8) is approximately 2000 times larger than that of the nuclear magneton defined in Equation (2.9).

In this classical picture, relativistic effects are not accounted for, so the gyromagnetic ratio of the electron and nuclei are somewhat different and are defined as

$$\gamma_e = g_e \frac{q_e}{2m_e} = 2.002318 \dots \frac{q_e}{2m_e} \quad (2.10)$$

$$\gamma_N = g_N \frac{q_N}{2m_N} \quad (2.11)$$

Here, g_e and g_N are defined as the Lande g-factors for the free electron and nuclei, respectively. The Lande g-factor for each nucleus is somewhat different since, γ_N , depends upon charge and mass of the nucleus under question.

Since, EPR is inherently based on the Zeeman splitting of the energy of an electron's magnetic moment in a magnetic field as well as the property of the spin angular momentum of the electron, these numbers will arise frequently in the analysis of the EPR spectra as discussed in the following sections.

2.3 Energy of an Electron in a Magnetic Field

During an EPR experiment, paramagnetic electron spins are polarized by a quasi-static magnetic field. Then the spin angular momenta of these electrons are altered with the

application of a perturbation to the system in the form of microwave radiation. Thus, it is important to discuss the energy of the electron in an applied magnetic field.

The energy of an electron within a quasi-static applied magnetic field, $\mathbf{B}_0$, is given by the spin Hamiltonian

$$\mathcal{H} = -\boldsymbol{\mu}_e \cdot \mathbf{B}_0. \quad (2.12)$$

Utilizing Equation (2.10), the magnetic moment of the electron can be defined as

$$\boldsymbol{\mu}_e = -g_e \mu_B \mathbf{S} \quad (2.13)$$

where

$$\mathbf{S} = \hat{S}_x \hat{i} + \hat{S}_y \hat{j} + \hat{S}_z \hat{k}. \quad (2.14)$$

Projected along the quantization axis (the z-axis), the electron spin angular moment is defined as

$$\hat{S}_z |s, m_s\rangle = m_s |s, m_s\rangle. \quad (2.15)$$

Here, it should be noted that $|s, m_s\rangle$ represents the spin states (spin-up or spin-down) of the electron:

$$\left| \frac{1}{2}, m_s \right\rangle = \left| \frac{1}{2}, +\frac{1}{2} \right\rangle = |\uparrow\rangle \quad (2.16)$$

$$\left| \frac{1}{2}, m_s \right\rangle = \left| \frac{1}{2}, -\frac{1}{2} \right\rangle = |\downarrow\rangle \quad (2.17)$$

If the magnetic field, $\mathbf{B}_0$, is aligned along the z-axis, the Zeeman Hamiltonian can then be defined as

$$\mathcal{H}_Z = g_e \mu_B B_{0,z} \hat{S}_z. \quad (2.18)$$

Substituting Equation (2.15) in Equation (2.18), we get

$$\mathcal{H}_Z|s, m_s\rangle = g_e\mu_B B_{0,z}|s, m_s\rangle. \quad (2.19)$$

Substituting Equations (2.16) and (2.17) into Equation (2.19), we find that

$$\mathcal{H}_Z \left| \frac{1}{2}, +\frac{1}{2} \right\rangle = +\frac{1}{2} g_e\mu_B B_z \left| \frac{1}{2}, +\frac{1}{2} \right\rangle = E_{\uparrow} \left| \frac{1}{2}, +\frac{1}{2} \right\rangle, \quad (2.20)$$

$$\mathcal{H}_Z \left| \frac{1}{2}, -\frac{1}{2} \right\rangle = -\frac{1}{2} g_e\mu_B B_z \left| \frac{1}{2}, -\frac{1}{2} \right\rangle = E_{\downarrow} \left| \frac{1}{2}, -\frac{1}{2} \right\rangle. \quad (2.21)$$

The change in energy required to transition between the spin states is then given by the difference in energy of the spin states. That is,

$$\Delta E = E_{\uparrow} - E_{\downarrow} = g_e\mu_B B_z. \quad (2.22)$$

In exposure to microwave radiation of frequency, ν , with photon energy of

$$E = h\nu \quad (2.23)$$

the spin angular momentum of the polarized electrons is altered, inducing a transition between the spin-up and spin-down states. Thus, the resonance condition occurs when

$$h\nu = g_e\mu_B B_z. \quad (2.24)$$

In real materials systems, perturbations from the local environment will influence the resonance condition. Measures of the changes to the resonance response due to the local environment of the paramagnetic defects (unpaired spins residing in real materials) gives rise to EPR's unrivaled analytical power in studying paramagnetic defects. The two most common influences to the resonance condition are spin-orbit coupling and electron-nuclear hyperfine interactions.

2.4 Spin-Orbit Coupling

The most common alteration to the resonance condition is spin-orbit coupling.

Consider an oversimplified picture of reality in which an electron with quantized orbital angular momentum ($L = n\hbar$) orbits a (positively charged) nucleus of atomic number Z . Then consider this same picture from the reference frame of the electron. From the prospective of the electron, it would appear as if the positive nucleus was rotating around it in a current loop:

$$I = \frac{Zqv}{2\pi r} = \frac{Zq}{2\pi r} \frac{L}{mr}. \quad (2.25)$$

If we consider the Biot Savart Law,

$$\mathbf{B}(r) = \frac{\mu_0}{4\pi} \int I \frac{d\mathbf{I} \times \mathbf{r}}{|\mathbf{r}|^3} \quad (2.26)$$

the current loop generated by the nucleus would create an applied magnetic field which the electron would then experience:

$$B = \frac{\mu_0}{4\pi} I \frac{2\pi r}{r^2} = \frac{\mu_0}{4\pi} \left(\frac{Zq}{2\pi r} \frac{L}{mr} \right) \frac{2\pi r}{r^2} = \frac{q\mu_0 ZL}{4\pi mr^3}. \quad (2.27)$$

The field generated by the current loop of the nucleus is a strong function of the charge of the nucleus, qZ , the mass of the nucleus, m , the distance between the nucleus and the electron, r , and the orbital angular momentum L . Equation (2.27) only provides qualitative understanding of the effects of spin-orbit coupling.

A full quantum mechanical description of spin orbit coupling requires second-order perturbation theory [1], [2]. Ultimately, the results of the derivation allow for the computation of the individual elements of the second rank $\mathbf{g}$ -tensor such that

$$\mathbf{g} = \begin{pmatrix} g_{xx} & g_{xy} & g_{xz} \\ g_{yz} & g_{yy} & g_{yz} \\ g_{zx} & g_{zy} & g_{zz} \end{pmatrix}. \quad (2.28)$$

For a simple case, such as an unpaired electron in a dangling bond,

$$g_{ij} = g_e - 2\lambda \sum_{i,j,n \neq 0} \frac{\langle 0|L_i|n\rangle\langle n|L_j|0\rangle}{E_n - E_0}. \quad (2.29)$$

Here, λ , represents the spin orbit coupling constant which again depends upon the charge of the nucleus, the mass of the nucleus, and the distance between the electron and the nucleus. The effect of spin-orbit coupling on the spin system is orientation dependent. Rotation of the sample within the field lead to changes in the resonance condition. The Hamiltonian of the system is typically expressed as:

$$\mathcal{H} = \mu_B (\mathbf{B} \cdot \mathbf{g} \cdot \mathbf{S}) \quad (2.30)$$

When only relatively small atoms are involved, the effect of spin-orbit coupling in solids are typically small.

2.5 Electron-Nuclear Hyperfine Interactions

The second most common alteration to the resonance condition is the effect of electron-nuclear hyperfine interactions. Some nuclei, like electrons, have an intrinsic angular momentum which results in a magnetic moment. These select nuclei only have a magnetic moment if the nucleus has an odd number of protons and/ or neutrons. When an additional magnetic moment is introduced with the introduction of one of these magnetic nuclei, an additional magnetic field is introduced to the system which can interact with the electron and further split its energy levels.

There are two classes of hyperfine interactions: the Fermi contact interaction and the dipole-dipole interaction. The Fermi contact interaction is completely isotropic and arises only when the probability density of the electron is non-zero at the nucleus ($\langle \psi | \psi \rangle \neq 0$). Therefore, a Fermi contact picture only applies to s-orbitals since s-orbitals have a non-zero probability

density at the nucleus. Although, the contact interaction is independent of the orientation of the applied fields, it is an interaction between the magnetic moments of nuclear spin and the electron spin and thus depends on the relative orientation between the electron and nuclear moments. That is

$$\mathcal{H}_{hf,Fermi} = a_s(\mathbf{S} \cdot \mathbf{I}) \quad (2.31)$$

where the coupling constant of a particular s-orbital electron is

$$a_s = \frac{8}{3} \mu_B \mu_N g_e g_N \langle \psi | \psi \rangle_s. \quad (2.32)$$

For orbitals which possess no probability density for the electron at the nucleus ($\langle \psi | \psi \rangle = 0$), such as p , d , and f orbitals, the Fermi contact interaction is not relevant. Their nuclear interactions are anisotropic and arise from the magnetic field at the nucleus due to the magnetic dipole field interactions with the electron-spin moment [1]. As long as the nuclear and electron moments are far enough apart, we can expect their interaction to be that of a pair of magnetic dipoles, the Hamiltonian being

$$\mathcal{H}_{hf} = \mu_B \mu_N g_e g_N \left(\frac{\mathbf{S} \cdot \mathbf{I}}{r^3} - \frac{3(\mathbf{S} \cdot \mathbf{r})(\mathbf{I} \cdot \mathbf{r})}{r^5} \right). \quad (2.33)$$

Since the hyperfine Hamiltonian is highly dependent upon direction, this Hamiltonian is often expressed as

$$\mathcal{H}_{hf} = \mathbf{S} \cdot \mathbf{A} \cdot \mathbf{I} \quad (2.34)$$

where, $\mathbf{A}$, hyperfine coupling is expressed as a second rank tensor:

$$\mathbf{A} = \begin{pmatrix} A_{xx} & A_{xy} & A_{xz} \\ A_{yx} & A_{yy} & A_{yz} \\ A_{zx} & A_{zy} & A_{zz} \end{pmatrix}. \quad (2.35)$$

In real crystals, electron-nuclear hyperfine interactions can result from central atoms (which host the dangling bond), nearest-neighbors (NN), next-nearest neighbors, and so-on. Accounting for all of the neighbors further out, the hyperfine interactions result in an isotropic hyperfine *bath*.

Hyperfine interactions will result in side peaks about the resonance condition which are split by the hyperfine coupling. The number of these resulting peaks depends upon the spin of the magnetic nuclei interacting with the electron. Given fixed distances between the atoms in a crystal, the relative amplitudes of the side peaks which result from hyperfine interactions depend upon the distance between the electron and magnetic nucleus and the probability of the occurrence of a magnetic so many neighboring sites away. If the positions of nearby magnetic nuclei are all equivalent, the relative amplitudes of the side peaks can be calculated via a binomial distribution since neighboring magnetic nuclei can occur in multiple combinations.

Ultimately, the Hamiltonian which contains the two most common alteration to the resonance condition, both spin-orbit coupling and electron-nuclear hyperfine interactions, can be expressed as:

$$\mathcal{H} = \mu_B (\mathbf{B} \cdot \mathbf{g} \cdot \mathbf{S}) + \sum_i \mathbf{S} \cdot \mathbf{A}_i \cdot \mathbf{I}_i . \quad (2.36)$$

2.6 Measuring Unpaired Electron Spins - The EPR Spectrometer

Figure 2.1 provides a schematic illustration of a typical EPR spectrometer. In order to perform an EPR measurement, a sample containing paramagnetic defects is placed in a high-Q microwave cavity connected to a microwave bridge which is tuned to the cavity's resonance frequency. The construction of the microwave bridge and its relevant circuitry is outlined in Figure 2.2. The cavity sits in an electromagnet which slowly varies the magnetic field across

the resonance condition.

To “tune” the cavity, one measures the reflected microwave power with a microwave detector diode out of port 3 of the circulator (outlined in Figure 2.2) as function of frequency. As the frequency is either increased or decreased, a “dip” in minimum in reflected power will be measured. The coupling frequency at which the cavity is tuned is the frequency of minimum reflected power. The cavity itself, contains a smaller set of coils which provide magnetic field modulation. The output of the microwave bridge (port IF on the mixer) is passed to a lock-in amplifier coupled to the modulation circuitry. The spin-dependent transitions are measured as a change in reflected microwave power as a function of the magnetic field. Due to the use of lock-in amplification, the observed EPR spectrum is an approximate derivative of the change microwave power with respect to the magnetic field.

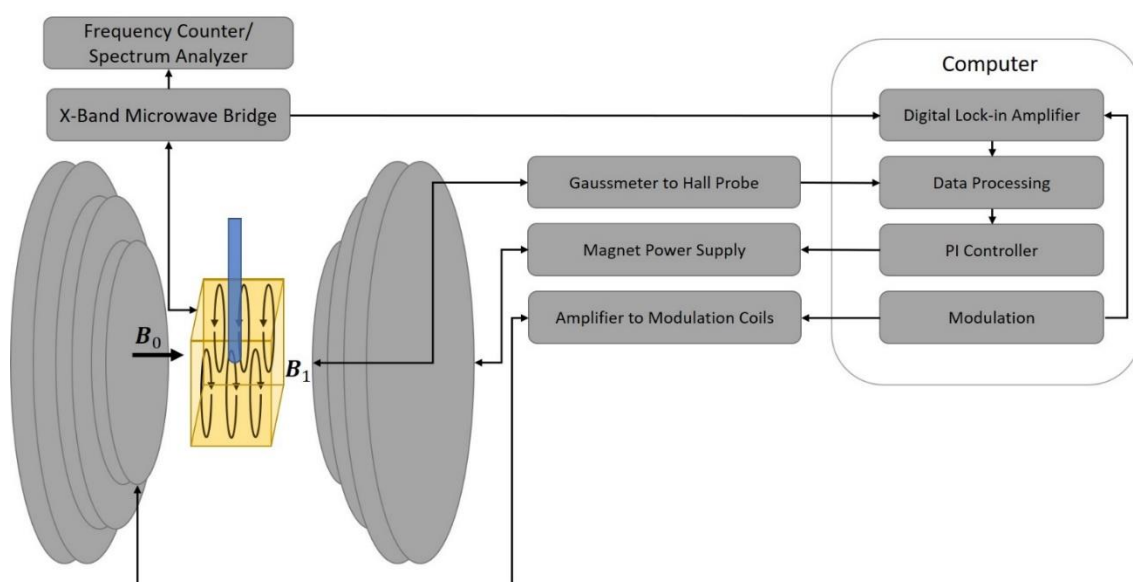


Figure 2.1 The EPR spectrometer.

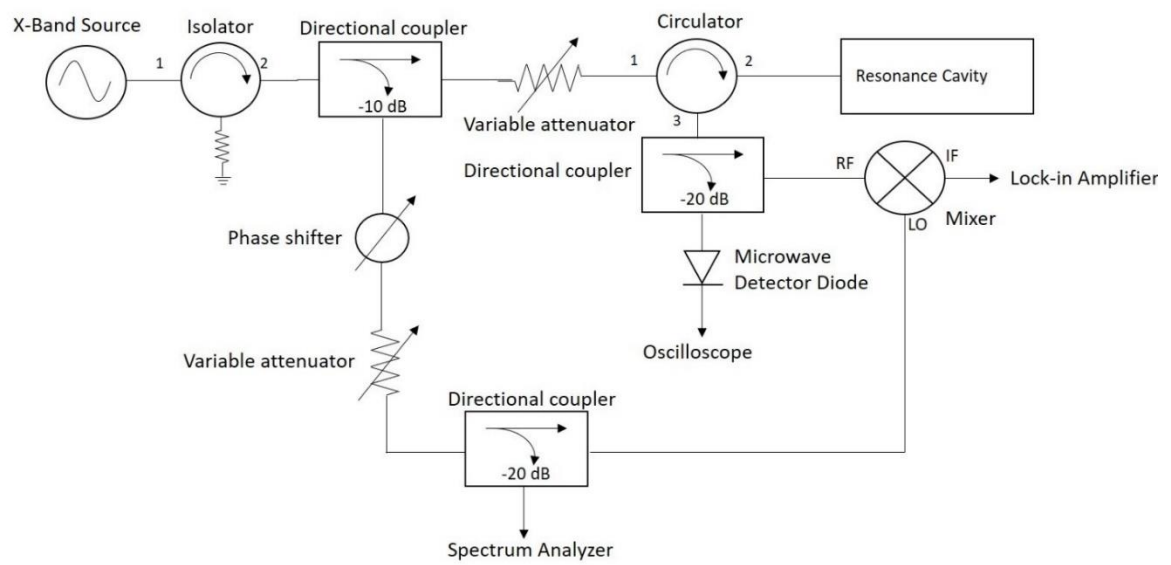


Figure 2.2 The EPR microwave bridge.

Chapter 3 The Theory of Spin-Dependent Transport

3.1 Introduction

Two notable models were produced to explain the spin-dependent transport phenomena observed in EDMR measurements. The first, produced in 1972 by Lepine, sought to explain the large changes in current observed in spin-dependent photoconductivity measurements [21]. The second, more widely accepted and more nearly correct model produced in 1978 by Kaplan, Solomon, and Mott (KSM) sought to explain the high sensitivities observed in spin-dependent recombination measurements [8]. Although the KSM were qualitatively correct in basing their model on the interactions between spin-pairs, the model is a quasi-classical model which does not consider the fundamentals of quantum mechanics. The KSM model can be extended to qualitatively explain the NZFMR response, but it cannot be extended quantitatively to do the same.

Several mechanisms have been explored to explain similar magnetic-field effects on current in organic semiconductors [22]–[27]. The proposed models for these similar effects focus on the possible spin-spin interactions present in bipolaron transport [24], [28], electron-hole (e-h) pair recombination [25], [26], or detrapping of triplet excitons [27]. These models have had considerable success in describing the spin-spin interactions present in these organic systems. Although these models have focused largely on both qualitative and quantitative explanations of organic magnetoresistance (OMAR), a phenomenon analogous to NZFMR, their models can be both qualitatively and quantitatively extended to explain the EDMR response as well.

Unfortunately, the organic systems for which the models were constructed contained

an abundance of nuclear spins with transport phenomena not completely understood. Multiple mechanisms could have conceivably been involved in the observed responses making it difficult to develop models from which OMAR/ NZFMR line shapes can be directly linked to defect structures. Inorganic crystalline semiconductors and some inorganic amorphous semiconductors offer substantial advantages in analysis. First, long range order is present in crystalline semiconductors. Second, short range order is often present in amorphous inorganic semiconductors. Third, widely utilized constituent elements silicon and carbon contain few nuclear magnetic moments. Additionally, transport phenomena in inorganic semiconductor devices are generally well understood.

This chapter will build from the Schrödinger through the relevant mathematics and physics of density matrix theory to the explain both the quantum Liouville equation (LE) and the stochastic quantum Liouville (SLE). These two fundamental equations are the backbone to the two models that best align with our understanding of the spin-spin interactions happening in inorganic semiconductor materials and devices during the EDMR and NZFMR measurements. These models include an adaptation of the e-h pair model [25], [26] for the case of the SDR and an adaptation of the two-site model [29] for the case of SDTAT.

3.2 The Schrodinger Equation

The Schrödinger equation is a linear partial differential equation that governs the wave function of a quantum-mechanical system:

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{\mathcal{H}} |\Psi(t)\rangle. \quad (3.1)$$

Here, $\hbar$, represents Planck's constant, $\hat{\mathcal{H}}$ represents some Hamiltonian operator, $|\Psi(t)\rangle$, in bracket notation represents the wave function as a function of time. In order to solve the

Schrödinger equation, the introduction of a propagator operator, $\hat{U}(t)$, that evolves the initial state with time is useful:

$$|\Psi(t)\rangle = \hat{U}(t)|\Psi\rangle \quad (3.2)$$

where

$$\hat{U}(t)|_{t=0} = 1. \quad (3.3)$$

Substituting the propagator operator into the Schrödinger equation, Equation (3.1) becomes

$$i\hbar \frac{\partial}{\partial t} (\hat{U}(t)|\Psi\rangle) = \hat{H}(\hat{U}(t)|\Psi\rangle) \quad (3.4)$$

$$i\hbar \frac{\partial \hat{U}(t)}{\partial t} |\Psi\rangle = \hat{H}\hat{U}(t)|\Psi\rangle. \quad (3.5)$$

This applies for any initial state and since the time dependence is now contained within the operator. Equation (3.5) can be written as operator equation that is

$$i\hbar \frac{\partial \hat{U}(t)}{\partial t} = \hat{H}\hat{U}(t) \quad (3.6)$$

Note that Equation (3.32) is in the same functional form as the Schrödinger Equation. This operator equation is in the form of a first-order linear differential equation:

$$i\hbar \frac{\partial \hat{U}(t)}{\partial t} - \hat{H}\hat{U}(t) = 0. \quad (3.7)$$

The solution of Equation (3.7) is

$$\hat{U}(t) = \exp\left(-\frac{i}{\hbar}\hat{H}t\right). \quad (3.8)$$

Note that $\hat{U}(t)$ is also a unitary operator such that

$$\hat{U}^\dagger \hat{U} = I \quad (3.9)$$

where I represents an identity matrix. Thus, the conjugate transpose of the unitary propagator can be defined as

$$\hat{U}^\dagger(t) = \exp\left(\frac{i}{\hbar}\hat{H}t\right) \quad (3.10)$$

$\hat{U}(t)$ can be substituted into Equation (3.2) such that the solution to the Schrödinger equation can be expressed as

$$|\Psi(t)\rangle = \exp\left(-\frac{i}{\hbar}\hat{H}t\right)|\Psi\rangle. \quad (3.11)$$

3.3 Density Matrix Theory

In order to derive the quantum Liouville equation, some formalisms of density matrix theory needs to be introduced. Density matrix theory is an invaluable tool used to describe the statistical nature of states within a quantum system. It is most valuable in describing the nature of mixed states. Mixed states are a combination of pure states which arise when the experimenter does not know which particular states are being manipulated. If a system has entangled subsystems, then each subsystem can be treated as a mixed state even if the complete system is comprised of only pure states.

The density operator operates on an ensemble of wave functions to describe the ensembles statistical nature. The density operator is expressed as

$$\hat{\rho}(t) = \sum_i W_i |\Psi_i(t)\rangle \langle \Psi_i(t)| \quad (3.12)$$

where, W_i , represents the statistical weight or probability of being in a certain state within the system. In a complete orthonormal basis, ϕ_i , the density-matrix can be defined as

$$\rho_{i,j} = \langle \phi_i | \hat{\rho} | \phi_j \rangle, \quad (3.13)$$

Substituting Equation (3.12) into Equation (3.13),

$$\rho_{i,j} = \langle \phi_i | \left(\sum_i W_i |\Psi_i(t)\rangle \langle \Psi_i(t)| \right) | \phi_j \rangle \quad (3.14)$$

which expands to

$$\rho = \begin{pmatrix} W_1 \langle \phi_1 | \Psi_1(t) \rangle \langle \Psi_1(t) | \phi_1 \rangle & W_1 \langle \phi_1 | \Psi_1(t) \rangle \langle \Psi_1(t) | \phi_2 \rangle & \cdots & W_1 \langle \phi_1 | \Psi_1(t) \rangle \langle \Psi_1(t) | \phi_n \rangle \\ W_2 \langle \phi_2 | \Psi_2(t) \rangle \langle \Psi_2(t) | \phi_1 \rangle & W_2 \langle \phi_2 | \Psi_2(t) \rangle \langle \Psi_2(t) | \phi_2 \rangle & \cdots & W_2 \langle \phi_2 | \Psi_2(t) \rangle \langle \Psi_2(t) | \phi_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ W_i \langle \phi_i | \Psi_i(t) \rangle \langle \Psi_i(t) | \phi_1 \rangle & \cdots & \cdots & W_i \langle \phi_i | \Psi_i(t) \rangle \langle \Psi_i(t) | \phi_n \rangle \end{pmatrix} \quad (3.15)$$

Here, the elements of ρ describe the probability of finding the spin system in a corresponding basis state. The density-matrix, ρ , completely describes the state of the ensemble that includes both electronic and or nuclear spins.

The density matrix has a host of properties which make it a particularly useful mathematical tool. The density matrix is Hermitian, that is,

$$\langle \phi_i | \hat{\rho} | \phi_j \rangle = \langle \phi_j | \hat{\rho} | \phi_i \rangle^* . \quad (3.16)$$

The diagonal elements are always positive,

$$\rho_{mm} \geq 0. \quad (3.17)$$

The probability of being in a certain state after a measurement can be defined as

$$W(\Psi) = \langle \Psi | \hat{\rho} | \Psi \rangle. \quad (3.18)$$

Expectation values can be found by taking the trace of the operator on the density matrix,

$$\langle \Psi | Q \hat{\rho} | \Psi \rangle = \langle Q \rangle = \text{tr}(\rho Q) = \text{tr}(Q \rho). \quad (3.19)$$

3.4 Derivation of the Quantum Liouville Equation

The density operator can also be rewritten in terms of the propagator operator that evolves some initial state independent of time with time:

$$\hat{\rho}(t) = \sum_i w_i \hat{U}(t) |\psi_n\rangle \langle \psi_n| \hat{U}^\dagger(t) \quad (3.20)$$

If we again consider a complete orthonormal basis and use a more simplified notation, the density matrix becomes

$$\boldsymbol{\rho}(t) = \hat{\mathbf{U}}(t) \boldsymbol{\rho} \hat{\mathbf{U}}^\dagger(t) \quad (3.21)$$

Recalling and substituting the definitions of $\hat{U}(t)$ and $\hat{U}^\dagger(t)$ defined in Equations (3.8) and (3.10), respectively, the density matrix can be expanded to

$$\boldsymbol{\rho}(t) = \left(\exp\left(-\frac{i}{\hbar} \hat{\mathcal{H}} t\right) \right) \boldsymbol{\rho} \left(\exp\left(\frac{i}{\hbar} \hat{\mathcal{H}} t\right) \right) \quad (3.22)$$

The quantum Liouville equation (LE) which describes the time evolution of that ensemble can be found by taking derivative with respect to time:

$$i\hbar \frac{\partial}{\partial t} \boldsymbol{\rho}(t) = i\hbar \frac{\partial \hat{\mathbf{U}}(t)}{\partial t} \boldsymbol{\rho} \hat{\mathbf{U}}^\dagger(t) + i\hbar \hat{\mathbf{U}}(t) \boldsymbol{\rho} \frac{\partial \hat{\mathbf{U}}^\dagger(t)}{\partial t} \quad (3.23)$$

$$i\hbar \frac{\partial}{\partial t} \boldsymbol{\rho}(t) = i\hbar \frac{\partial \left(\exp\left(-\frac{i}{\hbar} \hat{\mathcal{H}} t\right) \right)}{\partial t} \boldsymbol{\rho} \hat{\mathbf{U}}^\dagger(t) + i\hbar \hat{\mathbf{U}}(t) \boldsymbol{\rho} \frac{\partial \left(\exp\left(\frac{i}{\hbar} \hat{\mathcal{H}} t\right) \right)}{\partial t} \quad (3.24)$$

$$i\hbar \frac{\partial}{\partial t} \boldsymbol{\rho}(t) = \hat{\mathcal{H}} \left(\hat{\mathbf{U}}(t) \boldsymbol{\rho} \hat{\mathbf{U}}^\dagger(t) \right) + \left(\hat{\mathbf{U}}(t) \boldsymbol{\rho} \hat{\mathbf{U}}^\dagger(t) \right) \hat{\mathcal{H}} \quad (3.25)$$

$$i\hbar \frac{\partial}{\partial t} \boldsymbol{\rho}(t) = \hat{\mathcal{H}}(\boldsymbol{\rho}(t)) + (\boldsymbol{\rho}(t))\hat{\mathcal{H}} \quad (3.26)$$

$$i\hbar \frac{\partial}{\partial t} \boldsymbol{\rho}(t) = [\hat{\mathcal{H}}, \boldsymbol{\rho}(t)]. \quad (3.27)$$

Here, brackets denote the commutator operator, that is $[a, b] = ab - ba$. The LE represented in Equation (3.27) represents the non-dissipative, coherent evolution of the spin ensemble with respect to time. The LE does not take interactions with the environment into account. In order

to account for dissipative interactions with the environment, we need to use the stochastic quantum Liouville equation (SLE).

3.5 The Stochastic Quantum Liouville Equation

The SLE was first proposed by Scully and Lamb [30]. Sometimes referred to as the *master equation*, the SLE has had considerable success across different fields of research, ranging from delayed fluorescence in organic semiconducting crystals to laser theory.

The LE represents the non-dissipative, coherent evolution of the spin ensemble through time, without taking interactions with the environment into account. However, in order to properly describe both the coherent evolution and the environmentally dissipative, incoherent evolution of the spin ensemble, we turn to the SLE,

$$\frac{d\boldsymbol{\rho}}{dt} = -\frac{i}{\hbar}[\mathcal{H}, \boldsymbol{\rho}] - \frac{k}{2}\{\boldsymbol{\Lambda}, \boldsymbol{\rho}\} + p\boldsymbol{\Gamma} \quad (3.28)$$

The first term on the right side of the SLE describes the coherent evolution of spin pairs in the absence of interactions with the environment, exactly as the LE would. The second term introduces a dissipative effect that selectively removes spin pairs from the system at a rate of k . The projection operator, $\boldsymbol{\Lambda}$, projects onto the spin subspace from which the projected interactions are removed. The braces in the second term represent the anticommutator. The third term represents a source term that adds spin pairs in random orientations to the system at a rate of p and is typically proportional to an identity matrix, $\boldsymbol{\Gamma}$, in non-magnetic systems [22].

Although the SLE was introduced on a phenomenological basis, the incoherent spin-dependent reactions via interactions with the thermal bath are taken into account properly by the anticommutator of $\boldsymbol{\Lambda}$ and $\boldsymbol{\rho}$ [31]. Strictly speaking, $\boldsymbol{\rho}$ is not a density matrix since the trace of matrix is not equal to one. Rather, $\boldsymbol{\rho}$ is merely a measure for the ammount of spin pairs in

the system since the reaction products are neglected. The statistical interpretation of ρ can be obtained by calculating the reduced density operator for the spin pairs by simply dividing ρ by its trace.

Equation (3.28) is able to describe the generation, annihilation, coherent and non-evolution of spin-spin interactions through time. It will also serve as a starting point for the calculations of the different spin-spin reactions assumed to be responsible for the observed EDMR and NZFMR responses. By simply using the correct spin Hamiltonian and implementing small adaptations to the SLE, different models can be implemented and described [22]. This equation also serves as the basis from which the adaptation of the e-h pair model [25], [26] and the adaptation of two-site model [29] are formed that best describe SDR and SDTAT, respectively.

3.6 Adaptation of the Electron-Hole Pair Model to Describe Spin-Dependent Recombination

The e-h pair model put forth by Prigodin *et al.*[26] and the trap-induced magnetoresistance model of Harmon and Flatté [25] serve as the basis from which we constructed our model that best aligns with SDR taking place at a deep level defect. In this model, depicted in Figure 3.1, we consider an electron in the conduction band, a hole in the valence band, and an unpaired electron in a defect such as a dangling bond associated with a nuclear spin and hyperfine field present. The electron in the conduction band drops to an intermediate state. In the absence of a magnetic field, the singlet and triplet spin state between the electron in the intermediate state and the electron in the deep level defect site will be mixed by the hyperfine field of the nucleus. In the case of a singlet pair the electron in the intermediate

state will fall to the defect and recombine with the hole in the valence band. Recombination removes these carriers, and therefore spins, from the ensemble and reduces device current. In the case of a triplet pair, the electron in the conduction band has only the option to dissociate from the intermediate state and return to the conduction band. Since the electron is unable to recombine with the hole, device current is not altered in the system. The application of a modest external magnetic field suppresses the mixing caused by the hyperfine fields [22]. As the modest external field is reduced, the recombination rate increases.

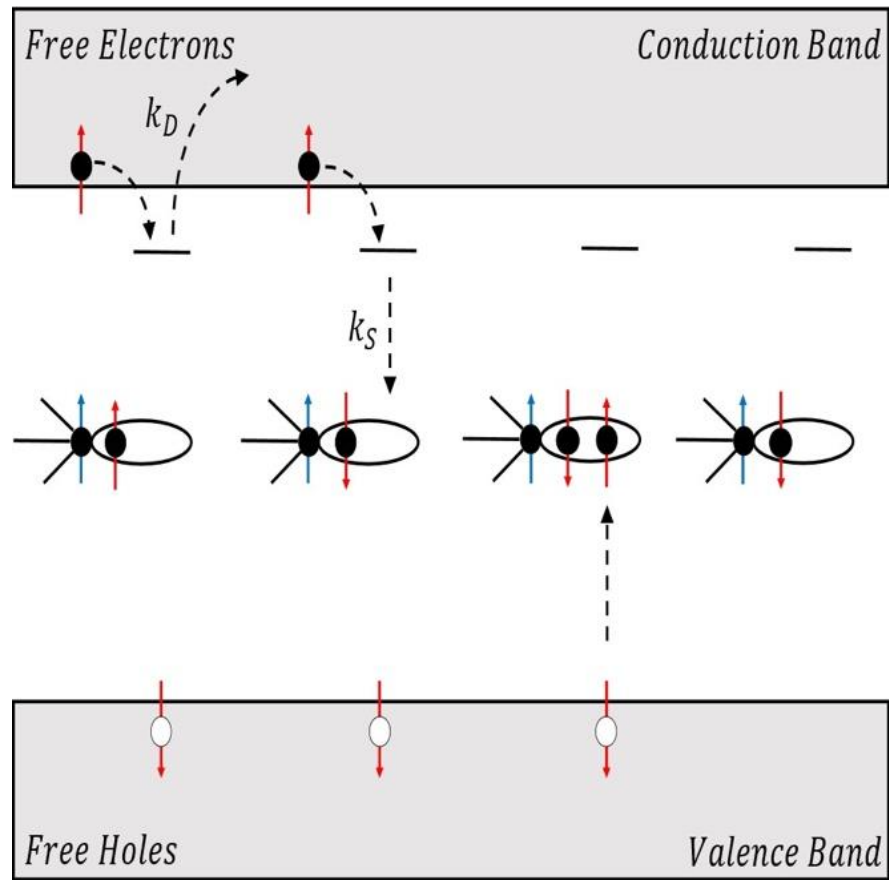


Figure 3.1 Spin dependent recombination model.

The time evolution of a density-matrix that can best describe this spin system can be represented by an adaptation of the SLE proposed by Hansen and Pedersen [32]:

$$\frac{d\boldsymbol{\rho}}{dt} = -\frac{i}{\hbar}[\mathcal{H}, \boldsymbol{\rho}] - \frac{1}{2}(k_S + k_D)\{\Lambda_S, \boldsymbol{\rho}\} - \frac{k_D}{2}\{\Lambda_T, \boldsymbol{\rho}\} + \frac{1}{8}p\Gamma. \quad (3.29)$$

The first term on the right side describes the coherent evolution of the spin pairs in the absence of the interaction with the environment - the mixing of the states caused by the hyperfine fields. The second term is a dissipative effect that selectively removes singlets (e-h pair) from the system at a rate of k_S and the third term is also a dissipative term that represents the selective removal of triplets that have dissociated to the conduction band at a rate of k_D . Λ_S and Λ_T denote the singlet and triplet projection operators, respectively. The fourth term represents a source of carriers to the system that are generated at a rate of p while Γ represents an identity matrix. Since carriers are being generated and annihilated at constant rates, we are only concerned with the steady state form of the SLE ($d\boldsymbol{\rho}/dt = 0$). The resulting form of the SLE is then

$$0 = -\frac{i}{\hbar}[\mathcal{H}, \boldsymbol{\rho}] - \frac{1}{2}(k_S + k_D)\{\Lambda_S, \boldsymbol{\rho}\} - \frac{k_D}{2}\{\Lambda_T, \boldsymbol{\rho}\} + \frac{1}{8}p\Gamma. \quad (3.30)$$

The factor of $\frac{1}{8}$ in the last term exists because the dimension of Γ is 8 such that the total generation rate of the spin pairs is $\frac{1}{8}p\text{Tr}(\Gamma) = p$ which is independent of the spin dimension.

The Hamiltonian for a single deep level recombination site within the bandgap can be represented by

$$\mathcal{H} = g\mu_B(\mathbf{S}_1 + \mathbf{S}_2) \cdot \mathbf{B} + a(\mathbf{S}_2 \cdot \mathbf{I}), \quad (3.31)$$

where the first term represents the Zeeman splitting of the electron spins, $\mathbf{S}_1$ (electron in the conduction band) and $\mathbf{S}_2$ (electron in the defect site), in the magnetic field while the second term represents an isotropic electron-nuclear hyperfine interaction, of strength a , between the

electron, $\mathbf{S}_2$, and the nucleus, $\mathbf{I}$. Of course, $\mathbf{S}_1$, could also represent the spin of a hole in the valance band if the recombination process involved a hole capture by a paramagnetic deep level defect. Explicit representation of the spin operators can be found in the appendix. This Hamiltonian considers isotropic interactions but can be adapted to accommodate anisotropic interactions. In this model, we assume that the electron in the intermediate state is not interacting with any nuclear spin or hyperfine field.

3.7 The Adaptation of the Two-Site Model to Describe Spin-Dependent Trap-Assisted Tunneling

The two-site model for organic semiconductors put forth by Wagemans *et al.* that considers only two characteristic sites best aligns with our current understanding of SDTAT in dielectrics and insulators [29]. SDTAT is enabled through variable range hopping and exploits the conservation of spin angular moment from trap-to-trap tunneling events [10]. Despite the existence of many sites that a carrier may occupy while hopping through the disordered system, the resistance is dominated by one bottleneck pair which the two-site model appropriately describes [24]. Figure 3.2, depicts the spin-dependence of the trap-assisted tunneling event. In order for the electron to successfully tunnel or hop to the other site, the two electron spins must be in a singlet configuration. If they are in a triplet configuration, the hopping event is forbidden. As the resonance condition is reached and SDTAT is detected through EDMR, the transition from triplet to singlet configurations is forced, allowing for previously forbidden tunneling events to take place, and thus a change in leakage current is measured. SDTAT detected though NZFMR does not involve a resonance induced transition of the spin state but rather a mixing of states [13], [33], [34].

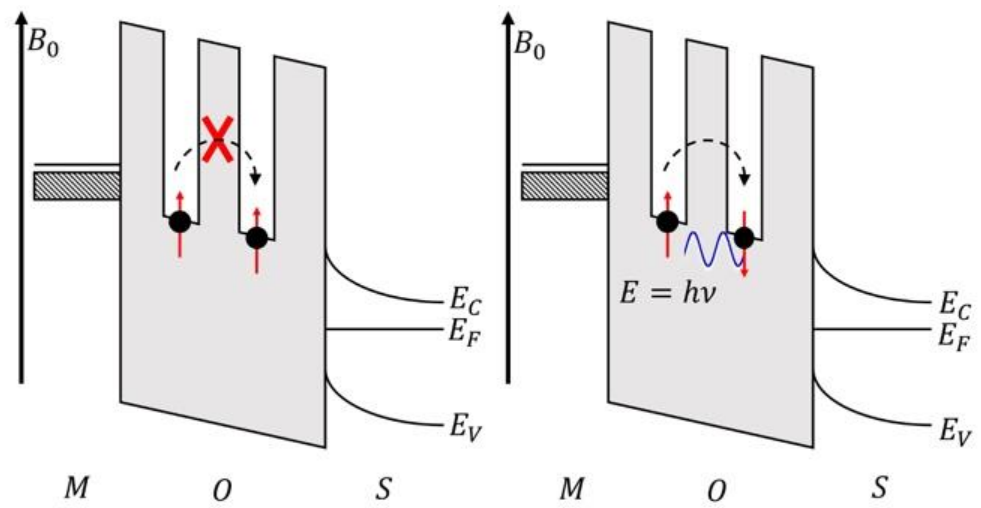


Figure 3.2 Depiction of SDTAT. Tunneling between traps is forbidden in the triplet configuration (left). Tunneling is allowed in the singlet configuration (right).

The adapted two-site model, as depicted in Figure 3.3, consists of two nuclear sites, α and β . Both sites have their own respective nuclear spins (both spins assumed $I = \frac{1}{2}$), and hyperfine fields and thus coupling constants, a_α and a_β , when singly occupied with an electron. If site α is unoccupied, an electron can hop from the environment, e , to site α at a rate of r_{ea} . Upon the occupation of site α , the electron will precess about the hyperfine field of the present nucleus. At this point, the electron can either hop directly to the environment at a rate of r_{ae} or hop to site β at a rate of $P_s r_{ab}$, where P_s is the probability of the electron at site α and the electron at site β being in a singlet configuration. After occupation of site β , the electron can dissociate back into the environment at a rate of r_{be} . In this model, the current flow is defined as the total flow of electrons from the environment to site α . In order to greatly simplify the arithmetic, we assume that the current limiting rate is not the electron hopping from site β into the environment, therefore we can assume $r_{be} = \infty$ [22].

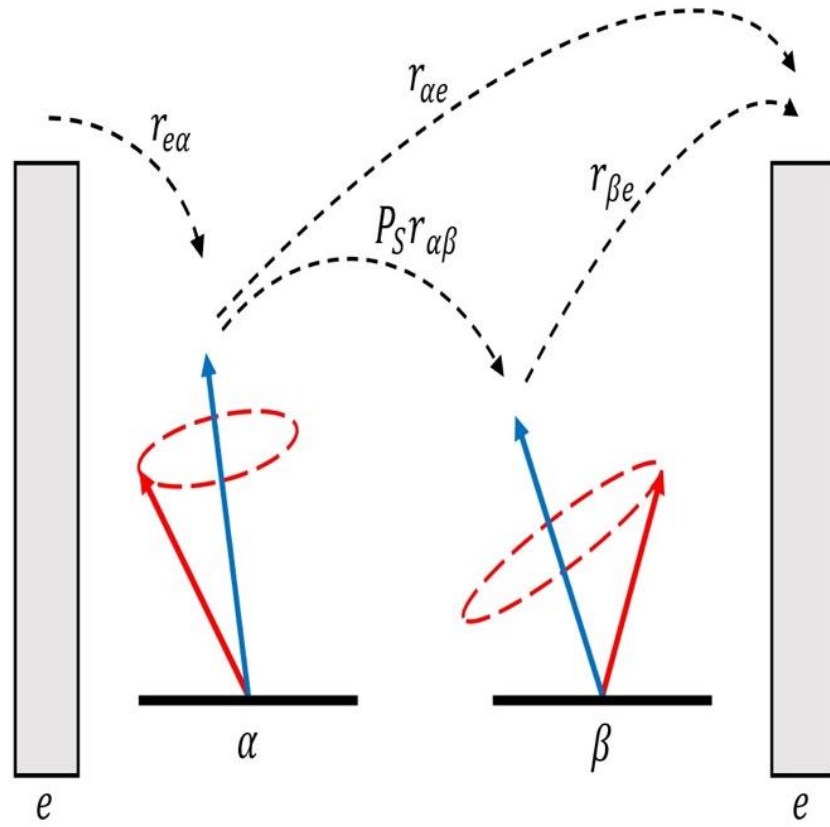


Figure 3.3 Depiction of the two-site model proposed by Wegemans *et al.* [29] Image adapted from Schellekens *et al.* [22]

The density-matrix representation of this ensemble can be represented by an adapted SLE as originally proposed by Wagemans *et al.* [29],

$$\begin{aligned} \frac{d\boldsymbol{\rho}}{dt} = & -\frac{i}{\hbar}[\mathcal{H}, \boldsymbol{\rho}] - \frac{1}{2}(r_{ab} + r_{ae})\{\Lambda_S, \boldsymbol{\rho}\} - \frac{1}{2}r_{ae}\{\Lambda_T, \boldsymbol{\rho}\} + \frac{1}{16}r_{ea}(1 \\ & - \text{Tr}(\boldsymbol{\rho}))\boldsymbol{\Gamma}, \end{aligned} \quad (3.32)$$

where Λ_S and Λ_T are the singlet and triplet projection operators, respectively, and $\boldsymbol{\Gamma}$ is an identity matrix that corresponds to a generation of spins at a random orientation. The current through the system is the current through site α and is defined as: $I = r_{ea}(1 - \text{Tr}(\boldsymbol{\rho}))$. Since product states (e.g. the doubly occupied state of β) are not accounted for, $\text{Tr}(\boldsymbol{\rho}) \leq 1$ [35]–[39]. Although the traps should be within the vicinity of the Fermi level, for simplification of the calculation we assume the traps remain mostly unoccupied ($1 \gg \text{Tr}(\boldsymbol{\rho})$). Therefore, the current through the system can be defined as the rate at which carriers are introduced to the system: $I = r_{ea}$. Assuming steady state conditions ($d\boldsymbol{\rho}/dt = \mathbf{0}$ at $t = \infty$) in addition to these simplifications, Equation (3.32) becomes

$$\mathbf{0} = -\frac{i}{\hbar}[\mathcal{H}, \boldsymbol{\rho}] - \frac{1}{2}(r_{ab} + r_{ae})\{\Lambda_S, \boldsymbol{\rho}\} - \frac{1}{2}r_{ae}\{\Lambda_T, \boldsymbol{\rho}\} + \frac{1}{16}r_{ea}\boldsymbol{\Gamma}. \quad (3.33)$$

Since there are now four spins included, the dimension of $\boldsymbol{\Gamma}$ is 16 which leads to the factor of $\frac{1}{16}$ in the last term in the same way we had for spin dependent recombination involving three spins.

The Hamiltonian of this system can be represented by

$$\mathcal{H} = g\mu_B(\mathbf{S}_1 + \mathbf{S}_2) \cdot \mathbf{B} + a_\alpha(\mathbf{S}_1 \cdot \mathbf{I}_\alpha) + a_\beta(\mathbf{S}_2 \cdot \mathbf{I}_\beta), \quad (3.34)$$

where the first term represents the Zeeman splitting of the electron spin, $\mathbf{S}_1$ and $\mathbf{S}_2$, while the second and third term represents an isotropic electron-nuclear hyperfine interaction, of strength a , between the nucleus and the electron occupying the site. Explicit representation of the spin

operators can be found in the appendix. Again, this Hamiltonian considers isotropic interactions but can be adapted to accommodate anisotropic interactions.

3.8 Modeling Spin-Dependent Trap-Assisted Tunneling in $^{28}\text{Si}/^{28}\text{SiO}_2$ Metal-Insulator Capacitors

The NZFMR response involves the mixing of singlet and triplet states of spin pairs involved in the trap-assisted tunneling response. The bulk of literature and the models outlined in sections 3.6 and 3.7 suggest that electron-nuclear hyperfine interactions are the predominant mechanism responsible for the mixing of spin-states. The exact understanding of what leads to the NZFMR response in the absence of electron-nuclear hyperfine interactions is somewhat limited both theoretically and experimentally. This section discusses the relevant theory which aligns with observations made from the SDTAT measurements made on $^{28}\text{Si}/^{28}\text{SiO}_2$ MIS capacitors outlined in Chapter 4

The pioneering organics work used to study the effects spin-spin interactions on magnetoresistance responses have involved highly complex systems with an abundance of nuclear spins and complex transport phenomena which are incompletely understood. Unpassivated $^{28}\text{Si}/^{28}\text{SiO}_2$ metal-insulator-semiconductor (MIS) capacitors are an excellent system in which to study the effects that lead to the NZFMR response in the absence of electron-nuclear hyperfine interactions. First, ^{29}Si magnetic nuclei which result in electron-nuclear hyperfine interactions will not be present. Second, long range order is present on the semiconductor side of the semiconductor-insulator interface while short range order is present in the amorphous $^{28}\text{SiO}_2$ dielectric. Third, transport phenomena Si/SiO_2 are fairly well understood.

This adaptation of the two-site model, as depicted in Figure 3.4, is similar to outlined in Figure 3.3. The two-site model consists of two singly occupied sites, α and β , which contain their own respective spins. The unpaired spin at site α represents a paramagnetic oxide trap while the unpaired spin at site β represents a paramagnetic an interface trap. If site α is unoccupied, the electron spin can tunnel from the environment, e , to site α at a rate of $r_{e\alpha}$. Upon the occupation of site α , the electron will precess about the externally applied magnetic field. At this point, the electron can either hop directly to the environment at a rate of $r_{\alpha e}$ or hop to site β at a rate of $P_s r_{\alpha\beta}$, where P_s is the probability of the electron at site α and the electron at site β being in a singlet configuration. After occupation of site β , the electron can dissociate back into the environment at a rate of $r_{\beta e}$. In this model, the current flow is defined as the total flow of electrons from the environment to site α . In order to greatly simplify the arithmetic, we assume that the current limiting rate is not the electron hopping from site β into the environment, therefore we can assume $r_{\beta e} = \infty$ [22]. The spin at site β is experiencing a magnetic dipolar interaction via an unpaired spin residing in a P_b center a distance r_{jk} away.

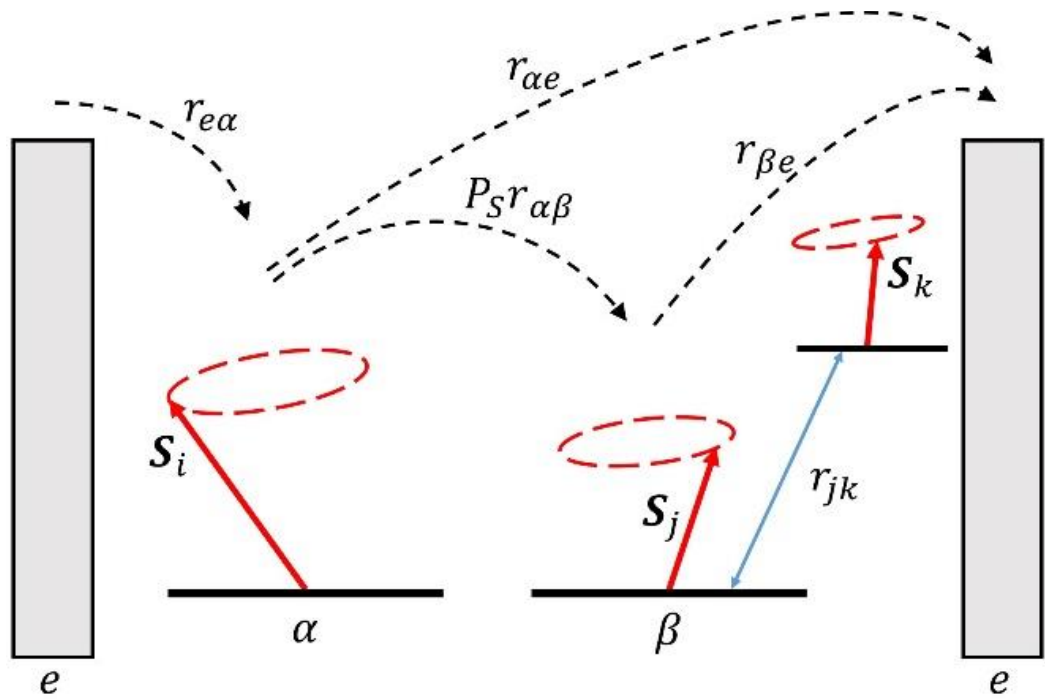


Figure 3.4 Adaptation two-site model proposed by Wegemans *et al.* [29] in the absence of nuclear spins.

The density-matrix representation of this ensemble can be represented by an adapted stochastic quantum Liouville equation (SLE) as originally proposed by Wagemans *et al.* [29],

$$\frac{d\boldsymbol{\rho}}{dt} = -\frac{i}{\hbar}[\mathcal{H}, \boldsymbol{\rho}] - \frac{1}{2}(r_{ab} + r_{ae})\{\mathbf{A}_S, \boldsymbol{\rho}\} - \frac{1}{2}r_{ae}\{\mathbf{A}_T, \boldsymbol{\rho}\} + \frac{1}{8}r_{ea}(1 - \text{Tr}(\boldsymbol{\rho}))\mathbf{\Gamma}, \quad (3.35)$$

where $\mathbf{A}_S$ and $\mathbf{A}_T$ are the singlet and triplet projection operators, respectively, and $\mathbf{\Gamma}$ is an identity matrix that corresponds to a generation of spins at a random orientation. The current through the system is the current through site α and is defined as: $I = r_{ea}(1 - \text{Tr}(\boldsymbol{\rho}))$. Since product states (e.g. the doubly occupied state of β) are not accounted for, $\text{Tr}(\boldsymbol{\rho}) \leq 1$ [35]–[39]. It should be noted that since we are interested in steady state conditions, the current through the system is constant and therefore the current, I , is treated as a constant scalar value. Assuming steady state conditions ($d\boldsymbol{\rho}/dt = 0$) and that the traps remain mostly unoccupied ($1 \gg \text{Tr}(\boldsymbol{\rho})$), equation (10) becomes

$$\mathbf{0} = -\frac{i}{\hbar}[\mathcal{H}, \boldsymbol{\rho}] - \frac{1}{2}(r_{ab} + r_{ae})\{\mathbf{A}_S, \boldsymbol{\rho}\} - \frac{1}{2}r_{ae}\{\mathbf{A}_T, \boldsymbol{\rho}\} + \frac{1}{8}r_{ea}\mathbf{\Gamma}. \quad (3.36)$$

The factor of $\frac{1}{8}$ in the last term exists because the dimension of $\mathbf{\Gamma}$ is of dimension 8 such that the total generation rate of the spin pairs is $\frac{1}{8}r_{ea}\text{Tr}(\mathbf{\Gamma}) = r_{ea}$ which is independent of the spin dimension.

When simulating the NZFMR response, the Hamiltonian of this system can be represented by

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_{D,jk}, \quad (3.37)$$

where the first term

$$\mathcal{H}_0 = g\mu_B \mathbf{B}_0 \cdot (\mathbf{S}_{z,i} + \mathbf{S}_{z,j} + \mathbf{S}_{z,k}) \quad (3.38)$$

represents the Zeeman splitting of the spin system under a quasi-static magnetic of strength $\mathbf{B}_0$, while the second term represents a magnetic dipolar interaction between two closely neighboring spins and is expressed as

$$\mathcal{H}_{D,jk} = \mathbf{S}_j \mathbf{D}_{jk} \mathbf{S}_k = \frac{1}{r_{jk}^3} \frac{\mu_0}{4\pi} g_i g_j \mu_B^2 \left(\mathbf{S}_j \cdot \mathbf{S}_k - 3(\mathbf{S}_j \cdot \hat{\mathbf{r}}_{jk})(\mathbf{S}_k \cdot \hat{\mathbf{r}}_{jk}) \right) \quad (3.39)$$

Here, μ_0 represents the permeability of free space ($4\pi * 10^{-7} T^2 m^3 J^{-1}$), μ_B represents the Bohr magneton ($9.247 * 10^{-24} JT^{-1}$), r_{jk} is the separation distance between the two spins, g represents the g-factor of the respective spins, and $\mathbf{S}_j$ and $\mathbf{S}_k$ represent the spin angular momentum operators of the spins at their respective sites. If we express the x - and y -components of the spin operators in terms of the raising and lowering operators as well as express the rectangular coordinates in terms of spherical coordinates, the Hamiltonian can be conveniently expressed as

$$\mathcal{H}_{D,ij} = \mathbf{S}_j \mathbf{D}_{jk} \mathbf{S}_k = \frac{1}{r_{jk}^3} \frac{\mu_0}{4\pi} g_i g_j \mu_B^2 (\mathbf{A} + \mathbf{B} + \mathbf{C} + \mathbf{D} + \mathbf{E} + \mathbf{F}) \quad (3.40)$$

where,

$$\mathbf{A} = \mathbf{S}_{z,j} \mathbf{S}_{z,k} (1 - \cos^2 \theta_{jk}) \quad (3.41)$$

$$\mathbf{B} = -\frac{1}{4} (\mathbf{S}_{+,j} \mathbf{S}_{-,k} + \mathbf{S}_{-,j} \mathbf{S}_{+,k}) (1 - \cos^2 \theta_{jk}) \quad (3.42)$$

$$\mathbf{C} = -\frac{3}{2} (\mathbf{S}_{+,j} \mathbf{S}_{z,k} + \mathbf{S}_{z,j} \mathbf{S}_{+,k}) \sin \theta_{jk} \cos \theta_{jk} e^{-i\phi_{jk}} \quad (3.43)$$

$$\mathbf{D} = -\frac{3}{2} (\mathbf{S}_{-,j} \mathbf{S}_{z,k} + \mathbf{S}_{z,j} \mathbf{S}_{-,k}) \sin \theta_{jk} \cos \theta_{jk} e^{i\phi_{jk}} \quad (3.44)$$

$$\mathbf{E} = -\frac{3}{4} \mathbf{S}_{+,j} \mathbf{S}_{+,k} \sin^2 \theta_{jk} e^{-i2\phi_{jk}} \quad (3.45)$$

$$\mathbf{F} = -\frac{3}{4} \mathbf{S}_{-,j} \mathbf{S}_{-,k} \sin^2 \theta_{jk} e^{i2\phi_{jk}} \quad (3.46)$$

Here, θ_{jk} and ϕ_{jk} are defined as the polar and azimuthal angles, respectively, between spins $\mathbf{S}_j$ and $\mathbf{S}_k$. Terms $\mathbf{E}$ and $\mathbf{F}$ are responsible specifically for NZFMR response [2].

3.9 Extension of the Models to Explain Electrically Detected Magnetic Resonance

3.9.1 The Rotating Reference Frame

If we consider the spin ensemble's interaction with a perpendicular oscillating magnetic field which gives rise to the EDMR response, a secondary Zeeman term

$$\mathcal{H}_1 = 2g\mu_B \mathbf{B}_1 \cos(\omega t) \cdot \left(\sum_i \mathbf{S}_{x,i} \right). \quad (3.47)$$

must be introduced to the Hamiltonian. Here, $\mathbf{B}_1$ represents the strength of the perpendicular oscillating magnetic field and ω represents the angular frequency of the field.

The NZFMR response is independent of the perpendicular oscillating magnetic field and therefore the term $\mathcal{H}_1$ and its time-dependence can be eliminated from the Hamiltonian when simulating just the NZFMR response. When simulating the EDMR response, the time-dependent Hamiltonian and the SLE needs be expressed in terms of rotating reference frame. The motivation behind this transformation is to make the Hamiltonian time-independent which makes it easy to solve for steady state conditions (where $d\rho/dt = \mathbf{0}$ at $t \rightarrow \infty$) of the spin ensemble with the inclusion of an oscillating magnetic field.

In transforming to the rotating reference frame, $\mathcal{H}_1$ can be approximated as

$$\mathcal{H}_1 \approx \mathcal{H}_a + \mathcal{H}_b \quad (3.48)$$

where,

$$\mathcal{H}_a = g\mu_B \mathbf{B}_1 \cdot \left(\sum_i \mathbf{S}_{x,i} \right) \cos(-\omega t) + g\mu_B \mathbf{B}_1 \cdot \left(\sum_i \mathbf{S}_{y,i} \right) \sin(-\omega t) \quad (3.49)$$

$$\mathcal{H}_b = g\mu_B \mathbf{B}_1 \cdot (\sum_i \mathbf{S}_{x,i}) \cos(\omega t) + g\mu_B \mathbf{B}_1 \cdot (\sum_i \mathbf{S}_{y,i}) \sin(\omega t). \quad (3.50)$$

Either $\mathcal{H}_a$ or $\mathcal{H}_b$ become time independent while the other term obtains a relative 2ω precession rate. This term is routinely dropped within the rotating-wave approximation (RWA) [2], such that

$$\mathcal{H}_1 \approx \mathcal{H}_a = g\mu_B \mathbf{B}_1 \cdot \left(\sum_i \mathbf{S}_{x,i} \right) \cos(-\omega t) + g\mu_B \mathbf{B}_1 \cdot \left(\sum_i \mathbf{S}_{y,i} \right) \sin(-\omega t) \quad (3.51)$$

where the spin operators are dimensionless and $\mathbf{B}_1 \ll \mathbf{B}_0$.

The density-matrix in the rotating reference frame can then be defined as

$$\tilde{\rho} = \mathbf{R}^{-1} \rho \mathbf{R} \quad (3.52)$$

where,

$$\mathbf{R} = e^{i\omega(\sum_i S_{x,i})}. \quad (3.53)$$

Substituting the new definition of the density-matrix in the rotating reference, the left-hand side of the SLE becomes

$$\frac{\partial \tilde{\rho}}{\partial t} = \frac{\partial (\mathbf{R}^{-1} \rho \mathbf{R})}{\partial t} = \mathbf{R}^{-1} \rho \frac{\partial \mathbf{R}}{\partial t} + \frac{\partial \mathbf{R}^{-1}}{\partial t} \rho \mathbf{R} + \mathbf{R}^{-1} \frac{\partial \rho}{\partial t} \mathbf{R} \quad (3.54)$$

where the three terms on the right are defined as

$$\mathbf{R}^{-1} \rho \frac{\partial \mathbf{R}}{\partial t} = \mathbf{R}^{-1} \rho \mathbf{R} \left(i\omega \left(\sum_i \mathbf{S}_{x,i} \right) \right) \quad (3.55)$$

$$\frac{\partial \mathbf{R}^{-1}}{\partial t} \rho \mathbf{R} = \left(-i\omega \left(\sum_i \mathbf{S}_{x,i} \right) \right) \mathbf{R}^{-1} \rho \mathbf{R} \quad (3.56)$$

$$\begin{aligned}
\mathbf{R}^{-1} \frac{\partial \boldsymbol{\rho}}{\partial t} \mathbf{R} &= -\frac{i}{\hbar} \mathbf{R}^{-1} [\mathcal{H}, \boldsymbol{\rho}] \mathbf{R} \\
&= -\frac{i}{\hbar} \left((\mathbf{R}^{-1} \mathcal{H} \mathbf{R}) (\mathbf{R}^{-1} \boldsymbol{\rho} \mathbf{R}) - (\mathbf{R}^{-1} \boldsymbol{\rho} \mathbf{R}) (\mathbf{R}^{-1} \mathcal{H} \mathbf{R}) \right) \\
&= -\frac{i}{\hbar} [\widetilde{\mathcal{H}}, \widetilde{\boldsymbol{\rho}}].
\end{aligned} \tag{3.57}$$

The remaining stochastic terms are determined similarly. The final expression of the SLE in the rotating reference frame can then be expressed as

$$\frac{d\widetilde{\boldsymbol{\rho}}}{dt} = -\frac{i}{\hbar} [\widetilde{\mathcal{H}}, \widetilde{\boldsymbol{\rho}}] - \frac{1}{2} (c_1 + c_2) \{\boldsymbol{\Lambda}_S, \widetilde{\boldsymbol{\rho}}\} - \frac{1}{2} c_2 \{\boldsymbol{\Lambda}_T, \widetilde{\boldsymbol{\rho}}\} + g \boldsymbol{\Gamma} \tag{3.58}$$

Here c_1 and c_2 represent the respective rate constants that depend upon the form of the SLE being solved and g represents the generation rate at which spins are being introduced into the system. The $\left(i\omega(\sum_i \mathbf{S}_{x,i})\right)$ terms in Equations 23 and 24 can be regrouped with the other Zeeman terms in the Hamiltonian such that the effective Hamiltonian in the rotating reference frame becomes

$$\widetilde{\mathcal{H}} = g\mu_B \left(\mathbf{B}_0 + \frac{\hbar\omega}{g\mu_B} \right) \left(\sum_i \mathbf{S}_{z,i} \right) + g\mu_B \mathbf{B}_1 \left(\sum_i \mathbf{S}_{x,i} \right) + \widetilde{\mathcal{H}}_{other} \tag{3.59}$$

where $\widetilde{\mathcal{H}}_{other}$ represents other possible spin-interactions in the rotating reference frame which can be added to the Hamiltonian.

3.9.2 Hyperfine Interactions in the Rotating Reference Frame

For the RWA approximation to eliminate the time dependence of the hyperfine interaction term, some restrictions must be assumed. All the hyperfine coupling tensors must be isotropic:

$$\mathbf{A} = \begin{pmatrix} a_{xx} & 0 & 0 \\ 0 & a_{yy} & 0 \\ 0 & 0 & a_{zz} \end{pmatrix} \quad (3.60)$$

where $a_{xx} = a_{yy} = a_{zz}$. When these conditions are satisfied, the hyperfine interaction in the rotating reference frame can be expressed as,

$$\widetilde{\mathcal{H}}_{hf} = \mathbf{S} \cdot \mathbf{A} \cdot \mathbf{I}. \quad (3.61)$$

3.9.3 Dipolar Interactions in the Rotating Reference Frame

The rotating reference frame eliminates the time dependence of secular terms ($\mathbf{A}$ and $\mathbf{B}$) of the dipolar interaction defined in Equation (3.40). The dipolar interaction in the rotating reference frame can then be expressed as,

$$\widetilde{\mathcal{H}}_{D,jk} = \frac{1}{r_{jk}^3} \frac{\mu_0}{4\pi} g_i g_j \mu_B^2 (\widetilde{\mathbf{A}} + \widetilde{\mathbf{B}}) \quad (3.62)$$

where

$$\widetilde{\mathbf{A}} = \mathbf{S}_{z,j} \mathbf{S}_{z,k} (1 - \cos^2 \theta_{jk}) \quad (3.63)$$

$$\widetilde{\mathbf{B}} = -\frac{1}{4} (\mathbf{S}_{+,j} \mathbf{S}_{-,k} + \mathbf{S}_{-,j} \mathbf{S}_{+,k}) (1 - \cos^2 \theta_{jk}). \quad (3.64)$$

The time dependence of the pseudo-secular terms ($\mathbf{C}$ and $\mathbf{D}$) and the non-secular terms ($\mathbf{E}$ and $\mathbf{F}$) of the dipolar interaction cannot be eliminated in the rotating reference frame. However, the net effects of the pseudo-secular terms $\mathbf{C}$ and $\mathbf{D}$ and non-secular terms $\mathbf{E}$ and $\mathbf{F}$ give rise to the half-field EDMR response and NZFMR response, respectively. Therefore, they may be disregarded for our purposes of simulating the steady state conditions (where $d\boldsymbol{\rho}/dt = \mathbf{0}$ at $t \rightarrow \infty$) of the spin ensemble about the resonance condition, specifically [2].

3.10 Solving the Stochastic Quantum Liouville Equation

Since we are interested in solving for the steady state conditions (where $d\rho/dt = \mathbf{0}$ at $t \rightarrow \infty$) Equations (3.30) and (3.33) can be algebraically manipulated to take the form:

$$\mathcal{L}\rho + \rho\mathcal{L}^\dagger = g, \quad (3.65)$$

Where

$$\mathcal{L} = \frac{i}{\hbar}\mathcal{H} + \frac{(c_1 + c_2)}{2}\Lambda_S + \frac{c_2}{2}\Lambda_T. \quad (3.66)$$

Here, $\mathcal{H}$ represents the Hamiltonian expressed in either the static reference frame or the rotating reference frame. Additionally, c_1 and c_2 represent the respective rate constants that depend upon the form of the SLE being solved and g represents the generation rate at which spins are being introduced into the system. Equation (3.66) is in the form of the time-continuous Lyapunov equation. The solution of Equation (3.66) and the density-matrix at steady state can be represented as

$$\rho = \int_0^\infty e^{-i\mathcal{L}t} g e^{i\mathcal{L}^\dagger t} dt. \quad (3.67)$$

In either the SDR or SDTAT model, the measurable-observable change in current is due to singlet-related events, which makes us concerned with the singlet projection of the density-matrix. The measurable SDR current is therefore proportional to the product of the probability of the formation of a singlet and the trace of the singlet population of the density-matrix,

$$I_{SDR} \propto Z \left(\frac{k_S}{p} \right) Tr(\Lambda_S \rho). \quad (3.68)$$

Here the constant Z describes the relative contribution due a signal in terms of an amplitude scaling factor. The measurable SDTAT current is proportional to product of the rate at which the sites are filled and the probability of occupation of the sites,

$$I_{SDTAT} \propto r_{e\alpha}(1 - Tr(\rho)). \quad (3.69)$$

3.11 Observing Spin-Dependent Transport

With the leading theories of the spin-dependent transport outlined, a discussion of the experimental techniques utilized to observe spin-dependent transport is useful. Two major methods of observing spin-dependent transport which will be discussed include EDMR and NZFMR. Both techniques are experimentally derived from EPR. Where EPR measures changes in microwave power, both EDMR and NZFMR measures changes in device current or voltage. This leads to a sensitivity more than ten million times greater than that of classical EPR, allowing observations of 10^3 and possibly fewer defects. Since the measurement involves a voltage or current within the active volume of the device structure under study, it is exclusively sensitivity to defects within the active volume of the device. Since this volume can often be varied via the application of a voltage or voltages, it is also often possible to alter and crudely identify the physical locations of the defects under study.

3.11.1 The Electrically Detected Magnetic Resonance Spectrometer

Since EDMR measures changes in device current rather than changes in microwave absorption, the technique does not necessarily require a microwave resonator and can be performed at a multitude of fields and frequencies. Since the EDMR detection does not depend upon the polarization of a spin system, it has a sensitivity which is nearly independent of the magnetic field and resonance frequency. In later chapters, EDMR measurements presented were collected by performing the measurement at high- frequencies (X-band $\nu = 9.5 \text{ GHz}$) and fields (350 mT) in addition to low- frequencies ($\nu = 151 \text{ MHz}$) and fields (5 mT).

3.11.1.1 *High-Frequencies & High-Fields*

In the EDMR measurement performed at high- frequencies and fields, a sample containing paramagnetic defects is mounted to a custom printed circuit board (PCB) where electrical contacts to the device are made. The sample/PCB assembly is then placed in a high-Q microwave cavity which is connected to a EDMR microwave bridge as presented in Figure 3.5. The EDMR bridge, as presented in Figure 3.6, is then tuned to the cavity's resonance frequency.

To tune the cavity, one measures the reflected microwave power with a microwave detector diode out of port 3 of the circulator (outlined in Figure 3.6) as function of frequency. As the frequency is either increased or decreased, a “dip” in minimum in reflected power should be measured. The coupling frequency at which the cavity is tuned is the frequency corresponding to the minimum in reflected power.

The cavity itself sits in an electromagnet which slowly varies the magnetic field across the resonance condition. A second, smaller set of coils in the cavity provide magnetic field modulation. The device is biased to create a current that is fed to a current to voltage preamplifier and then to a lock-in amplifier coupled to the modulation circuitry. The spin-dependent current through the device is measured as a function of the magnetic field. Due to the use of lock-in amplification, the observed EDMR spectrum is an approximate derivative of the change in spin-dependent current [40].

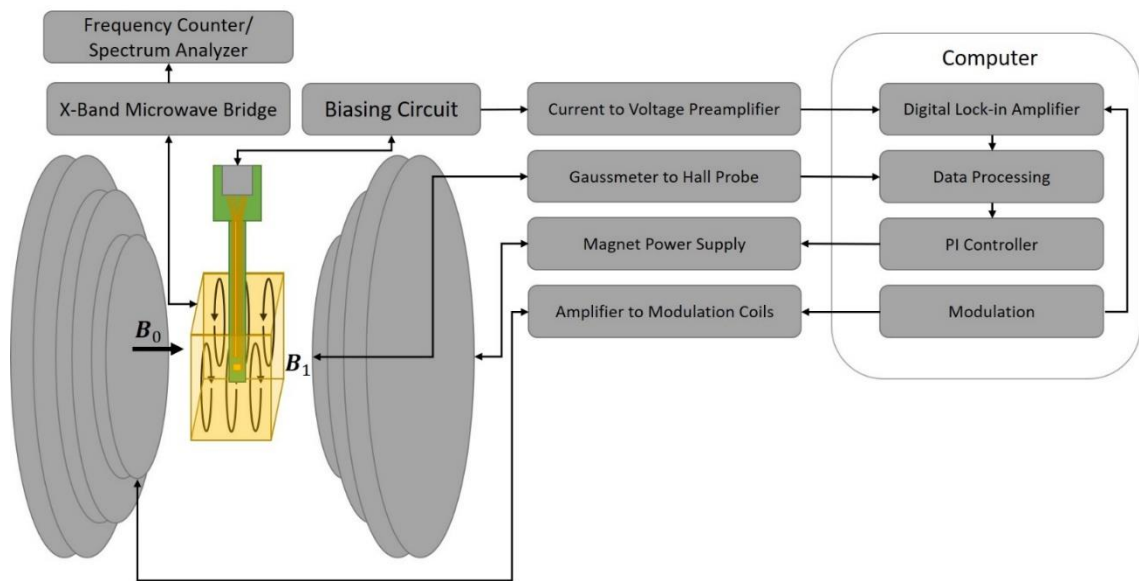


Figure 3.5 X-Band EDMR Spectrometer.

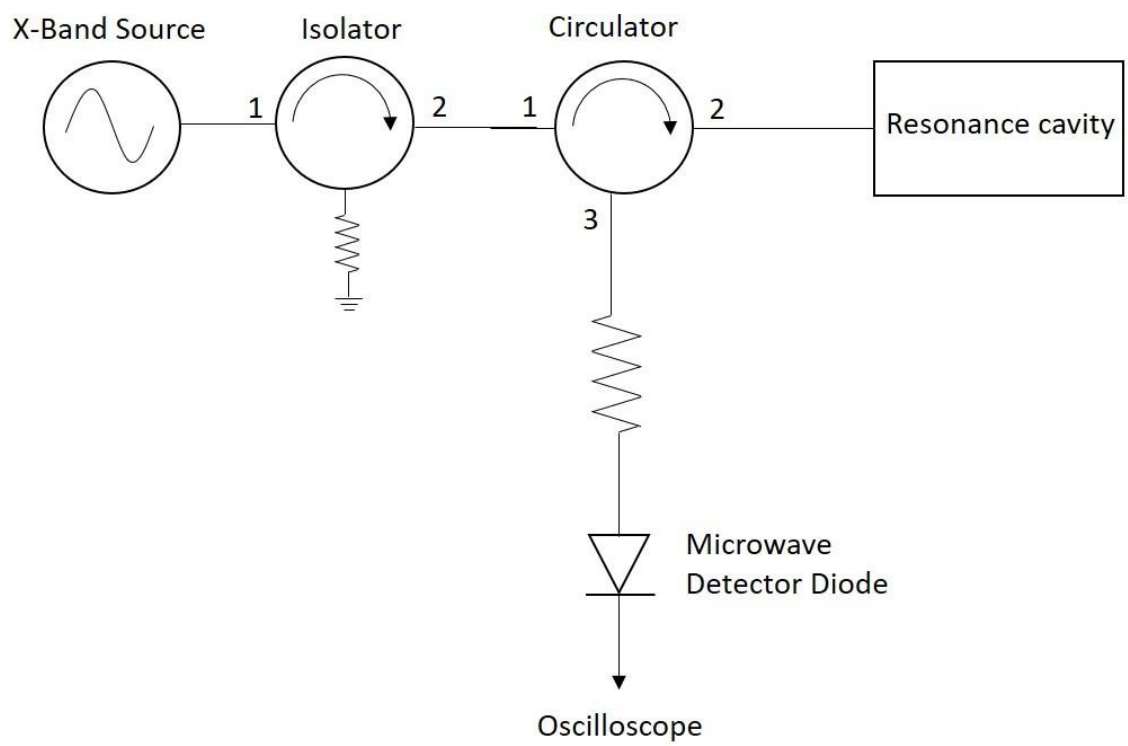


Figure 3.6 EDMR Microwave Bridge.

3.11.1.2 *Low-Frequencies & Low-Fields*

In the EDMR measurement performed at low- frequencies and fields, a sample containing paramagnetic defects is mounted to a custom printed circuit board (PCB) where electrical contacts to the device are made. The sample/PCB assembly is then placed in a RF resonator coil with an impedance matching LRC circuit typically tuned to an RF frequency of a few hundred megahertz. As outlined in Figure 3.7, the RF resonator coil and sample sits in an electromagnet which slowly varies the magnetic field across the resonance condition. A second, smaller set of coils provide a magnetic field modulation. The device is biased to create a current that is fed to a current to voltage preamplifier and then to a lock-in amplifier coupled to the modulation circuitry. The spin-dependent current through the device is measured as a function of the magnetic field. Due to the use of lock-in amplification, the observed EDMR spectrum is an approximate derivative of the change in spin-dependent current.

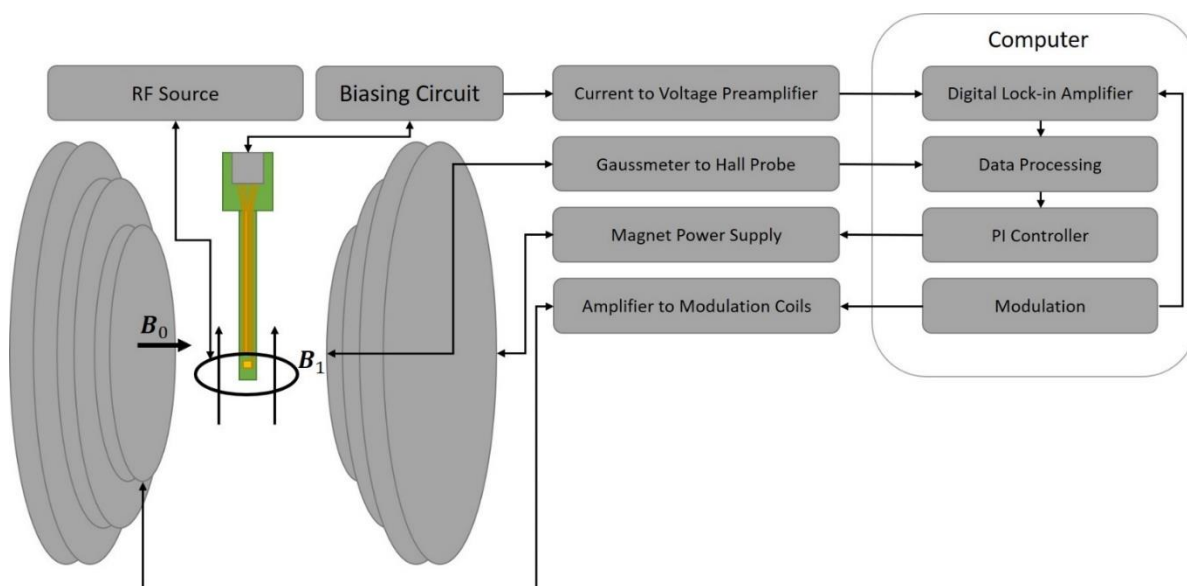


Figure 3.7 The Low-Frequency, Low Field EDMR Spectrometer.

3.11.2 The Near-Zero Field Magnetoresistance Spectrometer

During a standard EDMR measurement, a change in current can also be measured when the magnetic field is swept through zero: the NZFMR response. The NZFMR response occurs with or without the presence of the oscillating magnetic field. The elimination of the oscillating magnetic field makes the measurement much simpler than conventional EPR and EDMR. The small magnetic fields utilized in NZFMR also simplify the measurement. An experimental setup for the NZFMR measurement is shown in Figure 3.8. The device under study is placed between a set of electromagnetic coils which provide slowly varying magnetic field that is swept through zero. A second, smaller set of coils provide a magnetic field modulation. The device is biased to create a current that is fed to a current to voltage preamplifier and then to a lock-in amplifier coupled to the modulation circuitry. The spin-dependent current through the devices is measured as a function of the magnetic field. Due to the use of lock-in amplification, the observed NZFMR spectrum is an approximate derivative of the change in spin-dependent current. The NZFMR experimental setup is very similar to that of EDMR spectrometers, but lacks the resonant microwave or radio frequency source.

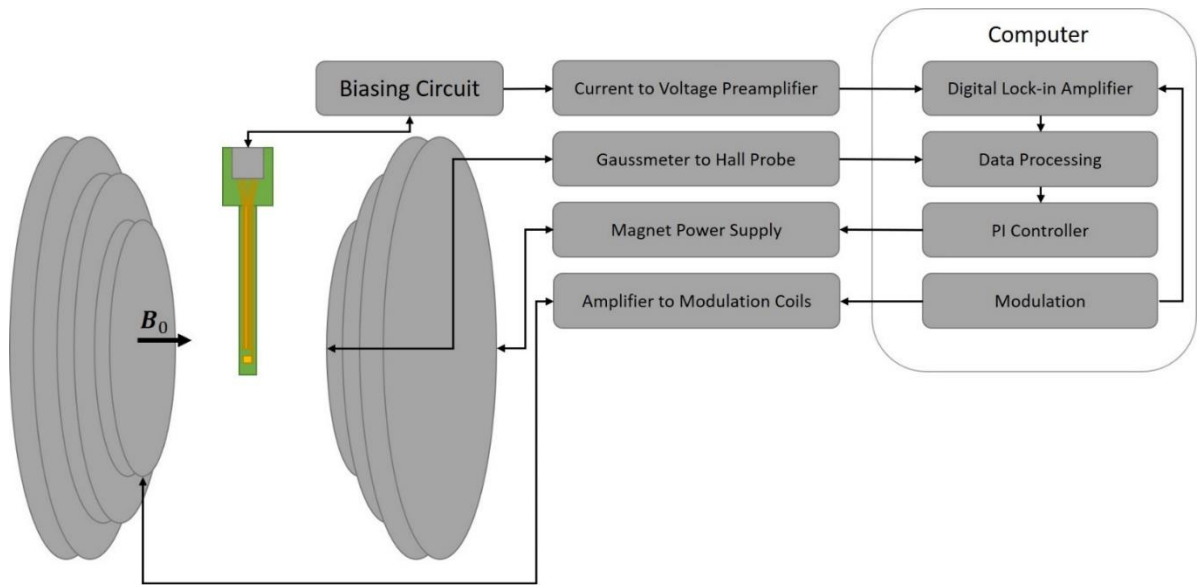


Figure 3.8 The NZFMR Spectrometer.

Chapter 4 Isotopic Dilution and Forming Gas Anneals Studies of Si/SiO₂

Metal-Insulator-Semiconductor Capacitors

4.1 Introduction

The metal-oxide-semiconductor field-effect transistor (MOSFET) is arguably the most important device in modern integrated circuitry due to its widespread use in both memory and logic applications. Most MOSFETs used in consumer and commercial applications are constructed from silicon and its oxide (silicon dioxide) making the Si/SiO₂ interface the most important material interface in modern electronics. This is so even in present day devices employing a high-k dielectric since there is still a thin silicon dioxide interfacial layer between the silicon and the high-k material. An understanding of the nature of defects at this interface and how they impact electrical transport is crucial for the development of modern micro- and nano-electronic devices. Electrically detected magnetic resonance studies of interface defects in Si/SiO₂ have been shown to be relevant to a variety of reliability issues in state the art MOS devices [41]–[45]. They are of particular importance in the process of time dependent dielectric breakdown (TDDB) [46]. The buildup of interface traps contributes to the formation of percolative leakage current paths. Upon formation of a path, the oxide resistance dramatically drops and the device breaks down [47].

4.2 Experimental Details

4.2.1 Device Details

The MIS capacitors studied were fabricated on highly conductive 300 mm (100)-oriented silicon substrates ($N_A \approx 10^{19} \text{ cm}^{-3}$) of natural (4.7%) ²⁹Si isotopic abundance. Atop

the substrate, a layer of p-doped silicon ($N_A \approx 10^{16} \text{ cm}^{-3}$) was deposited followed by the epitaxially growth of a $1 \mu\text{m}$ lightly doped ($N_A \approx 10^{14} \text{ cm}^{-3}$) silicon layer. In some samples, both layers were of natural isotopic abundance. On selected wafers, a 100 nm thick ^{29}Si -depleted epitaxially grown layer was deposited using a ^{28}Si silane precursor as described previously [48]. The residual ^{29}Si content in the ^{29}Si -depleted layer is measured to be about 800 ppm. Recall that silicon of natural isotopic abundance contains 47,000 ppm ^{29}Si . For the samples of natural isotopic abundance, no additional epitaxial growth was performed. A 7nm layer of silicon dioxide was grown using industry-standard dry thermal oxidation conditions. For samples where the top surface contained the ^{29}Si -depleted epi layer, the resulting thermal oxide was also depleted of ^{29}Si . Circular metallic contacts were deposited on top of the oxide across the wafer using shadow mask evaporation. The contacts consisted of 10 nm of titanium beneath 1 μm of aluminum. Samples that underwent a forming gas anneal (FGA) did so for 2 hours at 623K so after so after contact deposition.

4.2.2 Spectrometer Details

We have made our SDTAT EDMR measurements with a homemade EDMR spectrometer operating at X-band ($\nu \approx 9.25\text{GHz}$). The spectrometer consisted of a Hewlett-Packard 6268B power supply which supplied current to a 4-inch Varian electromagnet. This magnetic field was measured by a Lake Shore Cryotronics 475 DSP temperature-compensated Hall Gaussmeter connected to the computer. A Stanford Research Systems SR570 low-noise current-to-voltage preamplifier was used to bias the devices and measure device current which was passed through a National Instruments digital-to-analog data acquisition system connected to the computer. The computer communicated with the Gaussmeter and magnet power supply

allowing for magnetic field control. Additionally, the computer provided lock-in detection of the magnetic field modulation produced by a set of Helmholtz coils built in the walls of the T_{102} cavity where the sample was placed. The TE_{102} cavity was connected to a Resonance Instruments Model 8625 X-band microwave bridge which provided the oscillating magnetic field. The Low-Frequency EDMR and NZFMR spectrometer is similar in design, but instead of using a fairly high field electromagnet, it utilizes a custom-made electromagnet consisting of five nested pairs of Helmholtz coils, a lower power Kepco BOP 50-2M power supply. Low-Frequency EDMR measurements utilized a coil from Doty Scientific which operated at $\nu = 151\text{ MHz}$. As a consequence of lock-in detection in both systems, the measured spectra are approximate derivatives of their respective response. The magnitude of the modulation field and frequency used in all EDMR and NZFMR measurements was 0.2 mT at 1 kHz. The voltage applied to the device(s) during all the measurements was +2.0 V to the top metal contact; this puts the p-MIS device(s) into reverse bias. To achieve signal-to-noise ratio as reported in our results, several hundred scans were averaged for the unpassivated MIS capacitors while several thousand scans were averaged for the much smaller response in the passivated MIS capacitors. Each scan took two minutes.

4.3 High-Frequency Electrically Detected Magnetic Resonance Comparisons of Non-Forming Gas Annealed Si/SiO₂ & ²⁸Si/²⁸SiO₂ Metal-Insulator-Semiconductor Capacitors

Figure 4.1 shows the SDTAT response as measured through X-Band ($\nu \approx 9.25\text{ GHz}$) EDMR of the unpassivated Si/SiO₂ MIS capacitor containing silicon of natural isotopic abundance and silicon depleted of ²⁹Si with the B_0 field perpendicular and parallel to the (100)

interface. The unpassivated Si/SiO₂ MIS capacitor containing silicon of natural isotopic abundance exhibited an EDMR response with a zero-crossing $g_{\perp(100)}$ and $g_{\parallel(100)}$ of 2.0048 ± 0.0003 and 2.0042 ± 0.0003 , respectively, and linewidths in the perpendicular and parallel orientations of 1.0 ± 0.05 mT and 1.2 ± 0.05 mT, respectively. The unpassivated Si/SiO₂ MIS depleted of ²⁹Si exhibited zero-crossing $g_{\perp(100)}$ and $g_{\parallel(100)}$ of 2.0051 ± 0.0003 and 2.0042 ± 0.0003 , respectively, and linewidths in the perpendicular and parallel orientations of 1.1 ± 0.05 mT and 1.1 ± 0.05 mT, respectively.

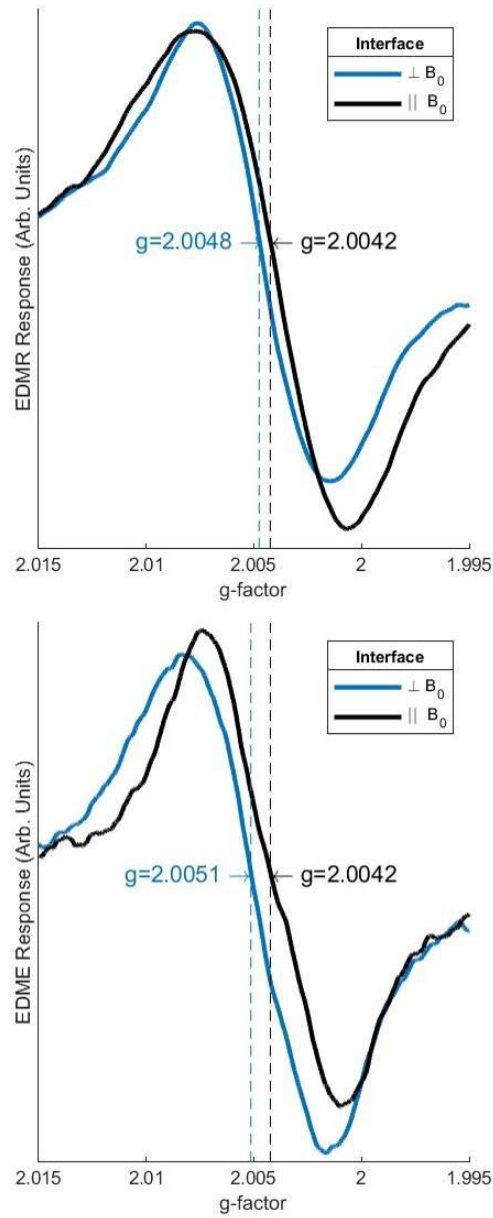


Figure 4.1 SDTAT response measured via high-field EDMR of unpassivated Si/SiO₂ MIS capacitors containing silicon of natural isotopic abundance (upper) and silicon depleted of ^{29}Si (lower) at orientations of the (100) interface perpendicular (blue) and parallel (black) to the B_0 field.

Previous reports on the (100) Si/SiO₂ interface have demonstrated that the dominating interface defects are the P_{b0} and P_{b1} centers [49], [50]. Both of these defects are Si dangling bonds with a central Si atom back bonded to three Si atoms. The g-values of these defects are orientation dependent.[49], [50] With the magnetic field oriented perpendicular to the (100) interface, we would expect to observe a single line with a zero-crossing $g = 2.0059$ for the P_{b0} center and a single line with $g = 2.0032$ for the P_{b1} center. This is because the g-tensor for the P_{b0} is $g_{||} = 2.0014$ and $g_{\perp} = 2.0082$ with axial symmetry about the [111] direction. The g-tensor for the P_{b1} is $g_1 = 2.0012$ about the (100) direction, $g_2 = 2.0076$ about the (111) direction, and $g_3 = 2.0052$ about the (011) direction.[51] When the magnetic field is oriented parallel to the interface, both the P_{b0} and P_{b1} signals split into multiple lines due to the multiple orientations of both defects possible on the (100) interface. Other studies of the centers present in the Si/SiO₂ system have reported on the observation of dominating oxide defects called E' centers [7], [52]. The E' center is a dangling bond on a silicon atom back bonded to three oxygen atoms. In conventional MOS devices, these defects are usually holes typically trapped in oxygen vacancies. Although there are several E' variants, none have g-tensors which match up with our EDMR spectrum. The most commonly observed E' centers which some call E'₁ has a “powder pattern” approximately corresponding to $g_{||} = 2.0018$ and $g_{\perp} = 2.0000$. The E' center magnetic resonance spectrum is an orientation independent powder pattern [7] because the defect exists in an amorphous matrix making all orientations of the defect equally likely.

Figure 4.2 illustrates a simulated EPR spectra of a mixture of P_{b0}, P_{b1}, and E' defects with the magnetic field oriented perpendicular and parallel to the (100) interface. Simulated spectra were generated via EasySpin [53] utilizing spectral parameters outlined by Lenahan and Conley [7] and Poindexter *et al.* [51] These simulated responses are composed of 79% P_{b0},

19% P_{b1} , and 2% E' center contribution. These simulations demonstrate the multiple overlapping lines present due to the P_{b0} and P_{b1} centers. Additionally, these simulations demonstrate a difference in the zero-crossing g-factor between the two orientations. Note that even with only a 2% E' center contribution, the corresponding E' signal is distinguishable against both P_{b0} and P_{b1} .

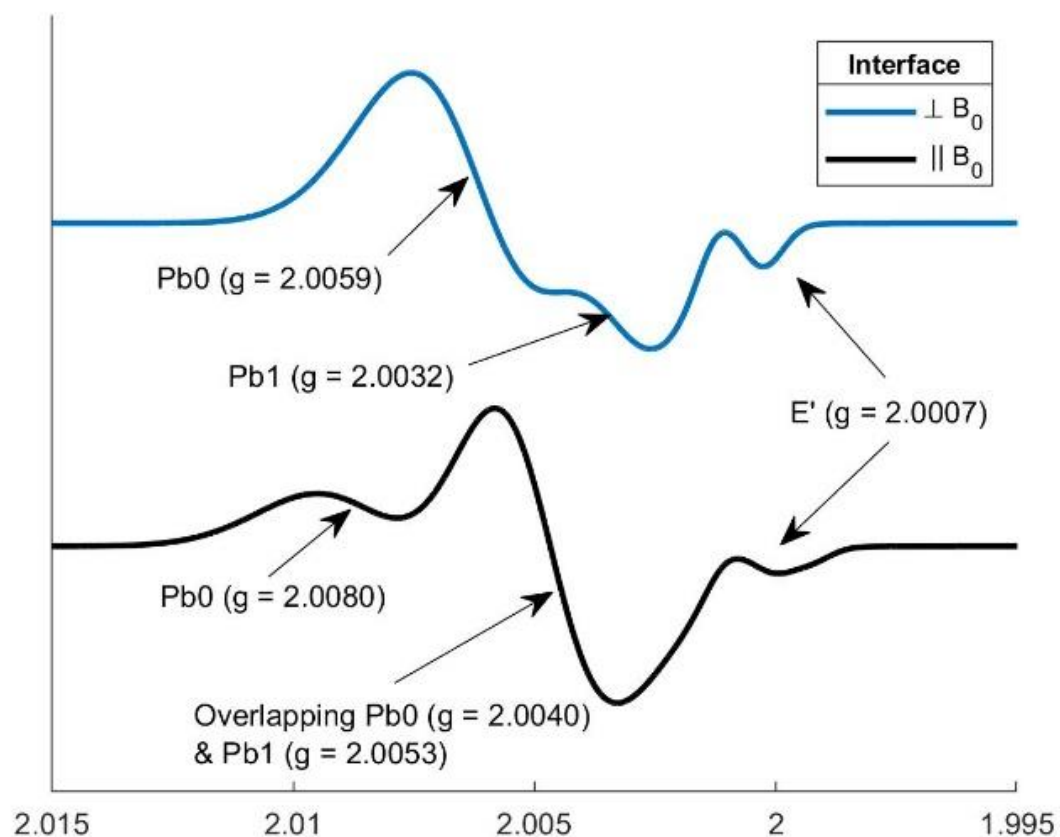


Figure 4.2 Simulated EPR response composed of 79% Pb₀, 19% Pb₁, and 2% E' center contribution with magnetic field both perpendicular and parallel to (100) interface. The origin and g-factor of each feature is indicated.

Although we do not observe multiple lines when the interface lies either parallel or perpendicular to the magnetic field, we do observe a consistent difference of the zero-crossing g -factor with respect to the B_0 field orientation. Additionally, the g -factors of both the P_{b0} and P_{b1} defects are within the linewidth of our measurement. [43], [54] The structure of the observed spectra is likely obscured at least in part by broadening typically observed in EDMR.

Figure 4.3a illustrates the simulated spectra of the same defects shown in Figure 4.2, but with broadening introduced to the P_{b0} , P_{b1} , and E' responses. Figure 5b illustrates the simulated spectra of the same defects shown in Figure 4.3a, but lacking an E' response. The broadened interface state spectra illustrated in Figure 4.3b exhibit a similar weak orientation dependence to the experimental EDMR data. It is also clear from these simulations that no discernable E' response is detected in our EDMR data. The EDMR data from the MIS capacitors of natural isotopic abundance and those depleted of ^{29}Si are repeated Figure 4.3c and Figure 4.3d, respectively, for comparison.

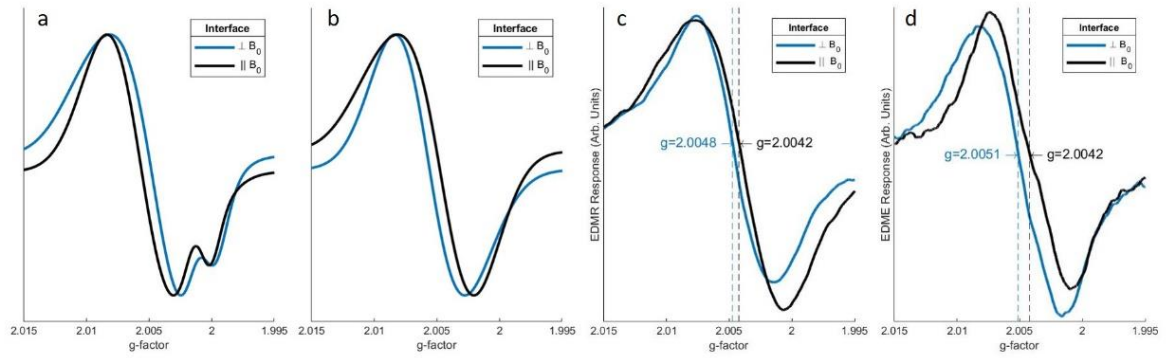


Figure 4.3 (a) Simulated EPR response composed of 79% P_{b0} , 19% P_{b1} , and 2% E' with magnetic field both perpendicular and parallel to (100) interface with broadening of the P_{b0} , P_{b1} , and E' responses. (b) Simulated EPR response composed of 80% P_{b0} and 20% P_{b1} , and 0% E' with magnetic field both perpendicular and parallel to (100) interface with broadening of the P_{b0} and P_{b1} responses. (c) Measured SDTAT response of unpassivated Si/SiO₂ MIS capacitors containing silicon of natural isotopic abundance. (d) Measured SDTAT response of Si/SiO₂ MIS capacitors depleted of ^{29}Si .

4.4 Near-Zero Field Magnetoresistance Comparisons of Non-Forming Gas Annealed Si/SiO₂ & ²⁸Si/²⁸SiO₂ Metal-Insulator-Semiconductor Capacitors

Figure 4.4 displays the integral of SDTAT response as measured through NZFMR of the unpassivated Si/SiO₂ MIS capacitors containing silicon of natural isotopic abundance and silicon depleted of ²⁹Si. The SDTAT response and line shape as measured through NZFMR has been explained both qualitatively and quantitatively via the two-site model and is a strong function of the hyperfine interactions present within the system [17], [22], [24], [29]. In fact, since the two sets of samples are virtually identical, apart from the ²⁹Si depletion, the line shape differences should be entirely due to the differences in hyperfine interactions. With the removal of ²⁹Si and therefore the elimination of nearly all the electron-nuclear hyperfine interactions from the system, it would be expected that the linewidth of the ²⁹Si depleted structures would be much narrower if the response is dominated by interface traps. We measure a linewidth of 0.567 mT and 0.254 mT for the unpassivated Si/SiO₂ MIS capacitor containing silicon of natural isotopic abundance and silicon depleted of ²⁹Si, respectively.

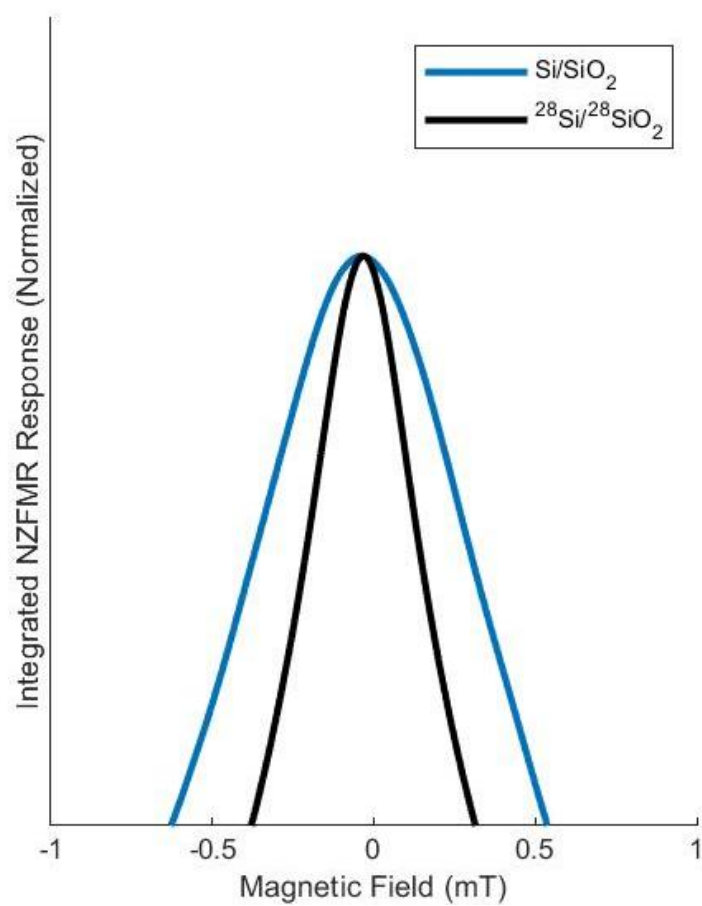


Figure 4.4 Integral of the SDTAT response measured via NZFMR from unpassivated Si/SiO₂ MIS capacitor containing silicon of natural isotopic abundance (blue) and silicon depleted of ²⁹Si (black).

The reduction of the NZMFR linewidth by a factor greater than two between samples provides further insight into the nature of the observed SDTAT response. On the Si side of the Si/SiO₂ interface containing silicon of natural isotopic abundance, the superhyperfine splitting due to a dangling bond's interaction with nearby ²⁹Si nuclei is reported to range between 1.4 mT and 1.6 mT [55]–[57]. The E' center silicon dangling bond defects in the oxide have a hyperfine splitting due to a dangling bond's interaction with nearby ²⁹Si and is reported to be about 0.12 mT; a full order of magnitude lower than that of the interface defects [58]–[61]. Thus, both the EDMR and NZFMR measurements strongly indicate the process of the electron tunneling across the oxide involves an interface defect to oxide defect tunneling step. Furthermore, for the relatively low leakage current densities observed in our measurements, this appears to be the rate limiting step.

4.5 Near-Zero Field Magnetoresistance Comparison of both Non-Forming and Forming Gas Annealed Si/SiO₂ & ²⁸Si/²⁸SiO₂ Metal-Insulator-Semiconductor Capacitors

Figure 4.5 displays the normalized integrals of SDTAT response as measured through NZFMR for both the unpassivated and passivated Si/SiO₂ and ²⁸Si/²⁸SiO₂ MIS capacitors. With the introduction of hydrogen at the interface from the FGA, the line shape and linewidth for both Si/SiO₂ and ²⁸Si/²⁸SiO₂ MIS capacitors are nearly identical. The linewidth after forming gas anneal for both Si/SiO₂ and ²⁸Si/²⁸SiO₂ MIS capacitors was 1.15 mT. This linewidth is nearly a factor of 2 greater in comparison to the unpassivated Si/SiO₂ MIS capacitor and nearly a factor of 5 greater in comparison to the unpassivated ²⁸Si/²⁸SiO₂ MIS capacitor. Figure 4.6 shows the non-normalized integrals of the SDTAT response as measured through NZFMR for both the unpassivated and passivated Si/SiO₂ and ²⁸Si/²⁸SiO₂ MIS capacitors. The amplitude

of the response was greatly reduced by the FGA. This would be anticipated if interface trap to oxide trap tunneling is the rate limiting step because the hydrogen passivates nearly all of the silicon dangling bonds at the interface. With the introduction of ^{29}Si magnetic nuclei to the interface, the signal broadened and the signal amplitude slightly decreased. Figure 4.7a shows the overlap of the second integrals of the both the Si/SiO_2 and $^{28}\text{Si}/^{28}\text{SiO}_2$ MIS capacitors which indicates these samples have similar interface state densities. In conventional EPR, the second integral of the lock-in response is proportional to the total number of paramagnetic spins within the sample [62]. While it is not yet clear how to calculate spin densities in NZFMR with this method, the nearly identical nature of the materials chemistry between the unpassivated Si/SiO_2 and $^{28}\text{Si}/^{28}\text{SiO}_2$ MIS capacitors suggests that a relative comparison of defect densities between capacitors scales with the second integral. Similarly displayed in Figure 4.7b, the overlap of the second integrals of the both the Si/SiO_2 and $^{28}\text{Si}/^{28}\text{SiO}_2$ MIS capacitors exposed to FGA suggest that these samples also have similar interface state densities. Again, the nearly identical nature of the materials chemistry between the passivated Si/SiO_2 and $^{28}\text{Si}/^{28}\text{SiO}_2$ MIS capacitors suggest that a relative comparison of defect densities between capacitors scales with the second integral as well. Interface state densities were measured directly using capacitance vs. voltage measurements before and after samples underwent FGA. Before the FGA, the interface state densities were found to be about $3 * 10^{12} \frac{1}{\text{cm}^2\text{V}}$ and $2 * 10^{12} \frac{1}{\text{cm}^2\text{V}}$ for the mid-half of the bandgaps of Si/SiO_2 and $^{28}\text{Si}/^{28}\text{SiO}_2$ MIS capacitors, respectively. After the FGA, interface densities were found to be to $9 * 10^9 \frac{1}{\text{cm}^2\text{V}}$ and $6 * 10^9 \frac{1}{\text{cm}^2\text{V}}$ for mid-half of the bandgaps of Si/SiO_2 and $^{28}\text{Si}/^{28}\text{SiO}_2$ MIS capacitors, respectively, consistent with about $10^{12} \frac{1}{\text{cm}^2}$ of ^1H atoms deposited at the interface. Note that the ratio of the interface trap density without and with the FGA is close to the ratio of NZFMR responses. Collectively, these results

suggest that hyperfine fields due the presence of ^1H magnetic nuclei dominate the local magnetic environment of the few remaining dangling bonds after FGA.

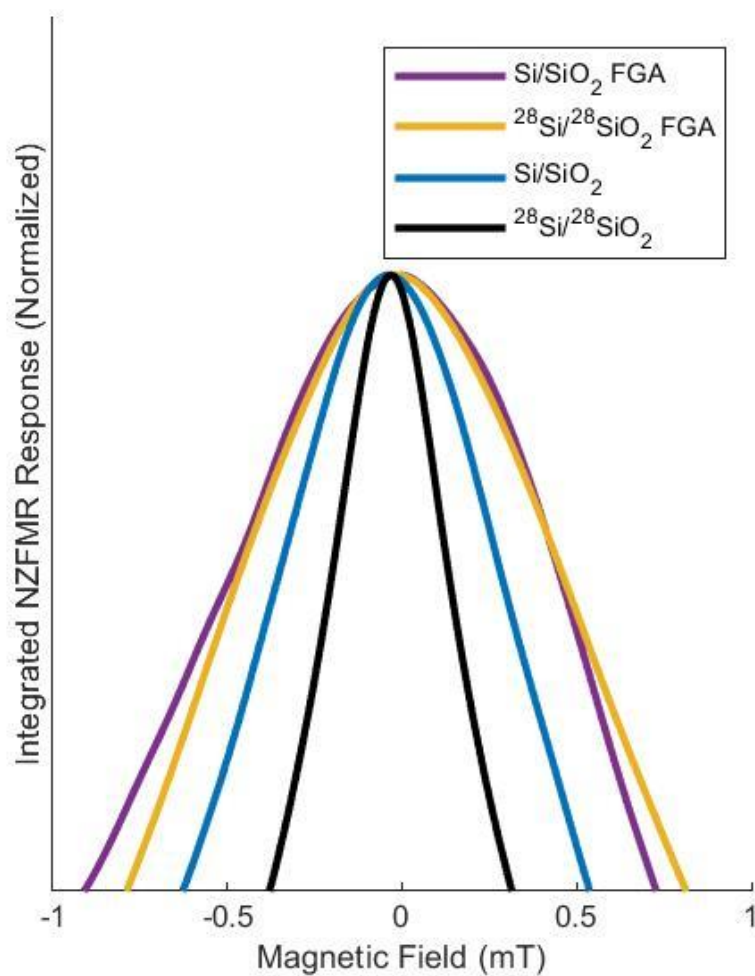


Figure 4.5 Normalized integral of the SDTAT response measured via NZFMR from unpassivated Si/SiO₂ MIS capacitor depleted of ²⁹Si (blue), unpassivated Si/SiO₂ MIS capacitor containing silicon of natural isotopic abundance (yellow), the passivated Si/SiO₂ MIS capacitor depleted of ²⁹Si (yellow), and the passivated Si/SiO₂ MIS capacitor containing silicon of natural isotopic abundance (purple).

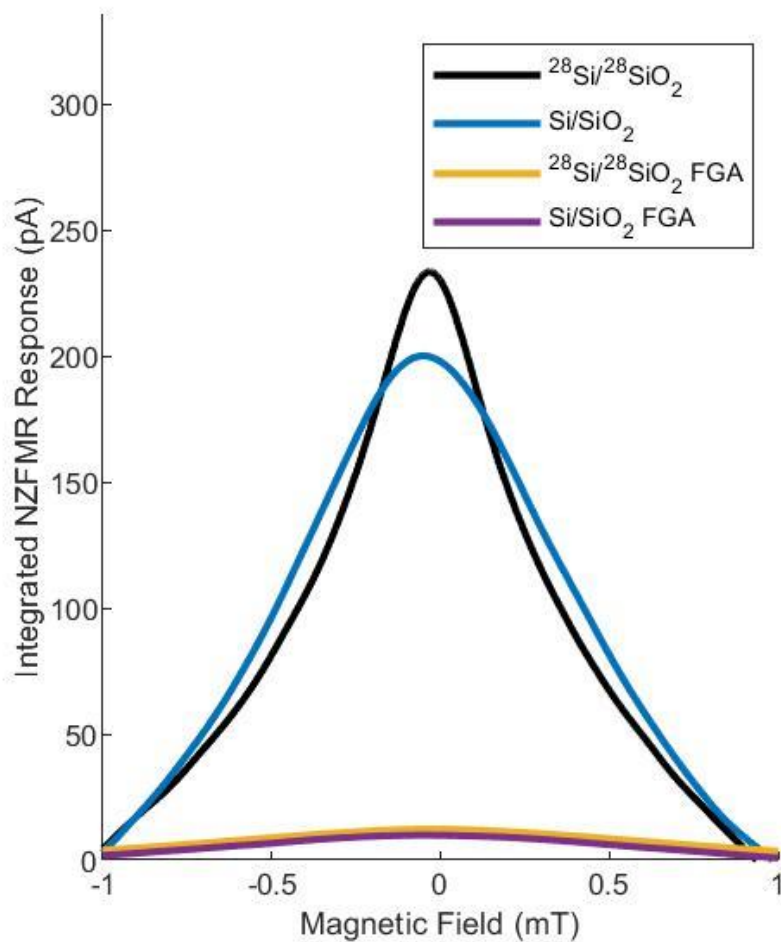


Figure 4.6 Non-Normalized integral of the SDTAT response measured via NZFMR from unpassivated Si/SiO₂ MIS capacitor depleted of ²⁹Si (blue), unpassivated Si/SiO₂ MIS capacitor containing silicon of natural isotopic abundance (yellow), the passivated Si/SiO₂ MIS capacitor depleted of ²⁹Si (yellow), and the passivated Si/SiO₂ MIS capacitor containing silicon of natural isotopic abundance (purple).

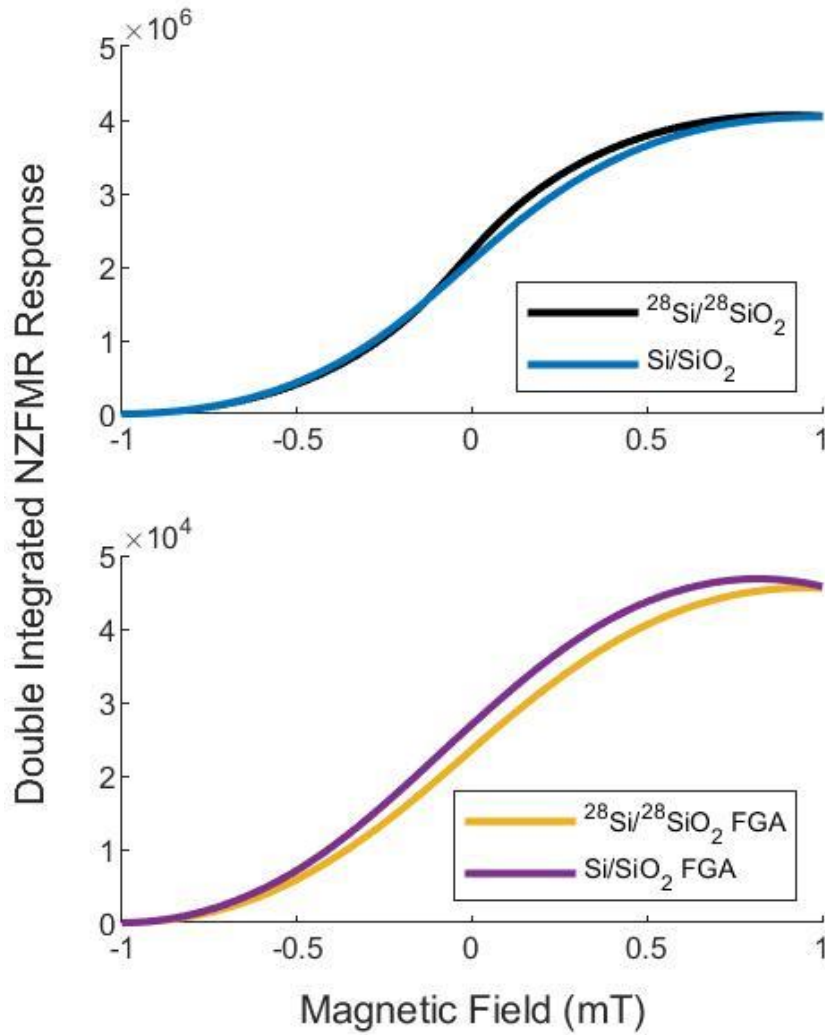


Figure 4.7 Double Integrated NZFMR Response **a.)** of non-passivated $^{28}\text{Si}/^{28}\text{SiO}_2$ (black) and Si/SiO_2 (blue) MIS capacitors **b.)** passivated $^{28}\text{Si}/^{28}\text{SiO}_2$ (yellow) and Si/SiO_2 (purple) MIS capacitors.

4.6 Low-Frequency Electrically Detected Magnetic Resonance and Near-Zero Field Magnetoresistance of Non-Forming Gas Annealed $^{28}\text{Si}/^{28}\text{SiO}_2$ Metal-Insulator-Semiconductor Capacitors

Figure 4 shows the experimental SDTAT response measured through NZFMR and low-frequency ($\nu = 151\text{MHz}$) EDMR for the unpassivated $^{28}\text{Si}/^{28}\text{SiO}_2$ MIS capacitor. Features are observed in the low-field EDMR response in addition to a “forbidden” half-field transition as displayed in Figure 5.

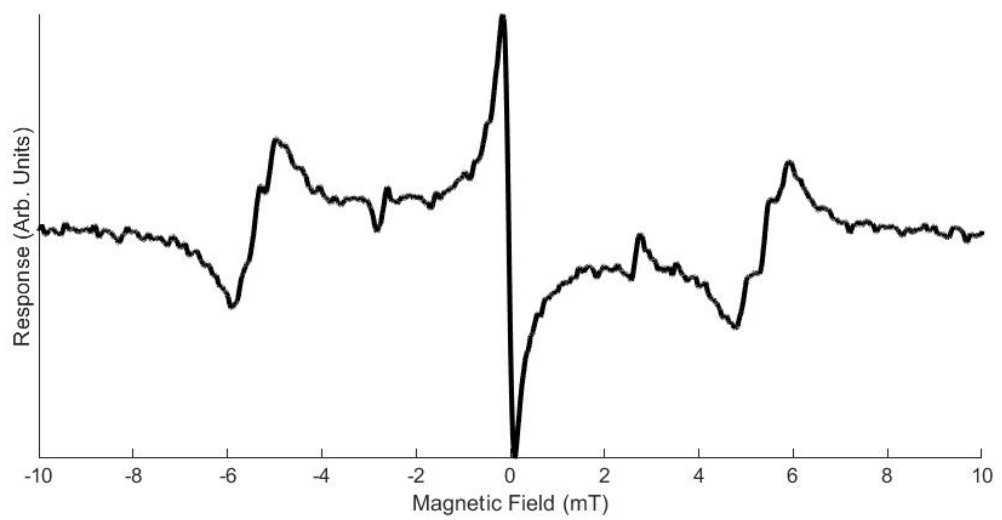


Figure 4.8 SDTAT response measured via low-frequency ($\nu = 151\text{MHz}$) EDMR and NZFMR of unpassivated $^{28}\text{Si}/^{28}\text{SiO}_2$ MIS capacitor. Note: in this wide scan, the spectrometer settings were optimized to include all of the relevant features which somewhat compromises the signal-to-noise ratio of the data presented.

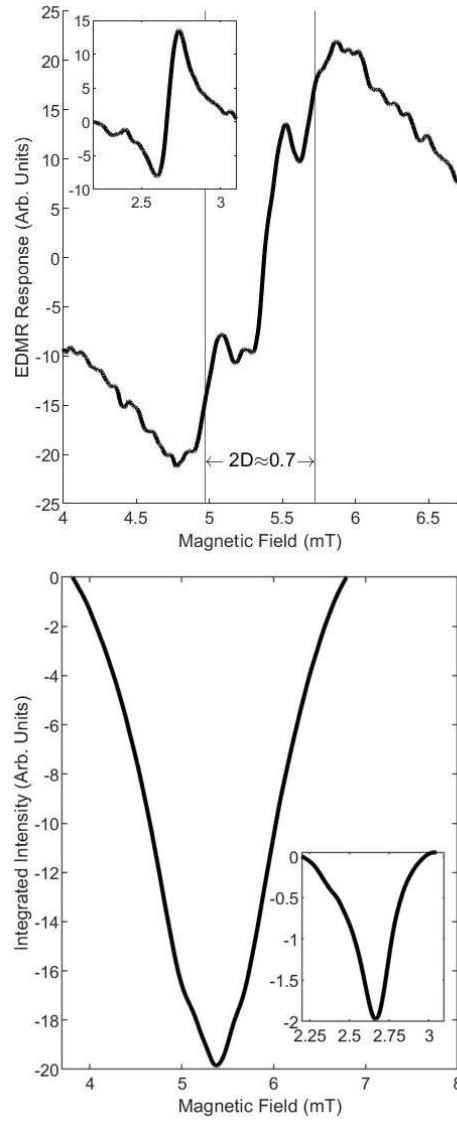


Figure 4.9 Low-Frequency ($\nu = 151\text{MHz}$) SDTAT EDMR responses of the unpassivated $^{28}\text{Si}/^{28}\text{SiO}_2$ MIS capacitor (upper, large) and its observed half-field response (upper, small). Integrated intensity of the main EDMR signal (lower, large) and integrated intensity of the half-field response (lower, small).

Half-field transitions are caused by a mixing of basis states due to magnetic dipolar interactions between closely neighboring unpaired spins [2], [63]–[67]. This allows for the transition between $|\uparrow\uparrow\rangle \leftrightarrow |\downarrow\downarrow\rangle$ levels at exactly half of the resonance field [2]. The Hamiltonian of the magnetic dipole-dipole interaction is given by Equations (3.39) and (3.40). A general expression for the approximate ratio of the half-field response to the full-field response (the response at the resonance condition) is given by [63]–[65]

$$\frac{I_{HF}}{I_{FF}} \approx \frac{9}{15} \left(\frac{g\mu_B D}{h\nu} \right)^2. \quad (4.1)$$

Here, I_{HF} represents the maximum of the integrated intensity of the half-field response while I_{FF} represents the maximum of the integrated intensity of the full-field response.

Additionally, μ_0 is the permeability of free space, μ_B represents the Bohr magneton, h is Planck's constant, g is the defect g-factor, ν is the resonance frequency, and D represents the dipolar coupling constant as determined by the half-field response, whereby

$$D \approx \frac{h\nu}{g\mu_B} \left(\frac{15}{9} \frac{I_{HF}}{I_{FF}} \right)^{\frac{1}{2}} \quad (4.2)$$

It is important to note that Equations (4.1) and (4.2) yield a precision in dipolar coupling strength which can be substantially degraded and subject to error, especially in the EDMR measurement. A discussion of several of these limitations is provide by Eaton *et al.* [64]. As referenced in Figure Figure 4.9, the ratio of the maximum integrated intensity of the half-field response to the maximum integrated intensity of the full-field response in the unpassivated $^{28}\text{Si}/^{28}\text{SiO}_2$ MIS capacitors is

$$\frac{I_{HF}}{I_{FF}} \approx 0.010 \quad (4.3)$$

which when solved for using Equation (4.2) yields a dipolar coupling strength of $D \approx 0.4\text{mT}$.

The dipolar coupling constant can also be estimated via the splitting observed in the top of Figure 4.9. From this splitting, we estimate $D \approx 0.35mT$ ($2D \approx 0.7mT$). We find good agreement between the two measures of the dipolar coupling constant.

Due to the negligible ^{29}Si content in the MIS capacitors, hyperfine interactions due to the interaction with ^{29}Si nuclei can be assumed to be absent or negligible. These MIS capacitors were also unpassivated, so hyperfine interactions with nearby ^1H nuclei at the interface are also absent or negligible. Additionally, results of sections 4.3, 4.4, and 4.5 also suggest that the oxide trap to interface trap tunneling event served as the rate limiting step in the transport across the device. Therefore, the observation of the half-field response in the experimental results suggests that magnetic dipolar interactions between paramagnetic interface states, P_b centers, are the dominant interactions in these MIS capacitors.

4.7 Conclusions

The results outlined in this chapter indicate that the SDTAT EDMR and NZFMR response was dominated by silicon dangling bonds back bonded to silicon at the interface, P_{b0} and P_{b1} centers. Our understanding of SDTAT is depicted in Figure 4.10. It includes both oxide and interface trap sites. The tunneling process observed in our leakage current is primarily rate limited by the step across the interface and not limited by trap-assisted-tunneling through oxide defects, a process which certainly plays an important role in leakage currents across some oxides. Despite the existence of the few potential sites that likely exist that could be occupied while tunneling through the oxide, the resistance at these low current levels is dominated by one bottleneck pair [24]. Faster transitions between spin states of just one of the sites in the pair open spin-blocked pathways becoming viable conduction channels for electronic-spin

transport [24]. This therefore limits the measurement to the site of the slower transition rate and not the multiple transitions that may exist with faster transition rates.

The devices studied in this section were processed nearly identically with the exception of the presence or absence of ^{29}Si magnetic nuclei, respectively. Additionally, both sets of devices underwent forming gas anneals in order to deposit ^1H magnetic nuclei at the interface by passivating interface states. Although the deposition and passivation of interface states with hydrogen reduces leakage currents across the interface, the chemical nature of the defects and the type of spin-transport remained generally unchanged. However, the magnitude of the NZFMR response is reduced by a factor of about 250 while the electrical measurement of interface trap density indicates a reduction of about 100. The substantial narrowing of the NZFMR response with the removal of ^{29}Si in the unpassivated MIS capacitors indicates that interface traps play an important role in tunneling across Si/SiO_2 and that interface trap to oxide trap tunneling event also serves as the rate limiting process in these phenomena. Additionally, with the introduction of ^1H to the interface after a forming gas anneal, linewidths and line shapes were nearly identical between Si/SiO_2 and $^{28}\text{Si}/^{28}\text{SiO}_2$ MIS capacitors suggesting the local magnetic environment of the remaining interface traps is dominated by hyperfine fields due to the abundance of ^1H magnetic nuclei at the interface.

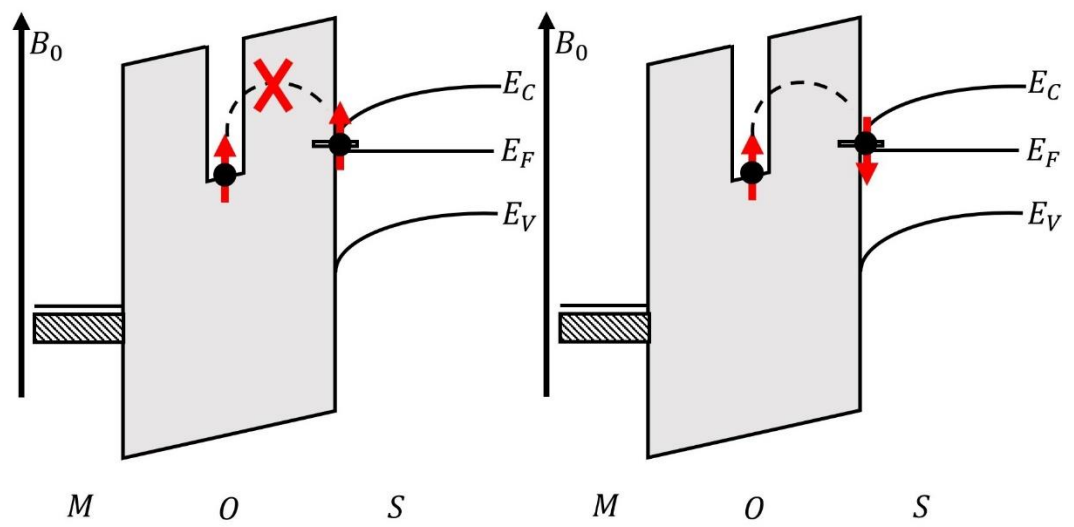


Figure 4.10 Depiction of rate limiting step across the dielectric/semiconductor interface observed via SDTAT. Tunneling between traps is forbidden due to the triplet configuration (left). Tunneling is allowed due to the singlet configuration (right).

Chapter 5 Least Squares Fitting of Models Developed from the Stochastic Quantum Liouville Equation

5.1 Introduction

Although some analytical results have been extracted from the NZFMR spectrum as outlined in Chapter 4 no systematic approach has been available with which to obtain information on the strength of the interaction given a defined spin cluster. This chapter outlines a systematic method with which to extract that information from the NZFMR response via a least squares fitting method using solutions of the SLE. We show that it is possible to extract information such as the strength of an interaction given a defined spin cluster. We utilize this approach in the three different material systems: a γ -irradiated Si/SiO₂ MOSFET, an a-Si:H MIS capacitor, and an unpassivated ²⁸Si/²⁸SiO₂ MIS capacitor. This method is then extended and applied to the low-frequency EDMR spectrum as measured through SDTAT of the unpassivated ²⁸Si/²⁸SiO₂ MIS capacitor.

5.2 Non-linear Least Squares Fitting Algorithm

When solving for the density-matrix as outlined in Equation (3.67), ρ is an inherent function of the rate constants, c_1 and c_2 , the generation rate, g , and the Hamiltonian, $\mathcal{H}$. The Hamiltonian is a function of the externally applied magnetic field(s) and respective coupling constant which depend upon the interactions present. Since the externally applied magnetic fields to the system are known at each point of the experimental sweep, the Hamiltonian simply becomes a function of the of the respective coupling constants, C_c . The density-matrix then becomes a function of coupling constants, C_c , rates c_1 and c_2 , and generation rate, g , so

$$\boldsymbol{\rho} = \boldsymbol{\rho}(C_c, c_1, c_2, g). \quad (5.1)$$

Since the measurable-observable current, I , through the system is proportional to the density-matrix, we find that the measurable-observable current is a function of those same constants,

$$I = I(C_c, c_1, c_2, g). \quad (5.2)$$

Given a measured NZFMR spectrum as function of magnetic field, we can then fit a simulated spectrum to the given data set using the appropriate form of the SLE via a non-linear weighted least squares method. The residual error, r^2 , is defined as

$$r^2(C_c, c_1, c_2, g) = \sum_i \left(w_i \left(I_{M,i} - I_{S,i}(C_c, c_1, c_2, g) \right)^2 \right), \quad (5.3)$$

where w_i is a weighting factor. Here the residual square error between the measured spectrum, I_M , and the simulated spectrum, I_S , is a function of the respective coupling constants, C_c , rates c_1 and c_2 , and generation, g . When the residual error is minimized at some set of values, that is when

$$\nabla(r^2(C_c, c_1, c_2, g)) = 0, \quad (5.4)$$

we can gain the information about the defects coupling constants and kinetic rate factors as a result of the value corresponding to the minimum.

In the fitting algorithm, w_i is defined from the curvature of the spectrum (second derivative of the absorption spectrum) such that

$$w_i = \left| \left(\frac{dI_M}{dB_0} \right)_i \right|. \quad (5.5)$$

The second derivative of the NZFMR spectrum highlights areas where important features generally appear. Weighting from the second derivative places higher residual error, or effectively higher importance, on these features for fitting. This forces the minimization

routine to fit these sections of the experimental data with greater accuracy instead of neglecting them.

Equation (5.3) is minimized through an interior point minimization routine outlined by Byrd *et al.* [68]. This type of algorithm is conveniently built into the function *fmincon* within the global optimization toolbox in MATLAB. When searching the parameter space for a minimum, the minimization routine can be left unconstrained or passed a window of constraints. Including a window of constraints based on physical parameters can cut down on computation time. Generally, rates c_1 and c_2 , and generation rate, g , are left unconstrained.

5.3 Near-Zero Field Magnetoresistance Spectral Fitting of γ -Irradiated Si/SiO₂ Metal-Oxide-Semiconductor Field-Effect Transistor

5.3.1 Experimental Details

The NZFMR measurement of the Si/SiO₂ device was conducted via the DCIV (gated-diode) biasing scheme where the source and drain are shorted together and forward biased and the gate is biased so that the channel resides in depletion [69]. When the gate bias is swept through depletion, a peak in substrate current is measured. This peak in substrate current is due to recombination at interfacial defects and is observed when there are roughly equal populations of electrons and holes are present at the interface. The DCIV NZFMR response is measured via the substrate current, with the gate bias held at the peak recombination current. This biasing scheme has been used in the past to explore the Si/SiO₂ interface via DCIV EDMR [52]. In our case, the gate bias was 0.3 V, and the source and drain bias was -0.33 V. This device was irradiated with a ⁶⁰Co source to a dose of 1 Mrad(Si) in order to generate interface states. The NZFMR DCIV measurement reported was made after irradiation. No detectable

NZFMR DCIV signal was present before irradiation. The Si/SiO₂ MOSFET used had an oxide thickness of 7.5 nm and a gate area of 41,000 μm^2 .

5.3.2 Fitting Results

The top of Figure 5.1 depicts a gated diode DCIV NZFMR measurement of a gamma irradiated Si/SiO₂ MOSFET and its fitted spectrum. Based on magnetic resonance measurements, the MOS interface defects involved are silicon dangling bonds, P_b centers [3], [4], [6], [7], [70]–[72]. We would also expect a fair amount of hydrogen in the interface region since the devices were annealed in forming gas (a mixture of H₂ and N₂). However, after irradiation damage we expect that hydrogen would still be present at the interface region but that the distribution would be somewhat different [73]. Therefore, we would also expect the possibility of a dangling bonds interaction with nearby hydrogen. We constructed our fitting algorithm to consider two separate recombination paths each with their own isotropic hyperfine coupling constant acting as an adjustable parameter with which to fit the experimental data. The first recombination path describes an electron falling from the intermediate state to a deep level P_b center that is interacting with distant ²⁹Si while the second path describes an electron falling from the intermediate state to a deep level P_b center that is interacting with nearby hydrogen. The recombination and dissociation rates were tethered together since we assume that these rates should be the same at all P_b centers regardless of the interaction taking place with the P_b center. The scaling factors that relate to the relative contributions were left as independent adjustable parameters. This brought the total number of adjustable parameters to six. The algorithm then fit these six parameters to the experimental data. The adjustable parameters of each path are represented in Table 5.1. Table 5.2 contains the isotropic hyperfine

coupling constants, a , singlet recombination rates, k_S , pair dissociation rates, k_D , and relative contribution scaling factor, Z , the fitting algorithm produced for the two recombination paths.

Our fitting algorithm generated the hyperfine coupling constants to be 1.4 mT and 0.2 mT. An interaction distant ^{29}Si interaction with a P_b center has an isotropic hyperfine coupling constant that ranges between 1.4-1.6 mT [55]–[57]. We find the 1.4 mT electron-nuclear hyperfine constant generated as result of the fit to be consistent with interactions between P_b centers with distant ^{29}Si while we attribute the 0.2 mT to an interaction with nearby hydrogen.

We are also confident of the parameters listed in Table 1 to be within roughly $\pm 20\%$. Figure 5.2 displays the percent difference from the minimum square error between the generated spectrum and the experimental spectrum as a range in colors within the parameter space. The axes of plot represent the parameter space and range from 80% to 120% of the values of best fit for the singlet recombination rates, k_S , pair dissociation rates, k_D , and isotropic hyperfine coupling constants, a , for the x, y, and z axes, respectively. Two plots were generated such that one hyperfine parameter of the two was fixed at its value of best fit while the second hyperfine parameters varied. The centers of the figures, (100%,100%,100%), are the values of best fit as presented in Table 1 and display the minimum square error. Points further from the center display increased minimum square error. The outer most corners of the plot on the top and bottom planes represent the extremes and of the parameter ranges and display differences from minimum square errors of 100% of the minimum. We attribute the larger range in the difference from the minimum square error to the modest signal-to-noise ratio (SNR) of the experimental spectrum. The algorithm's primary task is to reduce the residual error between the experimental data and its fit. Lower SNR inherently forces more residual error into the fit, decreasing the confidence of the values of best fit.

From the ratio of the relative contribution scaling factors, we found the relative abundances of the distant ^{29}Si and nearby hydrogen to be about 7% and 93% respectively. This is consistent with what we would expect at the interface; while the amount of hydrogen is not precisely known, it is reasonable to assume it to be much greater than the number of ^{29}Si sites.

In a recent publication [16], Si/SiO₂ NZFMR and EDMR were examined within a mean field model where hyperfine interactions were assumed to be due to Gaussian distribution of classical nuclear spin vectors at each of the electron sites (*i.e.* shallow and deep states). The work found a typical hyperfine field magnitude at the deep level defect to be 0.5 mT which falls within the range of values determined here. A complete comparison between the two approaches will require the model presented herein to be extended to include EDMR.

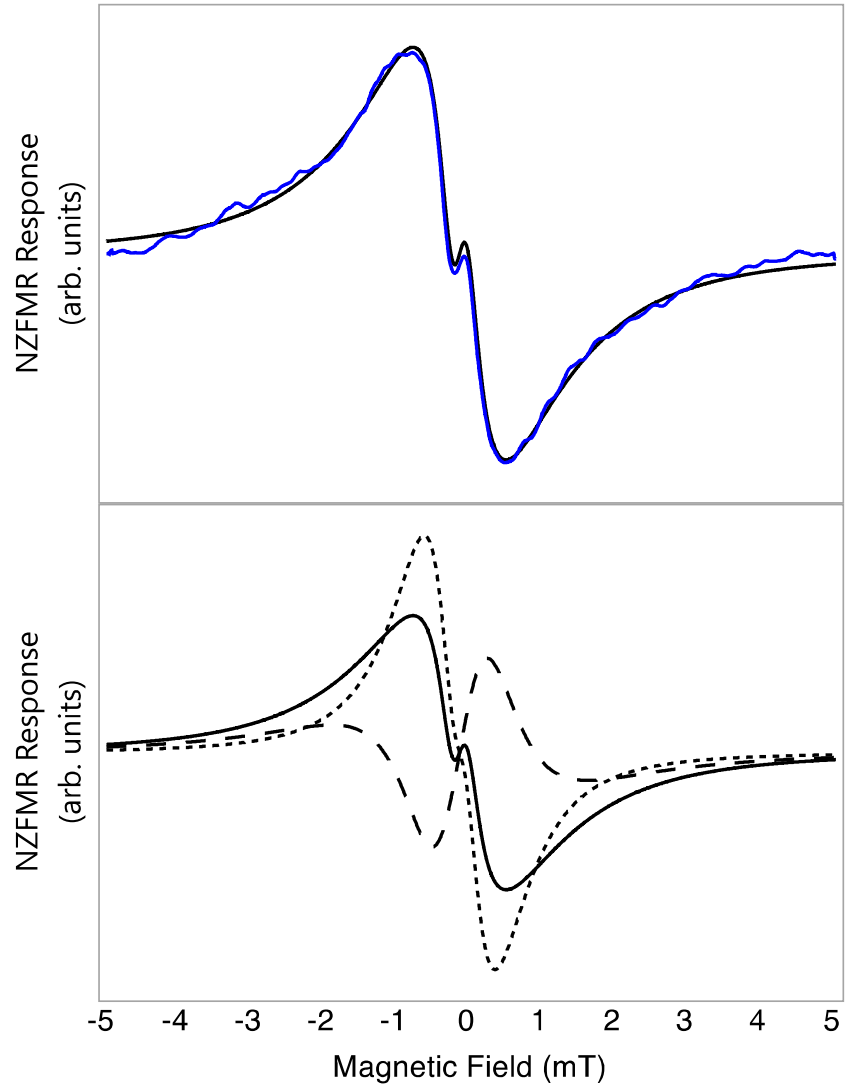


Figure 5.1 Si/SiO₂ MOSFET DCIV NZFMR experimental spectrum (blue). Fitted NZFMR spectrum (solid black) that is the superposition of the spectra due to a P_b center electron-nuclear hyperfine interactions with distant ²⁹Si (long dash) and nearby Hydrogen (short dash).

Table 5.1 Independent adjustable parameter, v_n , for each path of the simulated DCIV NZFMR measurement of the γ -Irradiated Si/SiO₂ Metal-Oxide-Semiconductor Field-Effect Transistor.

P_b Interactions with:	a (mT) [a (MHz)]	k_s (GHz)	k_D (GHz)	Z (arb. units)
<i>Distant ^{29}Si</i>	v_1	v_3	v_4	v_5
<i>Nearby Hydrogen</i>	v_2	v_3	v_4	v_6

Table 5.2 Fitted parameters of the SLE to the DCIV NZFMR measurement of the γ -Irradiated Si/SiO₂ Metal-Oxide-Semiconductor Field-Effect Transistor.

P_b Interactions with:	a (mT) [a (MHz)]	k_s (GHz)	k_D (GHz)	Z (arb. units)	Total Residual Error of the Normalized Spectra
<i>Distant ^{29}Si</i>	1.401 [39.263]	0.174	0.037	24.333	1.958
<i>Nearby Hydrogen</i>	0.206 [5.773]	0.174	0.037	1.704	

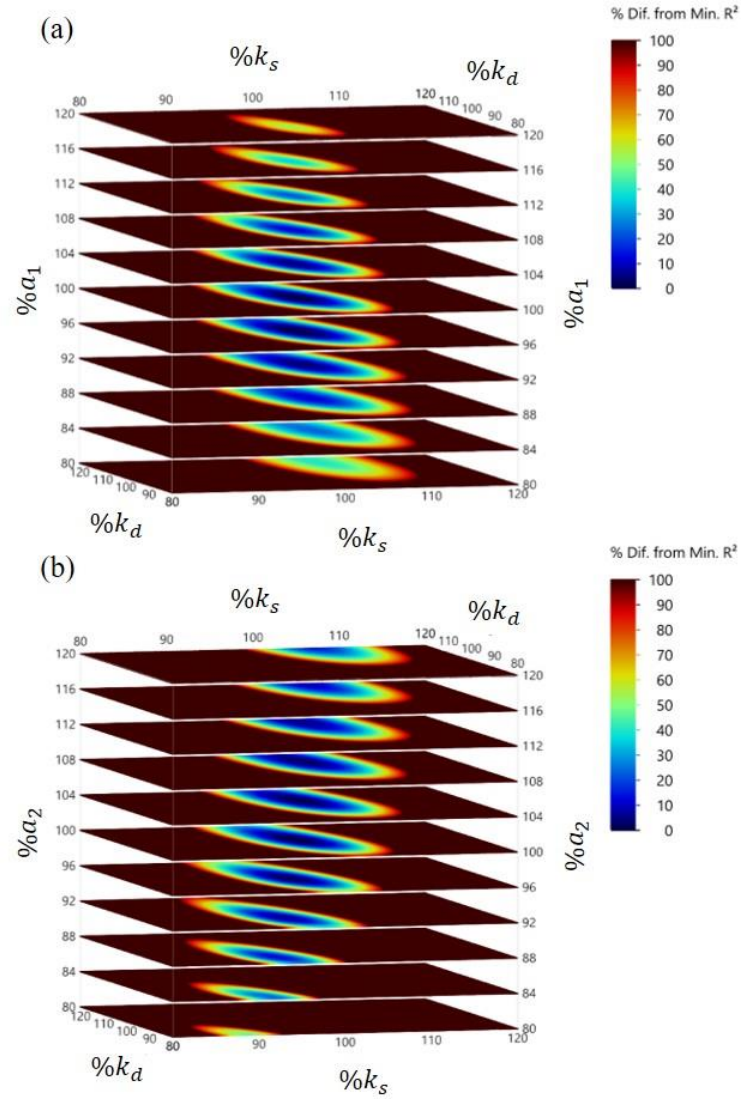


Figure 5.2 Percent difference from the minimum square error displayed as a color as a function of the parameter space for Si/SiO₂ MOSFET DCIV NZFMR measurement. (a) a_1 was varied while a_2 was held constant the value of best fit. (b) a_2 was varied while a_1 was held constant the value of best fit.

5.4 Near-Zero Field Magnetoresistance Spectral Fitting of a-Si:H Metal-Insulator-Semiconductor Capacitors

5.4.1 Experimental Details

The NZFMR measurements on a-Si:H MIS capacitor was made via SDTAT. The a-Si:H was biased with -2 V and the SDTAT NZFMR response was measured via the leakage current across the insulator region [74]. The MIS stack consisted of Ti/10 nm of a-Si:H/p-Si.

5.4.2 Fitting Results

The top of Figure 5.3 shows an NZFMR SDTAT measurement in a-Si:H along with a fitted spectrum. Based on previous high-field EDMR measurements and as well as EPR and EDMR literature, the dominating defect is the silicon dangling bond. As reported in the literature, a distant ^{29}Si interaction with a silicon dangling bond in a-Si:H has an isotropic hyperfine coupling constant of about 1.5 mT with an integrated intensity corresponding to a relative ratio of abundance of about 14% [75]. These particular a-Si:H samples also had a very high concentration of hydrogen of about 20-30%, so we expect the dangling bond to have some hyperfine interaction with nearby hydrogen. The coupling constant for hydrogen ranges between 0.35 mT and 1.0 mT (potentially higher) depending upon average distance between hydrogen and the dangling bond [76], [77]. Since interactions with nearby hydrogen are much more likely than interactions with distant ^{29}Si , the fitting algorithm was constructed to consider two separate hopping paths. The first path (labeled Path 1 in Table 5.3 and Table 5.4) describes an electron hopping from a dangling bond interacting with nearby hydrogen to another dangling bond also interacting nearby hydrogen. We would expect that the isotropic hyperfine coupling constant at both sites interacting with hydrogen would be somewhat similar, that is

we assume that $a_\alpha = a_\beta$. The second path (labeled Path 2 in Table 5.3 and Table 5.4) describes an electron hopping from a dangling bond interacting with nearby hydrogen to another dangling bond experiencing an interaction with distant ^{29}Si . These two sites in this path are experiencing two different hyperfine interactions, so we cannot assume their isotropic coupling constants would be the same, that is $a_\alpha \neq a_\beta$. Since all of the interactions with nearby hydrogen must have the somewhat similar isotropic hyperfine coupling constants, all three parameters were tethered together as one adjustable parameter. The isotropic hyperfine coupling constant for distant ^{29}Si was left as another adjustable parameter. We also assume the probability of hopping rates must be the same between paths. That is r_{ae} and r_{ab} must be the same for both paths, so r_{ae} values were tethered together as one adjustable parameter while r_{ab} values were tethered together as another adjustable parameter. The rate at which the two sites are filled, r_{ea} , were left as separate independent adjustable parameters. This brought the total number of adjustable parameters to six. The adjustable parameters of each path are represented in Table 5.3. The algorithm then fit these six parameters to the experimental data. Table 5.4 contains the isotropic hyperfine coupling constants, a_α and a_β , hopping rates, r_{ae} and r_{ab} , and the rate at which the two sites were filled, r_{ea} , that the fitting algorithm produced for the two hopping paths. Our fitting algorithm generated the hyperfine coupling constants for the two interactions present in the hopping paths to be about 1.7 mT and 0.3 mT. We find these isotropic electron-nuclear hyperfine interactions to be consistent with dangling bond interactions with distant ^{29}Si and nearby hydrogen.

Given that the electron starts at a site interacting with nearby hydrogen, the probability that the electron will hop to a site interacting with distant ^{29}Si is about 15%. The probability that the electron will hop to another site interacting nearby hydrogen is 20-30%, given that the

a-Si:H films is composed of 20-30% hydrogen. Therefore the probability of hopping to a site experiencing an interaction with nearby hydrogen is about 1.3-2 times more likely than hopping to a site experience an interaction with distant ^{29}Si . The ratio of rates at which the sites are filled from Table 4, $(r_{ea})_{Path\ 1}/(r_{ea})_{Path\ 2}$, indicate that the hop to a site interacting with nearby hydrogen is about 2.6 times more likely than a hop to a site experience an interaction with distant ^{29}Si . We find agreement between the ratio of the rates at which the sites are filled to be consistent with the two most likely hopping paths.

We are confident of the parameters listed in Table 5.4 to be within at least $\pm 10\%$. Figure 5.4 displays the percent difference from the minimum square error of a generated spectrum and the experimental data as a range in colors within the parameter space. The axes of plot represent the parameter space and range from -10% to +10% of hopping rates, r_{ae} and r_{ab} , isotropic hyperfine coupling constants, a , for the x, y, and z axes, respectively. Two plots were generated such that one hyperfine parameter of the two was fixed at its value of best fit while the second hyperfine parameters varied. The centers of the figures, (100%,100%,100%), are the values of best fit as presented in Table 5.4 and display the minimum square error. Points further from the center display increased minimum square error. The outer most corners of the plot on the top and bottom planes represent the extremes and of the parameter ranges and display differences from minimum square errors off 100% of the minimum. We attribute the smaller range in the difference from the minimum square error to the higher signal-to-noise ratio (SNR) of the experimental spectrum. Since SNR was significantly higher in for this experimental data, the confidence of the fit was higher as well.

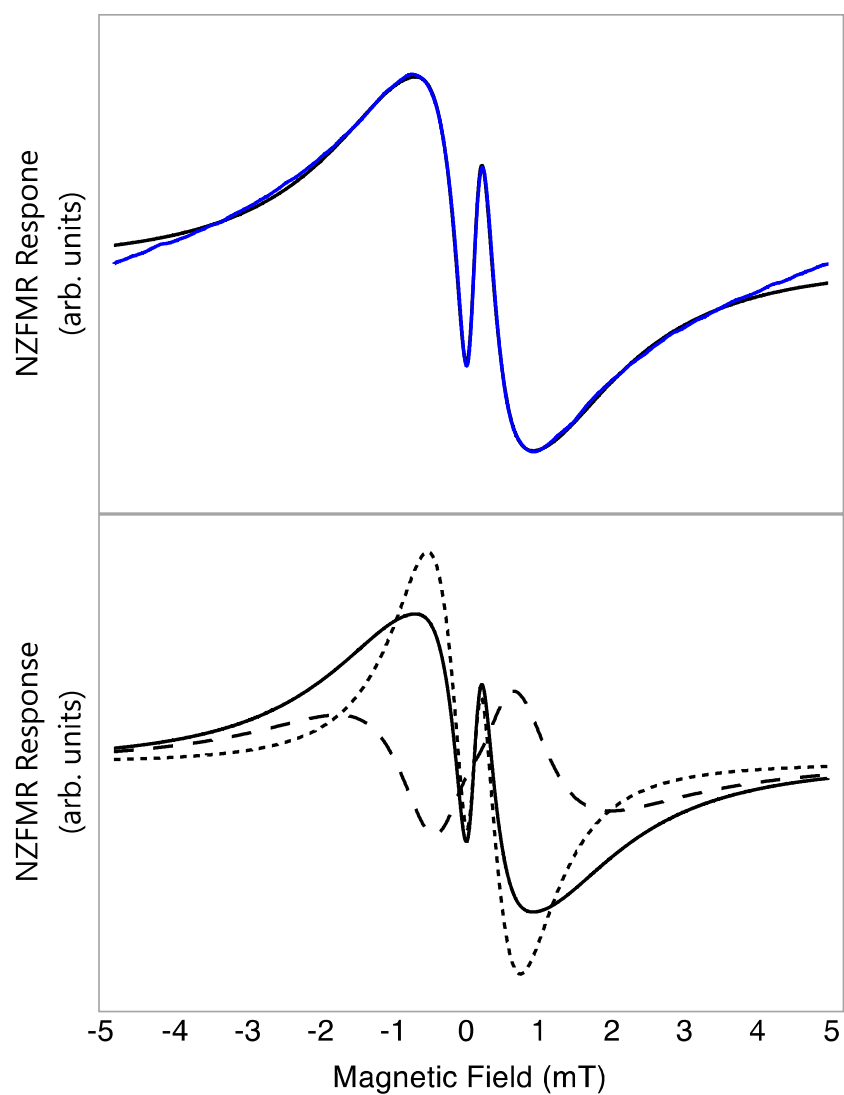


Figure 5.3 a-Si:H SDTAT NZFMR experimental spectrum (blue). Fitted NZFMR spectrum (solid black) that is the superposition of the spectra due to hopping from site A to site A (short dash) and hopping from site A to Site B (long dash).

Table 5.3 Independent adjustable parameter, v_n , for each path of the simulation of the SDTAT NZFMR Measurement of a-Si:H.

Hopping Paths	a_α (mT) [a_α (MHz)]	a_β (mT) [a_β (MHz)]	r_{ab} (GHz)	r_{ae} (GHz)	r_{ea} (GHz)
Path 1	v_1	v_1	v_3	v_4	v_5
Path 2	v_1	v_2	v_3	v_4	v_6

Table 5.4 Fitted parameters of the SLE to the SDTAT NZFMR Measurement of a-Si:H.

Hopping Paths	a_α (mT) [a_α (MHz)]	a_β (mT) [a_β (MHz)]	r_{ab} (GHz)	r_{ae} (GHz)	r_{ea} (GHz)	Total Residual Error of the Normalized Spectra
Path 1	0.332 [9.304]	0.332 [9.304]	0.128	0.048	616.488	0.990
Path 2	0.332 [9.304]	1.721 [48.231]	0.128	0.048	238.244	

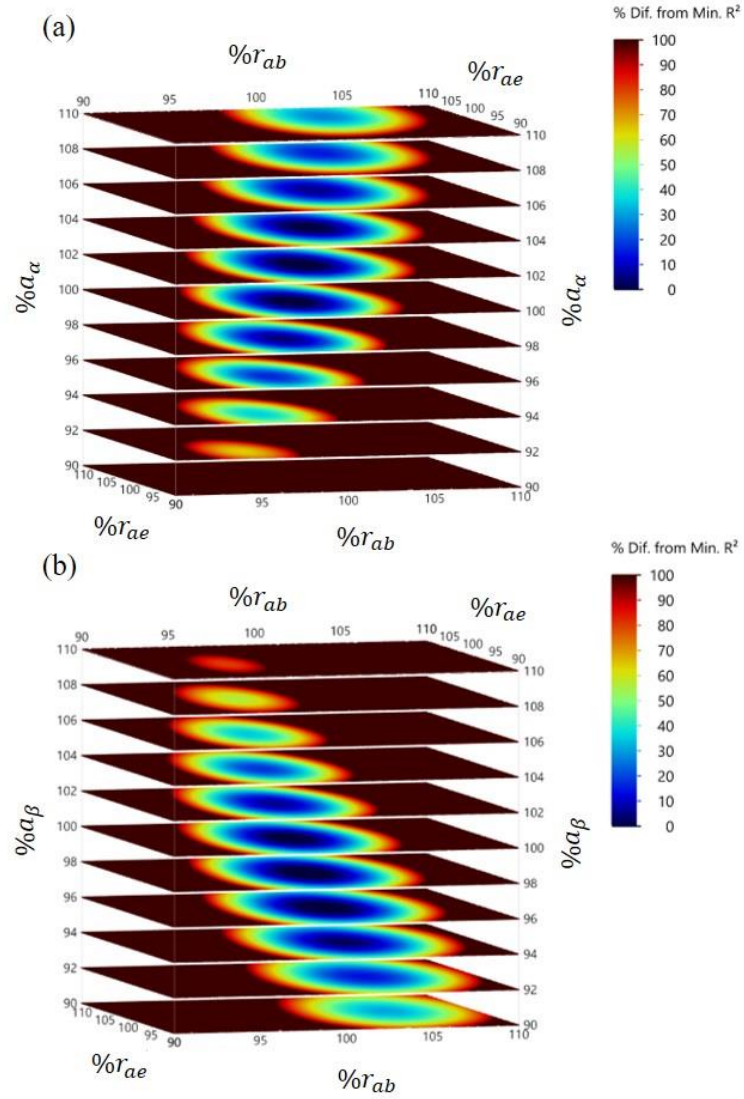


Figure 5.4 Percent difference from the minimum square error displayed as a color as a function of the parameter space for a-Si:H SDTAT NZFMR measurement. (a) a_α was varied while a_β was held constant the value of best fit. (b) a_β was varied while a_α was held constant the value of best fit.

5.5 Near-Zero Field Magnetoresistance Spectral & Electrically Detected Magnetic Resonance Spectral Fitting of $^{28}\text{Si}/^{28}\text{SiO}_2$ Metal-Insulator-Semiconductor Capacitors

5.5.1 Experimental Details

Experiments and further experimental details for these specific devices are outlined in Chapter 4

5.5.2 Fitting Results

As outlined in Chapter 4 the results of specific devices indicated that SDTAT EDMR and NZFMR response was dominated by silicon dangling bonds back bonded to silicon at the interface, $\text{P}_{\text{b}0}$ and $\text{P}_{\text{b}1}$ centers [78], [79]. Due to the negligible ^{29}Si content in the MIS capacitors, hyperfine interactions due to the interaction with ^{29}Si nuclei can be assumed to be absent or negligible. These MIS capacitors were also unpassivated, so hyperfine interactions with nearby ^1H nuclei at the interface are also absent or negligible. Additionally, the results also suggest that the oxide trap to interface trap tunneling event served as the rate limiting step in the transport across the device. Therefore, the observation of the half-field response in the experimental results suggests that magnetic dipolar interactions between paramagnetic interface states, P_{b} centers, are the dominant interactions in these MIS capacitors.

We constructed both our EDMR and NZFMR fitting algorithms to be consistent with our understanding of our experimental observations. We considered two characteristic tunneling sites: site α with spin $\mathbf{S}_i$ in the oxide and site β with spin $\mathbf{S}_j$ at interface. The rate limiting step in spin-transport is tunneling from site α to site β , tunneling from an oxide trap to interface trap. From the observation of our half-field response, we constructed a magnetic

dipolar interaction of strength, D , between spin $\mathbf{S}_j$ at site β and an unpaired interface spin $\mathbf{S}_k$ which we left as an adjustable parameter in our model. Since both spin $\mathbf{S}_j$ and spin $\mathbf{S}_k$ lie at the interface, the polar angle θ_{jk} was set to 89° such that the secular terms ($\mathbf{A}$ and $\mathbf{B}$) of the dipolar interaction do compute to $\mathbf{0}$. The azimuthal angle ϕ_{jk} was set to 0° . Tunneling rates $r_{\alpha\beta}$, $r_{\alpha e}$, and absolute scale were left as independent adjust parameters when performing separate NZFMR and EDMR fits with the algorithm. Fits were performed separately in order to eliminate the associated time-dependencies of the SLE when solving and simulating each respective response. This brought the total number of adjustable parameters to four for both the NZFMR fit and EDMR fit. The algorithm then fit these four parameters to the experimental NZFMR and EDMR data. Figure 5.5 and Figure 5.6 show the low-frequency NZFMR and EDMR spectra, respectively, and the simulated spectrum of least residual error. Table 5.5 contains the dipolar coupling constants, D , and the tunneling rates $r_{\alpha\beta}$ and $r_{\alpha e}$ that the fitting algorithm produced for the spectrum of best fit for the NZFMR response. Table 5.6 contains the dipolar coupling constants, D , and tunneling rates $r_{\alpha\beta}$ and $r_{\alpha e}$ that the fitting algorithm produced for the spectrum of best fit for the EDMR response. The resulting dipolar coupling constants from the independent fits were found to be consistent with one another: $D_{NZFMR} = D_{EDMR} \approx 0.280 \text{ mT}$. We also found dipolar constant resulting from the spectra of best fit to be in fair agreement with the measures of the dipolar coupling constant from the ratio of half-field response to the full-field response ($D \approx 0.4 \text{ mT}$) and the splitting about the full-field response ($D \approx 0.35 \text{ mT}$). Tunneling rates $r_{\alpha\beta}$ and $r_{\alpha e}$ from the independent fits were also found to be in agreement with one another.

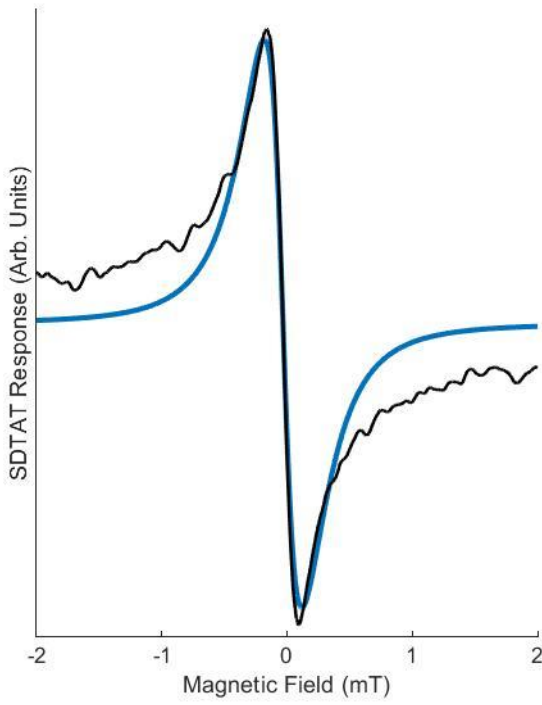


Figure 5.5 SDTAT response measured via NZFMR of unpassivated $^{28}\text{Si}/^{28}\text{SiO}_2$ MIS capacitor (black). Fitted spectrum of least residual error (blue).

Table 5.5 NZFMR Parameters of best fit for the unpassivated $^{28}\text{Si}/^{28}\text{SiO}_2$ MIS capacitor.

D (mT) [D (MHz)]	r_{ab} (GHz)	r_{ae} (GHz)
0.280 (7.86)	0.144	0.027

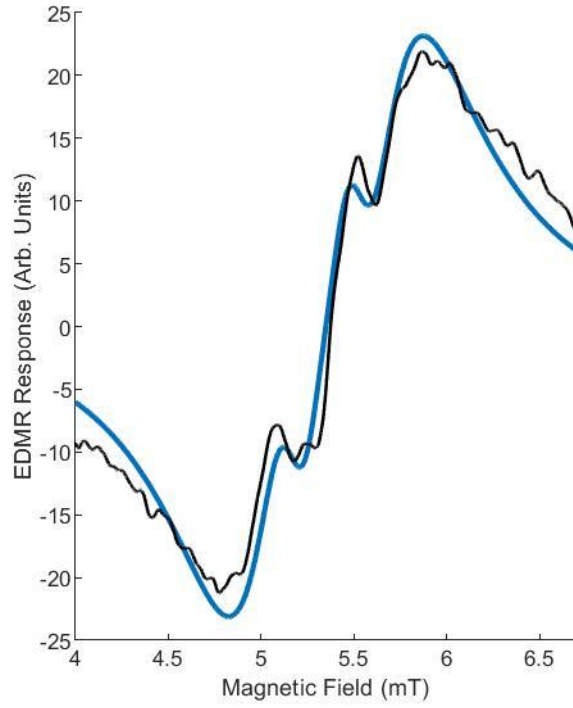


Figure 5.6 SDTAT response measured via low-frequency ($\nu = 151\text{MHz}$) EDMR of the unpassivated $^{28}\text{Si}/^{28}\text{SiO}_2$ MIS capacitor (lower, black). Fitted spectrum of least residual error (lower, blue). Half-field EDMR response measured of the unpassivated $^{28}\text{Si}/^{28}\text{SiO}_2$ MIS capacitor (upper, black). Calculated half-field response.

Table 5.6 EDMR Parameters of best fit for the unpassivated $^{28}\text{Si}/^{28}\text{SiO}_2$ MIS capacitor.

D (mT) [D (MHz)]	r_{ab} (GHz)	r_{ae} (GHz)
0.280 (7.86)	0.157	0.039

As mentioned previously, the time dependence of the pseudo-secular terms ($\mathbf{C}$ and $\mathbf{D}$) which give rise to the half-field response cannot be eliminated when transforming the static reference frame to the rotating reference. For same reason, we had to fit the NZFMR and EDMR responses separately since NZFMR terms rely on E and F which cannot be made time-independent by the RWA. In order to simulate the half response, the SLE can be treated as system of coupled differential equation and solved as a function of time where time is taken to $t = \infty$ such that $d\mathbf{\rho}/dt = \mathbf{0}$ for each point of the magnetic field sweep. In our model the density-matrix (an 8 by 8 matrix) represents 64 coupled differential equations. Solving as a function of time to $t \approx \infty$ becomes computationally demanding. Application of Floquet theory [80] may provide time-independent solutions to the SLE potentially allowing for methods to solve and simulate more complex resonance phenomena like half-field or multiple-photon transitions [81], [82], however, the application is not yet obvious or clear.

5.6 Conclusions

In this work we show that the NZFMR response contains some of the analytical power that has been demonstrated in EPR and EDMR measurements when determining the chemical and physical nature of point defects in semiconductor devices and insulator thin films. With relatively small computational effort, we fit experimental NZFMR responses utilizing the models based on the SLE outlined in Chapter 3. We performed the fit by utilizing a least squares fitting approach. We fit the from three distinct materials systems, all with different spin-dependent transport mechanism and different spin-spin interactions taking place. These spin-spin interaction included as isotropic electron-nuclear hyperfine coupling constants and magnetic dipolar coupling constants. Additionally, we extend the theory show the explicit connection of the defects observed between the NZFMR response and the EDMR which has

not yet been demonstrated. Our results are in good agreement with understanding of the defects present in the systems. Our work indicates that the NZFMR response and the low- frequency and field EDMR response and fitting of the those responses via the SLE is a simple yet powerful addition to the family of spin-based techniques used to explore the chemical and physical nature of point defects in semiconductor devices and insulator thin films.

Chapter 6 Conclusions & Future Work

6.1 Conclusions

The primary object of this research is to further develop and advance NZFMR as a spectroscopy technique that approaches the analytical power of EPR and EDMR at measuring and determining performance limiting atomic-scale defects in semiconductor devices, materials, and insulators. Although the primary object was not fully achieved and may not ever be fully achieved, much progress has been made. Chapter 3 provided a comprehensive overview of the recent developments in both the qualitative and quantitative models used to describe spin-dependent transport observed via both EDMR and NZFMR measurements. Chapter 3 also provided the theoretical connection between the defects observed in EDMR and NZFMR. Chapter 4 then demonstrates the experimental results of studies of isotopic dilution and forming gas anneals on Si/SiO₂ metal-insulator-semiconductor capacitors. These results showed the explicit, experimental connection between the defects observed in EDMR and NZFMR. Chapter 5 then presented a method by which to analyze and interpret NZFMR results and then extends that same method to analyze and further interpret EDMR results. Chapter 5 merged theory and experiment together demonstrating the explicit connection between the defects observed with the EDMR measurement, the NZFMR measurement, and both the qualitative and quantitative theory that describes their responses.

6.2 Future Work

Although these models constructed from the SLE are somewhat robust in their ability to both qualitatively and quantitatively calculate both NZFMR and EDMR spectra, further

experimentation and theoretical study is necessary to fully develop NZFMR as a spectroscopy tool.

6.2.1 Incorporation of Spin-Relaxation Mechanisms into the SLE

Experimentally, as the amplitude of $\mathbf{B}_1$ is increased, the EDMR response eventually saturates [83]. The point at which saturation takes effect is dependent upon the spin-lattice relaxation time, T_1 , and the spin-spin relaxation time, T_2 [2], [83]. In our quantitative model, the same saturation effects are not observed with increases in $\mathbf{B}_1$. For low levels of $\mathbf{B}_1$ in both our model and experiment, we find agreement between theoretical and experimental line shape. However, this agreement diverges with increases in $\mathbf{B}_1$. We speculate this inconsistency between experiment and the model to be a result the lack of incorporation of spin-relaxation mechanisms into the current model.

The incorporation of elements the Lindblad master equation to the models outlined in Chapter 3 may correct these inconsistencies. The Lindblad term in particular that may be necessary to be added to the models is defined as:

$$\mathbf{L} = \sum_{i=+, -, z} r_i \left(\mathbf{S}_i \rho \mathbf{S}_i - \frac{1}{2} \{ \mathbf{S}_i^\dagger \mathbf{S}_i, \rho \} \right). \quad (6.1)$$

Here, $\mathbf{L}$, represents the Lindblad term to the SLE, ρ , represents the density matrix, and $\mathbf{S}_i$, represents the respective spin operator ($\mathbf{S}_+$, $\mathbf{S}_-$, or $\mathbf{S}_z$) determined by index, i , under the sum. Here, r_i , represents a constant that incorporates the spin-relaxation times which are defined as:

$$r_+ = r_- = \frac{1}{\sqrt{2T_1}} \quad (6.2)$$

$$r_z = \sqrt{\frac{1}{2T_2} - \frac{1}{4T_1}} \quad (6.3)$$

Again, with the addition of the Lindblad term to the SLE, the time-dependence can no longer

be eliminated when solving for steady-state conditions of the spin ensemble. This requires the SLE to be treated as system of coupled differential equation and solved as a function of time where time is taken to $t = \infty$ such that $d\rho/dt = \mathbf{0}$ for each point of the magnetic field sweep. In our model the size of the density-matrix which represents a set of coupled differential equations scales by $2^{n_s} \times 2^{n_s}$ where, n_s , represents the number of spins included in the system. Solving as a function of time to $t \approx \infty$ becomes computationally demanding and can introduce some computational “noise” into the simulated spectrum. Therefore, new and better methods of solving coupled differential equations need to be researched.

6.2.2 Methods to Experimentally Determine the Kinetic Rate Factors.

In the NZFMR model, the kinetic rate factors associated with the respective introduction and removal of singlet and triplet pairs have significant influence over the line shape of the spectrum. However, there is no experimental method available yet with which to measure these experimental parameters. One promising technique, Rapid-Scan NZFMR, is a variation of the Rapid-Scan EPR technique which may be able to probe these rates [62]. However, further engineering of the physical hardware used in the detection scheme is required in addition to further experimentation.

6.2.3 Experimental NZFMR Studies of Systems of High-Spin Nuclei

Via use of the SLE, theory predicts resonance-like peaks should appear in the NZFMR spectrum with the incorporation of high-spin nuclei where $I > 1$. Figure 6.1 demonstrates an example of theoretical spectrum of a material with the incorporation of nuclei of $I = \frac{5}{2}$.

Preliminary SDTAT NZFMR measurements of 5nm Al₂O₃ MIS capacitors are

demonstrated in Figure 6.2. Aluminum nuclei are 100% abundant with a nuclear spin of $I = \frac{5}{2}$, however no resonance-like peaks are observed in Figure 6.1. However, the NZFMR spectrum presented is extremely broad and the side peaks are not observed. However, the spectrum may be characteristic of an amorphous or polycrystalline sample with a superposition of hyperfine side peaks which produce a broad signal.

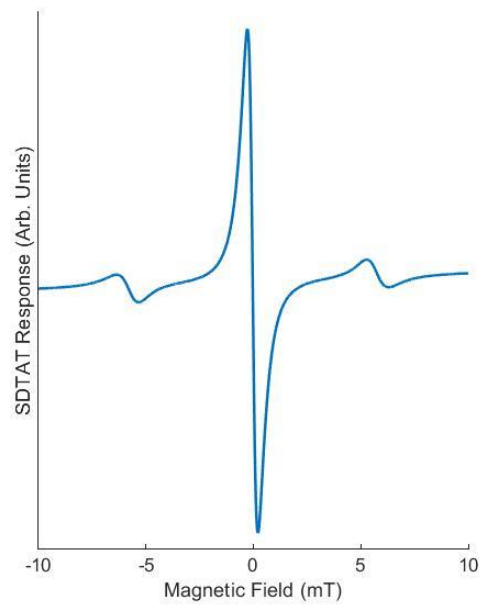


Figure 6.1 NZFMR simulation of with an isotropic hyperfine interaction with a single high-spin nucleus.

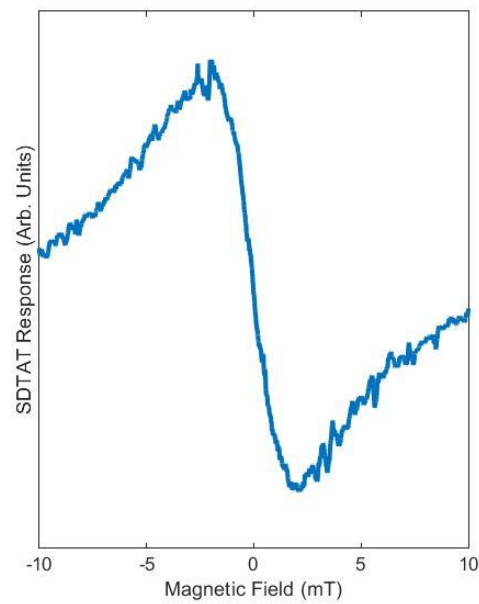


Figure 6.2 SDTAT as measured through NZFMR measurement of an Al_2O_3 MIS capacitor.

Figure 6.3 shows the SDTAT response as measured through X-Band ($\nu \approx 9.25$ GHz) EDMR of the Al_2O_3 MIS capacitors where the B_0 field lies perpendicular and parallel to the (100) interface. The Al_2O_3 MIS capacitors exhibited zero-crossing $g_{\perp(100)}$ and $g_{\parallel(100)}$ of 2.0066 ± 0.0003 and 2.0056 ± 0.0003 , respectively, and linewidths of both the perpendicular and parallel orientations were found to be 1.0 ± 0.05 mT. The consistent difference of the zero-crossing g-factor with respect to the B_0 field orientation in addition to the fact that g-factors of both the P_{b0} and P_{b1} defects are within the linewidth of our measurement lead us to believe we are again observing interface states, P_{b0} and P_{b1} centers, at the $\text{Al}_2\text{O}_3/\text{Si}$ interface. These results are extremely similar to what was observed in the (100) Si/SiO_2 MIS capacitors outlined in 4.3. Similar results have also been found using EPR which align with our observation [84], [85]. The tunneling process observed in these leakage currents must again be primarily rate limited by the step across the interface and not limited by trap-assisted-tunneling through oxide defects. Further experimentation with and theoretical study of high-spin nuclei are needed in order to understand the preliminary results presented.

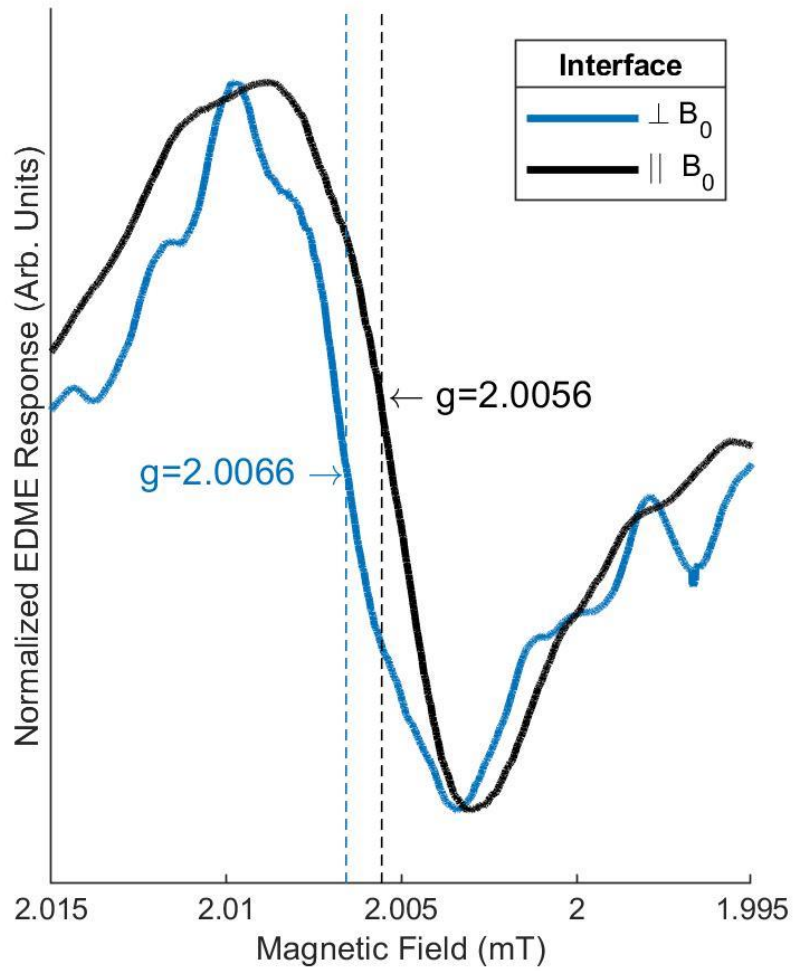


Figure 6.3 SDTAT response as measured through X-Band ($\nu \approx 9.25$ GHz) EDMR of an Al_2O_3 MIS capacitors where the B_0 field lies perpendicular (blue) and parallel (black) to the (100) Si interface.

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Appendix A: MATLAB Source Code: Near-Zero Field Magnetoresistance Spectral Fitting of γ -Irradiated Si/SiO₂ Metal-Oxide-Semiconductor Field-Effect Transistor

```
clc,clear

load('SiO2.mat')
Field= Field./10;
Filtered= Filtered;
clearvars -except Field Filtered

Hyperfine1= 1.56;
Hyperfine2= 0.215;
ks1= 0.25;
kd1= 0.036;
Gen1= .2;
Gen2= 4;

Parameters=
FitNZFMR(Field,Filtered,Hyperfine1,Hyperfine2,ks1,kd1,Gen1,Gen
2);

function [Parameters] =
FitNZFMR(Field,I,Hyperfine1,Hyperfine2,ks1,kd1,Gen1,Gen2)
%FitNZFMR Given input spectral data, finds the spectral
parameters that best match
%the spectra.
% Initial Guess Array
x0(1)= Hyperfine1;
x0(2)= Hyperfine2;
x0(3)= ks1;
x0(4)= kd1;
```

```

x0(5)= Gen1;
x0(6)= Gen2;
% Constraints
min= 0;
max= 1e16;
lb= [1.1,.05,min,min,min,min];
ub= [2.0,1.1,max,max,max,max];
% Arbitrary Weighting Function
w=
abs(smoothdata((gradient(I)./gradient(Field)),'sgolay',300))*1
00;
% Defining Misc. Global Variable
global Field I History
History= [];
% Least Squares Function
fun=@(x0) LeastSquares(x0,Field,I,w);
% Options for fmincon
Opts= optimset('Display','iter','TolFun',1e-7,'TolX',1e-
7,'MaxIter',5000,...

'MaxFunEvals',10000,'PlotFcns',@PlotMe,'UseParallel',true,'Tol
Con',1e-7,...
    'OutputFcn',@HistorySaver);%,'Hessian','lbfgs');
% Minimization Function
Parameters= fmincon(fun,x0,[],[],[],[],lb,ub,[],Opts);
% Save Output
save('Parameters.mat','Parameters');
save('History.mat','History');
% Clean Global workspace
clear global Field I History
end

function [stop] = PlotMe(x0,optimValues,state)
%PlotMe is called on by fminsearch within FitNZFMR.m. This
allows for an
%live plot of fminsearch's output parameters that up the

```

```

inputed spectra data.
% [stop] = PlotMe(x0,optimValues,state)
stop = false;
global Field I
ISpectraProducer=
SpectraCombiner(Field,x0(1),x0(2),x0(3),x0(4),x0(5),x0(6));
plot(Field,I,Field,ISpectraProducer,'linewidth',2)
grid on
xlabel('Field, B_z (mT)')
ylabel('NZFMR Response, \Delta I (pA)')
legend('Collected Data','Simulated
Data','location','northeast')
set(gca,'FontSize',18)
end

function [stop] = HistorySaver(x,optimvalues,state)
%HistorySaver is called on by fminsearch to save the history
of each
%iteration of the solver.
% [stop] = HistorySaver(x,optimvalues,state)
global History
stop = false;
if isequal(state,'iter')
    History = [History;x];
end
end

function [error] = LeastSquares(x0,Field,I,w)
%LeastSquares Is the type of minimization routine called on by
fminsearch
%to fit a spectra to the input spectra data.
% [error] = LeastSqaures(x0,Field,I)
error= sum(w.*(I -
SpectraCombiner(Field,x0(1),x0(2),x0(3),x0(4),x0(5),x0(6))).^2
);
end

```

```

function [I] =
SpectraCombiner(Field,Hyperfine1,Hyperfine2,ks1,kd1,Gen1,Gen2)
%SpectraCombiner Sums two individual spectra.
% [I] =
SpectraCombiner(Field,HyperfineValue1,HyperfineValue2,ks1,kd1,
ks2,kd2,Gen1,Gen2)
I1= SpectraProducer(Field,1/2,2.0028,Hyperfine1,ks1,kd1,Gen1);
I2= SpectraProducer(Field,1/2,2.0028,Hyperfine2,ks1,kd1,Gen2);
I= I1+I2;
end

```

```

function [I] =
SpectraProducer(Field,NuclearSpin,g1,HyperfineValue1,ks1,kd1,Gen1)
%PassSpectraFit is passed into fminsearch to extract the
%spectroscopic components of the NZFMR Spectra.
% Conditions for fminsearch
HyperfineValue1= abs(HyperfineValue1);
ks1= abs(ks1);
kd1= abs(kd1);
Gen1= abs(Gen1);
% Spin System
[sx,sy,sz]= SpinSystem([1/2,1/2,NuclearSpin]);
% Spin Operators
S_tot= sz{1}+sz{2};
% Electron-Electron Spin Product Projection Operators
S1S2= sx{1}*sx{2} + sy{1}*sy{2} + sz{1}*sz{2};
% Electron-Nuclear Spin Product Projection Operators
IS1= sx{3}*sx{1} + sy{3}*sy{1} + sz{3}*sz{1};
% Singlet & Triplet Projection Operators
Ident= eye(size(S1S2));
Ps= (1/4)*(Ident-4*S1S2);
Pt= (1/4)*(3*Ident+4*S1S2);
% Constants
uB= 5.788e-8; %eV/mT

```

```

h_bar= 6.582E-7; %eV*ns
% Hamiltonians + Solve
Hhf1= g1*uB*HyperfineValue1*IS1;
I1= zeros(length(Field),1);
G1= (1/8)*Gen1*Ident;
parfor k= 1:length(Field)
    B0= Field(k)+0.15995;
    H01= g1*uB*B0*(S_tot);
    H_NZFMR1= H01 + Hhf1;
    A= (-((1i)/h_bar)*H_NZFMR1-(ks1/2)*Ps-(kd1/2)*Pt);
    B= (((1i)/h_bar)*H_NZFMR1-(ks1/2)*Ps-(kd1/2)*Pt);
    P= lyap(A,B,G1);
    I1(k,1)= (real(trace(Ps*P)));
end
I1= I1-I1(1);
I= gradient(I1)./gradient(Field);
end

function [sx,sy,sz] = spinSystem(spins)
%SpinSystem creates cell arrays of sx, sy, and sz in the
appropriate spin
%bases given a vector of nuclear spins.
% [sx,sy,sz] = spinSystem(spins)
iterations= length(spins);

spinMat= [];
for l= 1:iterations
    spinMat= vertcat(spinMat,spins);
end

spinCellx{iterations,iterations}= [];
spinCelly{iterations,iterations}= [];
spinCellz{iterations,iterations}= [];

for m= 1:iterations
    for n= 1:iterations

```

```

        if n==m
            spinCellx{m,n}= Pauli(spinMat(m,n),1);
            spinCelly{m,n}= Pauli(spinMat(m,n),2);
            spinCellz{m,n}= Pauli(spinMat(m,n),3);
        else
            spinCellx{m,n}= eye(size(Pauli(spinMat(m,n),1)));
            spinCelly{m,n}= eye(size(Pauli(spinMat(m,n),2)));
            spinCellz{m,n}= eye(size(Pauli(spinMat(m,n),3)));
        end
    end
end

sx{iterations,1}= [];
sy{iterations,1}= [];
sz{iterations,1}= [];

for j= 1:iterations
    x= spinCellx{j,1};
    y= spinCelly{j,1};
    z= spinCellz{j,1};
    for k= 2:iterations
        x= kron(x,spinCellx{j,k});
        y= kron(y,spinCelly{j,k});
        z= kron(z,spinCellz{j,k});
    end
    sx{j,1}= x;
    sy{j,1}= y;
    sz{j,1}= z;
end
end

function [Out] = Pauli(Spin,xyz)
%Pauli2 creates a Pauli spin matrix (vector) given a given
Spin (S=1/2,S=1,S=3/2,...) and
%the vector direction (x,y, or z).
%    [Out] = Pauli2(Spin,xyz)

```

```

iterations= 2*Spin+1;
m= linspace(-Spin,Spin,iterations);
Out= zeros(iterations,iterations);
switch xyz
    case 1
        for j= 1:iterations
            for k= 1:iterations
                Out(j,k)= (DiracDelta(j,k+1) +
DiracDelta(j+1,k))*0.5*sqrt((Spin*(Spin+1)-m(j)*m(k)));
            end
        end
    case 2
        for j= 1:iterations
            for k= 1:iterations
                Out(j,k)= (DiracDelta(j,k+1) -
DiracDelta(j+1,k))*0.5i*sqrt((Spin*(Spin+1)-m(j)*m(k)));
            end
        end
    case 3
        for j= 1:iterations
            for k= 1:iterations
                Out(j,k)= -DiracDelta(j,k)*m(j);
            end
        end
end
end

function [Out] = DiracDelta(m,n)
%DiracDelta is a numeric equivalent of the MATLAB built in:
kroneckerDelta.
% [Out] = DiracDelta(m,n)
if m==n
    Out= 1;
else
    Out= 0;
end

```

end
end

Appendix B: MATLAB Source Code: Near-Zero Field Magnetoresistance

Spectral Fitting of a-Si:H Metal-Insulator-Semiconductor Capacitors

```
clc,clear
load('datadata1536.mat')
Field= datadata1536(:,1)./10;
I= -datadata1536(:,2);
clearvars -except Field I

%Defect 1
Hyperfine1= .2;
ks1= .2;
kd1= .017;
Gen1= 1e2;
%Defect 2
Hyperfine2= 1.4;
ks2= .2;
kd2= .019;
Gen2= 2.5e1;

Parameters=
FitNZFMR(Field,I,Hyperfine1,Hyperfine2,ks1,kd1,Gen1,Gen2)

function [Parameters] =
FitNZFMR(Field,I,Hyperfine1,Hyperfine2,ks1,kd1,Gen1,Gen2)
%FitNZFMR Given input spectral data, finds the spectral
parameters that best match
%the spectra.

x0(1)= Hyperfine1;
x0(2)= Hyperfine2;
```

```

x0(3)= ks1;
x0(4)= kd1;
x0(5)= Gen1;
x0(6)= Gen2;

min= 0;
max= 1e16;

lb= [.05,1.4,min,min,min,min];
ub= [2.0,1.6,max,max,max,max];

w= abs(smoothdata((diff(I)./diff(Field)),'sgolay',200));

global Field I History
History= [];
fun=@(x0) LeastSquares(x0,Field,I,w);
Opts= optimset('Display','iter','TolFun',1e-9,'TolX',1e-
9,'MaxIter',5000,...

'MaxFunEvals',10000,'PlotFcns',@PlotMe,'UseParallel',true,'Tol
Con',1e-9,...
    'OutputFcn',@HistorySaver);%,'Hessian','lbfgs');
Parameters= fmincon(fun,x0,[],[],[],[],lb,ub,[],Opts);
save('Parameters2.mat','Parameters');
save('History2.mat','History');
clear global Field I History
end

function [stop] = PlotMe(x0,optimValues,state)
%PlotMe is called on by fminsearch within FitNZFMR.m. This
allows for an
%live plot of fminsearch's output parameters that up the
inputted spectra data.
% [stop] = PlotMe(x0,optimValues,state)
stop = false;
global Field I

```

```

ISpectraProducer=
SpectraCombiner(Field,x0(1),x0(2),x0(3),x0(4),x0(5),x0(6));
plot(Field,I,Field(1:end-1),ISpectraProducer,'linewidth',2)
grid on
xlabel('Field, B_z (mT)')
ylabel('NZFMR Response, \Delta I (pA)')
legend('Collected Data','Simulated
Data','location','northeast')
set(gca,'FontSize',18)
end

function [stop] = HistorySaver(x,optimvalues,state)
%HistorySaver is called on by fminsearch to save the history
of each
%iteration of the solver.
% [stop] = HistorySaver(x,optimvalues,state)
global History
stop = false;
if isequal(state,'iter')
    History = [History;x];
end
end

function [error] = LeastSquares(x0,Field,I,w)
%LeastSquares Is the type of minimization routine called on by
fminsearch
%to fit a spectra to the input spectra data.
% [error] = LeastSqaures(x0,Field,I)
I= I(1:end-1);
error= sum(w.*(I -
SpectraCombiner(Field,x0(1),x0(2),x0(3),x0(4),x0(5),x0(6))) .^2
);
end

function [I] =
SpectraCombiner(Field,Hyperfine1,Hyperfine2,ks1,kd1,Gen1,Gen2)

```

```

I1= SpectraProducer(Field,[],2.0028,Hyperfine1,ks1,kd1,Gen1);
I2=
SpectraProducer2(Field,[],2.0028,Hyperfine1,Hyperfine2,ks1,kd1
,Gen2);
I= I1+I2;
end

function [I] =
SpectraProducer(Field,NuclearSpin,g1,HyperfineValue1,ks1,kd1,G
en1)
% Conditions for fminseach
HyperfineValue1= abs(HyperfineValue1);
ks1= abs(ks1);
kd1= abs(kd1);
Gen1= abs(Gen1);
% Spin System
[sx,sy,sz]= SpinSystem([1/2,1/2,1/2,1/2]);
% Spin Operators
S_tot= sz{1}+sz{2};
S1S2= sx{1}*sx{2} + sy{1}*sy{2} + sz{1}*sz{2};
Ident= eye(size(S1S2));

IS1= sx{3}*sx{1} + sy{3}*sy{1} + sz{3}*sz{1};
IS2= sx{4}*sx{2} + sy{4}*sy{2} + sz{4}*sz{2};

Ps= (1/4)*( Ident-4*S1S2);
Pt= (1/4)*(3*Ident+4*S1S2);
% Constants
uB= 5.788e-8; %eV/mT
h_bar= 6.582E-7; %eV*ns
% Hamiltonians + Solve
Hhf1= g1*uB*HyperfineValue1*(IS1 + IS2);
I1= zeros(length(Field),1);
G1= 0.25*Gen1*Ident;
parfor k= 1:length(Field)
    B0= Field(k)-0.037851;

```

```

H01= g1*uB*B0*(S_tot);
H_NZFMR1= H01 + Hhf1;
A= (-((1i)/h_bar)*H_NZFMR1-(ks1/2)*Ps-(kd1/2)*Pt);
B= (((1i)/h_bar)*H_NZFMR1-(ks1/2)*Ps-(kd1/2)*Pt);
P1= lyap(A,B,G1);
I1(k,1)= 1-real(trace(P1));
end
I1= I1-I1(1);
I= diff(I1)./diff(Field);
end

function [I] =
SpectraProducer2(Field,NuclearSpin,g1,HyperfineValue1,Hyperfin
eValue2,ks1,kd1,Gen1)
% Conditions for fminseach
HyperfineValue1= abs(HyperfineValue1);
HyperfineValue2= abs(HyperfineValue2);
ks1= abs(ks1);
kd1= abs(kd1);
Gen1= abs(Gen1);
% Spin System
[sx,sy,sz]= SpinSystem([1/2,1/2,1/2,1/2]);
% Spin Operators
S_tot= sz{1}+sz{2};
S1S2= sx{1}*sx{2} + sy{1}*sy{2} + sz{1}*sz{2};
Ident= eye(size(S1S2));

IS1= sx{3}*sx{1} + sy{3}*sy{1} + sz{3}*sz{1};
IS2= sx{4}*sx{2} + sy{4}*sy{2} + sz{4}*sz{2};

Ps= (1/4)*( Ident-4*S1S2);
Pt= (1/4)*(3*Ident+4*S1S2);
% Constants
uB= 5.788e-8; %eV/mT
h_bar= 6.582E-7; %eV*ns
% Hamiltonians + Solve

```

```

Hhf1= g1*uB*(HyperfineValue1*(IS1) + HyperfineValue2*(IS2));
I1= zeros(length(Field),1);
G1= 0.25*Gen1*Ident;
parfor k= 1:length(Field)
    B0= Field(k)-0.037851;
    H01= g1*uB*B0*(S_tot);
    H_NZFMR1= H01 + Hhf1;
    A= (-((1i)/h_bar)*H_NZFMR1-(ks1/2)*Ps-(kd1/2)*Pt);
    B= (((1i)/h_bar)*H_NZFMR1-(ks1/2)*Ps-(kd1/2)*Pt);
    P1= lyap(A,B,G1);
    I1(k,1)= (1-real(trace(P1)));
end
I1= I1-I1(1);
I= diff(I1)./diff(Field);
end

function [sx,sy,sz] = spinSystem(spins)
%SpinSystem creates cell arrays of sx, sy, and sz in the
appropriate spin
%bases given a vector of nuclear spins.
% [sx,sy,sz] = spinSystem(spins)
iterations= length(spins);

spinMat= [];
for l= 1:iterations
    spinMat= vertcat(spinMat,spins);
end

spinCellx{iterations,iterations}= [];
spinCelly{iterations,iterations}= [];
spinCellz{iterations,iterations}= [];

for m= 1:iterations
    for n= 1:iterations
        if n==m
            spinCellx{m,n}= Pauli(spinMat(m,n),1);

```

```

        spinCelly{m,n}= Pauli(spinMat(m,n),2);
        spinCellz{m,n}= Pauli(spinMat(m,n),3);
    else
        spinCellx{m,n}= eye(size(Pauli(spinMat(m,n),1)));
        spinCelly{m,n}= eye(size(Pauli(spinMat(m,n),2)));
        spinCellz{m,n}= eye(size(Pauli(spinMat(m,n),3)));
    end
end
end

sx{iterations,1}= [];
sy{iterations,1}= [];
sz{iterations,1}= [];

for j= 1:iterations
    x= spinCellx{j,1};
    y= spinCelly{j,1};
    z= spinCellz{j,1};
    for k= 2:iterations
        x= kron(x,spinCellx{j,k});
        y= kron(y,spinCelly{j,k});
        z= kron(z,spinCellz{j,k});
    end
    sx{j,1}= x;
    sy{j,1}= y;
    sz{j,1}= z;
end
end

function [Out] = Pauli(Spin,xyz)
%Pauli2 creates a Pauli spin matrix (vector) given a given
Spin (S=1/2,S=1,S=3/2,...) and
%the vector direction (x,y, or z).
% [Out] = Pauli2(Spin,xyz)
iterations= 2*Spin+1;
m= linspace(-Spin,Spin,iterations);

```

```

Out= zeros(iterations,iterations);
switch xyz
    case 1
        for j= 1:iterations
            for k= 1:iterations
                Out(j,k)= (DiracDelta(j,k+1) +
DiracDelta(j+1,k))*0.5*sqrt((Spin*(Spin+1)-m(j)*m(k)));
            end
        end
    case 2
        for j= 1:iterations
            for k= 1:iterations
                Out(j,k)= (DiracDelta(j,k+1) -
DiracDelta(j+1,k))*0.5i*sqrt((Spin*(Spin+1)-m(j)*m(k)));
            end
        end
    case 3
        for j= 1:iterations
            for k= 1:iterations
                Out(j,k)= -DiracDelta(j,k)*m(j);
            end
        end
end
end

function [Out] = DiracDelta(m,n)
%DiracDelta is a numeric equivalent of the MATLAB built in:
kroneckerDelta.
% [Out] = DiracDelta(m,n)
if m==n
    Out= 1;
else
    Out= 0;
end
end

```


**Appendix C: MATLAB Source Code: Near-Zero Field Magnetoresistance
Spectral & Electrically Detected Magnetic Resonance Spectral Fitting of
 $^{28}\text{Si}/^{28}\text{SiO}_2$ Metal-Insulator-Semiconductor Capacitors**

```
clear global
clc,clear
load('w081_1_s1_2v.mat')
I_Data= -w081_1_s1_2v(:,2);
Field= w081_1_s1_2v(:,1)/10;
[sx,sy,sz,sp,sm]= SpinSystem([.5 .5 .5]);

% Initial Guess
v= .151;
B1= .02;
D= .3;
theta= 89*pi/180;
phi= 1*pi/180;
r_ab= .15;
r_ae= .04;
r_ea1= .5e2;
r_ea2= 4e4;

% EDMR Fitting (Ignore Parameter 5)
Field_EDMR= Field;
Field_EDMR(1230:2867)= [];
I_Data_EDMR= I_Data;
I_Data_EDMR(1230:2867)= [];
[EDMR_Parameters] =
Fit(Field_EDMR,I_Data_EDMR,D,r_ab,r_ae,r_ea1,r_ea2,sx,sy,sz,sp
,sm);
```

```

% NZFMR Fitting (Ignore Parameter 4)
Field_NZFMR= Field(1946:2151);
I_Data_NZFMR= I_Data(1946:2151);
[NZFMR_Parameters] =
Fit(Field_NZFMR,I_Data_NZFMR,D,r_ab,r_ae,r_ea1,r_ea2,sx,sy,sz,
sp,sm);

function [Parameters] =
Fit(Field,I,D,r_ab,r_ae,r_ea1,r_ea2,sx,sy,sz,sp,sm);
x0(1)= D;
x0(2)= r_ab;
x0(3)= r_ae;
x0(4)= r_ea1;
x0(5)= r_ea2;

min= 0;
max= 1e16;

lb= [.08,min,min,min,min];
ub= [.5,max,max,max,max];

w= ones(size(Field));

global Field I History sx sy sz sp sm
History= [];
fun=@(x0) LeastSquares(x0,Field,I,w,sx,sy,sz,sp,sm);
Opts= optimset('Display','iter','TolFun',1e-9,'TolX',1e-
9,'MaxIter',5000,...
    'MaxFunEvals',10000,'PlotFcns',@PlotMe,...
    'UseParallel',true,'TolCon',1e-9,...
    'OutputFcn',@HistorySaver);%,'Hessian','lbfgs');
Parameters= fmincon(fun,x0,[],[],[],[],lb,ub,[],Opts);
save('Parameters.mat','Parameters');
save('History.mat','History');
clear global Field I History
end

```

```

function [stop] = PlotMe(x0,optimValues,state)
%PlotMe is called on by fminsearch within FitNZFMR.m. This
allows for an
%live plot of fminsearch's output parameters that up the
inputted spectra data.
% [stop] = PlotMe(x0,optimValues,state)
stop = false;
global Field I sx sy sz sp sm
ISpectraProducer=
SpectraCombiner(Field,x0(1),x0(2),x0(3),x0(4),x0(5),sx,sy,sz,s
p,sm);
plot(Field,I,Field,ISpectraProducer,'linewidth',2)
grid on
xlabel('Field, (mT)')
ylabel('Response (Arb. Units)')
legend('Collected Data','Simulated
Data','location','northeast')
set(gca,'FontSize',18)
end

```

```

function [stop] = HistorySaver(x,optimvalues,state)
%HistorySaver is called on by fminsearch to save the history
of each
%iteration of the solver.
% [stop] = HistorySaver(x,optimvalues,state)
global History
stop = false;
if isequal(state,'iter')
    History = [History;x];
end
end

```

```

function [error] = LeastSquares(x0,Field,I,w,sx,sy,sz,sp,sm)
%LeastSqaures Is the type of minimization routine called on by
fminsearch

```

```

%to fit a spectra to the input spectra data.
% [error] = LeastSqaures(x0,Field,I)
error= sum(w.*(I -
SpectraCombiner(Field,x0(1),x0(2),x0(3),x0(4),x0(5),sx,sy,sz,s
p,sm)).^2);
end

function [I] =
SpectraCombiner(Field,D,r_ab,r_ae,r_ea1,r_ea2,sx,sy,sz,sp,sm)
I1=
NZFMR_Dip(Field,D,89*pi/180,1*pi/180,r_ab,r_ae,r_ea1,sx,sy,sz,
sp,sm);
I2=
EDMR_Dip(Field,.151,.02,D,89*pi/180,1*pi/180,r_ab,r_ae,r_ea2,s
x,sy,sz,sp,sm);
I= I1+I2;
end

function [I] =
NZFMR_Dip(Field,D,theta,phi,r_ab,r_ae,Gen1,sx,sy,sz,sp,sm)
D= abs(D); %nm
r_ab= abs(r_ab);
r_ae= abs(r_ae);
Gen1= abs(Gen1);
g1= 2.005;
% Spin Oppoperators
S_tot= sz{1}+sz{2}+sz{3};
S1S2= sx{1}*sx{2} + sy{1}*sy{2} + sz{1}*sz{2};
Ident= eye(size(S1S2));
Ps= (1/4)*( Ident-4*S1S2);
Pt= (1/4)*(3*Ident+4*S1S2);
% Constants
uB= 5.788e-8; %eV/mT
h_bar= 6.582E-7; %eV*ns
[A13,B13,C13,D13,E13,F13] =
H_dip_NZFMR(1,3,pi/2,0,sx,sy,sz,sp,sm);

```

```

H_dipolar= D*g1*uB*(E13+F13);

% Hamiltonians + Solve
I1= zeros(length(Field),1);
G= 0.125*Gen1*Ident;
for k= 1:length(Field)
    B0= Field(k)+0.03175;
    H01= g1*uB*B0*(S_tot);
    H_NZFMR1= H01 + H_dipolar;
    A= (-((1i)/h_bar)*H_NZFMR1-((r_ab)/2)*Ps-(r_ae/2)*Pt);
    B= (((1i)/h_bar)*H_NZFMR1-((r_ab)/2)*Ps-(r_ae/2)*Pt);
    P= lyap(A,B,G);
    I1(k,1)= (1-real(trace(P)));
end
I1= I1-I1(1);
I= gradient(I1)./gradient(Field);
end

function [A,B,C,D,E,F] =
H_dip_NZFMR(SpinNO1,SpinNO2,theta,phi,sx,sy,sz,sp,sm)
A= sz{SpinNO1}*sz{SpinNO2}*(1-3*(cos(theta))^2);
B= (-
1/4)*(sp{SpinNO1}*sm{SpinNO2}+sm{SpinNO1}*sp{SpinNO2})*(1-
3*(cos(theta))^2);
C= (-
3/2)*(sp{SpinNO1}*sz{SpinNO2}+sz{SpinNO1}*sp{SpinNO2})*sin(the
ta)*cos(theta)*exp(-1i*phi);
D= (-
3/2)*(sm{SpinNO1}*sz{SpinNO2}+sz{SpinNO1}*sm{SpinNO2})*sin(the
ta)*cos(theta)*exp( 1i*phi);
E= (-3/4)*(sp{SpinNO1}*sp{SpinNO2})*((sin(theta))^2)*exp(-
2i*phi);
F= (-3/4)*(sm{SpinNO1}*sm{SpinNO2})*((sin(theta))^2)*exp(
2i*phi);
end

```

```

function [I] =
EDMR_Dip(Field,v,B1,D,theta,phi,r_ab,r_ae,Gen1,sx,sy,sz,sp,sm)
D= abs(D); %mT
r_ab= abs(r_ab);
r_ae= abs(r_ae);
Gen1= abs(Gen1);
g1= 2.005;
% Spin Operators
S_tot= sz{1}+sz{2}+sz{3};
S1S2= sx{1}*sx{2} + sy{1}*sy{2} + sz{1}*sz{2};
Ident= eye(size(S1S2));
Ps= (1/4)*( Ident-4*S1S2);
Pt= (1/4)*(3*Ident+4*S1S2);
% Constants
uB= 5.788e-8; %eV/mT
h_bar= 6.582E-7; %eV*ns
[A13,B13]= H_dip_EDMR_RWA(1,3,theta,phi,sx,sy,sz,sp,sm);
H_dipolar= D*g1*uB*(A13+B13);
H1= g1*uB*B1*(sx{1} + sx{2} + sx{3});

% Hamiltonians + Solve
I1= zeros(length(Field),1);
G= 0.125*Gen1*Ident;
for k= 1:length(Field)
    B0= Field(k)+0.03175;
    H01= g1*uB*(B0+((h_bar*v*2*pi)/(g1*uB)))*(S_tot);
    H02= g1*uB*(B0-((h_bar*v*2*pi)/(g1*uB)))*(S_tot);
    H_EDMR1= H01 + H1 + H_dipolar;
    H_EDMR2= H02 + H1 + H_dipolar;
    A= (-((1i)/h_bar)*H_EDMR1-((r_ab)/2)*Ps-(r_ae/2)*Pt);
    B= (((1i)/h_bar)*H_EDMR1-((r_ab)/2)*Ps-(r_ae/2)*Pt);
    C= (-((1i)/h_bar)*H_EDMR2-((r_ab)/2)*Ps-(r_ae/2)*Pt);
    D= (((1i)/h_bar)*H_EDMR2-((r_ab)/2)*Ps-(r_ae/2)*Pt);
    P1= lyap(A,B,G);
    P2= lyap(C,D,G);
    I1(k,1)= (1-real(trace(P1)));
end

```

```

        I2(k,1)= (1-real(trace(P2)));
    end
    I= I1+I2;
    I= I-I(1);
    I= gradient(I)./gradient(Field);
end

function [A,B] =
H_dip_EDMR_RWA(SpinNO1,SpinNO2,theta,phi,sx,sy,sz,sp,sm)
A= sz{SpinNO1}*sz{SpinNO2}*(1-3*(cos(theta))^2);
B= (-
1/2)*(sx{SpinNO1}*sx{SpinNO2}+sy{SpinNO1}*sy{SpinNO2})*(1-
3*(cos(theta))^2);
end

function [sx,sy,sz,sp,sm] = SpinSystem(spins)
%SpinSystem creates cell arrays of sx, sy, and sz in the
appropriate spin
%bases given a vector of nuclear spins.
%    [sx,sy,sz] = SpinSystem(spins)
iterations= length(spins);

spinMat= [];
for l= 1:iterations
    spinMat= vertcat(spinMat,spins);
end

spinCellx{iterations,iterations}= [];
spinCelly{iterations,iterations}= [];
spinCellz{iterations,iterations}= [];

for m= 1:iterations
    for n= 1:iterations
        if n==m
            spinCellx{m,n}= Pauli(spinMat(m,n),1);
            spinCelly{m,n}= Pauli(spinMat(m,n),2);

```

```

        spinCellz{m,n}= Pauli(spinMat(m,n),3);
    else
        spinCellx{m,n}= eye(size(Pauli(spinMat(m,n),1)));
        spinCelly{m,n}= eye(size(Pauli(spinMat(m,n),2)));
        spinCellz{m,n}= eye(size(Pauli(spinMat(m,n),3)));
    end
end
end

sx{iterations,1}= [];
sy{iterations,1}= [];
sz{iterations,1}= [];
sp{iterations,1}= [];
sm{iterations,1}= [];

for j= 1:iterations
    x= spinCellx{j,1};
    y= spinCelly{j,1};
    z= spinCellz{j,1};
    for k= 2:iterations
        x= kron(x,spinCellx{j,k});
        y= kron(y,spinCelly{j,k});
        z= kron(z,spinCellz{j,k});
    end
    sx{j,1}= x;
    sy{j,1}= y;
    sz{j,1}= z;
    sp{j,1}= x+1i*y;
    sm{j,1}= x-1i*y;
end
end

function [Out] = Pauli(Spin,xyz)
%Pauli2 creates a Pauli spin matrix (vector) given a given
Spin (S=1/2,S=1,S=3/2,...) and
%the vector direction (x,y, or z).

```



```

% [Out] = Pauli2(Spin,xyz)
iterations= 2*Spin+1;
m= linspace(-Spin,Spin,iterations);
Out= zeros(iterations,iterations);
switch xyz
    case 1
        for j= 1:iterations
            for k= 1:iterations
                Out(j,k)= (DiracDelta(j,k+1) +
DiracDelta(j+1,k))*0.5*sqrt((Spin*(Spin+1)-m(j)*m(k)));
            end
        end
    case 2
        for j= 1:iterations
            for k= 1:iterations
                Out(j,k)= (DiracDelta(j,k+1) -
DiracDelta(j+1,k))*0.5i*sqrt((Spin*(Spin+1)-m(j)*m(k)));
            end
        end
    case 3
        for j= 1:iterations
            for k= 1:iterations
                Out(j,k)= -DiracDelta(j,k)*m(j);
            end
        end
end
end

function [Out] = DiracDelta(m,n)
%DiracDelta is a numeric equivalent of the MATLAB built in:
kroneckerDelta.
% [Out] = DiracDelta(m,n)
if m==n
    Out= 1;
else
    Out= 0;
end

```

end
end

Appendix D: Non-Technical Abstract

As the semiconductor industry races to keep pace with Moore's law of doubling the number of transistors that can be placed on a chip, the requirements for advances in the downscaling of transistor technology becomes ever more challenging. Today, devices are fabricated on extremely small scales with new and more complex materials systems. At such small scales, individual atomic scale defects are responsible for the improper function of these devices. Still to this day, devices are characterized by basic electrical measurements and then optimized via fabrication trial and error. Despite being a relatively successful approach to design a working device, the approach can take several years to optimize the devices. To expedite this design process, it is best to understand the physical mechanisms and root-causes which lead to the formation of these atomic scale defects in the first place. With this physical understanding, process optimization could be accelerated while development costs of new technology could be reduced. This requires, new, innovative, and fast root-cause identification methods and tools which are easily can be easily integrated into current testing platforms. Several premier techniques based on the principles of magnetic resonance already exist which allow scientists and engineers to study and understand performance limiting defects in fully processed devices. However, a newer and simpler tool based on the principles of magnetoresistance has not yet been completely proven theoretically or experimentally. In this work, the magnetoresistance tool is used in conjunction with the premier magnetic resonance tools to study defects in a known material to prove its analytical capabilities. With small variations to the material system studied alongside the development of a theoretical understanding of the physical principles behind the operation of the tool, the magnetoresistance tool proves to be nearly as analytically powerful as the premier magnetic resonance while being

simpler to implement and use.

Elias B. Frantz

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EXPERIENCE

Graduate Technical Intern

Jan. 2021 – Sep. 2021

Intel Corporation

Hillsboro, OR

- Performed Electrically Detected Magnetic Resonance (EDMR) measurements on semiconductor quantum computing devices, BEOL dielectrics, & transistors.
- Designed & incorporated EDMR into a wafer probe station.
- Experimented with & implemented Rapid-Scan EDMR for increased rates of useful data acquisition.
- Mapped EDMR supply chain, market size, & ROI.
- Facilitated in bringing EDMR capabilities to Technology Development & Corporate Quality Network Reliability Labs.

Summer Scholar

May 2019 – Aug. 2019

The Air Force Research Laboratory

Albuquerque, NM

- Studied the effects of proton irradiation on technological semiconductor materials of interest.
- Acquired & maintained a DoD Secret Clearance (SF-86).
- Established experimental capabilities of photoluminescence under extreme hydrostatic pressure.
- Performed photoluminescence experiments on semiconductor materials as a function of laser power & temperature.
- Performed X-ray diffraction on semiconductor materials.

Manufacturing & Development Engineering Intern

May 2017 – Aug. 2017

TE Connectivity

Middletown, PA

- Designed tooling from concept through prototyping to high production capabilities.
- Mapped current & future value streams for previously existing processes of products.
- Created process flow diagrams to map production methods.
- Programmed original software to quickly & accurately analyze heat transfer data.
- Programmed Keyence inspection system for assembly line implementation.

Engineering Intern

May 2016 – Aug. 2016

Laird Technologies

Stroudsburg, PA

- Designed custom springs & electromagnetic shielding solutions for six customer companies.
- Performed Finite Element Analysis - FEA- for designed solutions, reinforcing customer requirements.
- Optimized existing solutions for more than 50 in house designs.
- Interacted with customers on a regular basis, determining criteria & redesigning when appropriate.

RESEARCH

Publications & Proceedings in Print

- E. B. Frantz, N. J. Harmon, D. J. Michalak, E. M. Henry, M. E. Flatte, S. W. King, J. S. Clarke, and P. M. Lenahan, "Extraction of Dipolar Coupling Constants from Low-Frequency Electrically Detected Magnetic Resonance and Near-Zero Field Magnetoresistance Spectra via Least Squares Fitting to Models Developed from the Stochastic Quantum Liouville Equation," *Journal of Applied Physics*, Accepted, 2021
- E. B. Frantz, D. J. Michalak, N. J. Harmon, E. M. Henry, M. E. Flatté, J. S. Clarke, and P. M. Lenahan, "Effects of ^{29}Si and ^1H on the Near-Zero Field Magnetoresistance Response of Si/SiO_2 Interface States: Implications for Oxide Tunneling Currents,," *Applied Physics Letters*, vol. 119, p. 184 101, 2021. DOI: 10.1063/5.0066640
- E. B. Frantz, D. J. Michalak, N. J. Harmon, E. M. Henry, S. J. Moxim, M. E. Flatté, S. W. King, J. S. Clarke, and P. M. Lenahan, "Electrically detected magnetic resonance and near-zero field magnetoresistance in $^{28}\text{Si}/^{28}\text{SiO}_2$," *Journal of Applied Physics*, vol. 130, p. 065 701, 2021. DOI: 10.1063/5.0057871
- E. B. Frantz, N. J. Harmon, S. R. McMillan, S. J. Moxim, M. E. Flatté, and P. M. Lenahan, "Extraction of Isotropic Electron-Nuclear Hyperfine Coupling Constants of Paramagnetic Point Defects from Near-Zero Field Magnetoresistance Spectra via Least Squares Fitting to Models Developed from the Stochastic Quantum Liouville Equation," *Journal of Applied Physics*, vol. 128, p. 124 504, 2020. DOI: 10.1063/5.0019875
- J. V. Logan, E. B. Frantz, L. K. Casias, M. P. Short, C. P. Morath, and P. T. Webster, "Potential for neutron and proton transmutation doping of GaN and Ga_2O_3 ," *Materials Advances*, vol. 1, p. 45, 2020. DOI: 10.1039/d0ma00017e
- R. J. Waskiewicz, E. B. Frantz, P. M. Lenahan, S. W. King, N. J. Harmon, and M. E. Flatte, "Identifying defects responsible for leakage currents in thin dielectric films," in *IEEE International Integrated Reliability Workshop Final Report*, IEEE, 2018, pp. 1–4. DOI: 10.1109/IIRW.2018.8727077

Presentations

- E. B. Frantz, D. J. Michalak, N. J. Harmon, E. M. Henry, S. J. Moxim, M. E. Flatté, S. W. King, J. S. Clarke, and P. M. Lenahan, "Electrically Detected Magnetic Resonance and Near-Zero Field Magnetoresistance in $^{28}\text{Si}/^{28}\text{SiO}_2$," in *American Vacuum Society 67*, 2021
- E. Frantz, R. Waskiewicz, B. Manning, and P. Lenahan, "Near Zero-Field Magnetoresistance and High and Ultra Low Frequency Electrically Detected Magnetic Resonance in 4H-SiC p-n Junctions," in *APS March Meeting*, Los Angeles, CA, 2018. [Online]. Available: <http://meetings.aps.org/link/BAPS.2018.MAR.E11.8>